

Foundations of Mathematical Reasoning for Economics

Logic, sets, real numbers, inequalities, sums, induction,
the binomial formula, functions, polynomials and limits of sequences

A complete textbook for Topic 1 and the opening of Topics 2–3.

Built from the ECN115 2026–27 lecture slides (Lectures 1 and 2), the Topic 1 study and revision guide, and the January 2024, 2025 and 2026 examination papers as summarised in that guide. Material that goes beyond these sources is marked *Beyond the course materials*; errors and ambiguities in the sources are marked *Note on the source material*.

Contents

	How to Use This Book	iv
	Module Overview	vii
1	The Language of Mathematics	1
1.1	Propositions	1
1.2	Proofs, axioms and theorems.	2
1.3	Implication	3
1.4	Equivalence.	4
1.5	Negation	5
1.6	Quantifiers: “for all” and “there exists”	6
1.7	How proofs are built	7
1.8	Chapter review	7
1.9	Practice questions	9
1.10	Solutions to practice questions	9
2	Sets	11
2.1	Sets and elements.	11
2.2	Equality of sets	13
2.3	Subsets	13
2.4	Intersection and union	14
2.5	Set-builder notation	15
2.6	Set difference	16
2.7	Sets in examination questions	17
2.8	Sets in economics.	18
2.9	Chapter review	18
2.10	Practice questions	19
2.11	Solutions to practice questions	20
3	Number Systems and the Properties of Real Numbers	22
3.1	The four number systems	22
3.2	The properties of the real numbers.	23
3.3	Consequences of the properties	26
3.4	The Least Upper Bound property	27

3.5	Intervals	27
3.6	Chapter review	29
3.7	Practice questions	30
3.8	Solutions to practice questions	30
4	Solving Inequalities	32
4.1	What it means to solve an inequality	32
4.2	Linear inequalities	33
4.3	Products: sign analysis	35
4.4	Quadratic inequalities	37
4.5	Inequalities involving fractions.	40
4.6	Inequalities in economics.	41
4.7	Chapter review	42
4.8	Practice questions	43
4.9	Solutions to practice questions	44
5	Absolute Value and the Triangle Inequality	46
5.1	Definition.	46
5.2	Properties	47
5.3	Absolute value as distance	48
5.4	The triangle inequality	48
5.5	Solving inequalities with absolute values	50
5.6	Chapter review	51
5.7	Practice questions	52
5.8	Solutions to practice questions	53
6	Summation Notation	55
6.1	The summation operator	55
6.2	Counting terms, one-term sums and empty sums	56
6.3	Properties of the summation operator	57
6.4	Computing sums with the properties	58
6.5	Sums in economics	60
6.6	Chapter review	60
6.7	Practice questions	61
6.8	Solutions to practice questions	62
7	Mathematical Induction	64
7.1	The principle of induction	64
7.2	The first example	65
7.3	The structure of an induction proof	67
7.4	An equivalent form of the step	67
7.5	More examples	68
7.6	The alternative (strong) formulation	69
7.7	What induction is not	70

7.8	Chapter review	70
7.9	Practice questions	72
7.10	Solutions to practice questions	72
8	The Binomial Formula	74
8.1	Powers of a sum	74
8.2	Factorials and binomial coefficients	75
8.3	The binomial formula	76
8.4	Using the binomial formula	78
8.5	The binomial formula in economics and finance	79
8.6	Chapter review	79
8.7	Practice questions	80
8.8	Solutions to practice questions	81
9	Functions and Polynomials	82
9.1	The concept of a function.	82
9.2	Polynomials	84
9.3	Factorising a polynomial through a root	84
9.4	Chapter review	87
9.5	Practice questions	88
9.6	Solutions to practice questions	88
10	Sequences and Limits	90
10.1	Sequences	90
10.2	Behaviour at infinity	91
10.3	The definition of a limit	92
10.4	Proving convergence	93
10.5	Showing that a sequence has no limit	94
10.6	Further properties	95
10.7	Limits in economics	96
10.8	Chapter review	96
10.9	Practice questions	98
10.10	Solutions to practice questions	98
	Final Master Review	100
	Module summary	100
	How the ideas connect	101
	Master formula sheet	102
	Glossary	103
	What do I actually need to know?	108
	Final practice paper	110
	Solutions to the final practice paper	111

How to Use This Book

This book is a self-contained textbook for the opening part of ECN115 *Mathematical Methods in Economics and Finance*. It covers everything presented in the first two lectures of the module, which together make up Topic 1 (*Introduction*) and the first steps into Topic 2 (*Functions*) and Topic 3 (*Limits and smooth functions*). It is written so that a student who has never seen the lecture slides can learn the material from these pages alone, and so that a student who has seen them can use it to understand every step properly rather than memorise it.

The slides present ideas in the order in which they were spoken. A textbook can do better: here the material has been reorganised so that every idea is introduced only after the ideas it depends on. For example, quantifiers (“for all” and “there exists”) appear in the second lecture but are needed much earlier to state definitions precisely, so they are taught in Chapter 1 alongside the rest of the language of mathematics.

How each chapter is organised

Every chapter follows the same pattern, so that you always know where to find things.

1. **Introduction and prerequisites.** A short paragraph saying what the chapter is about and why it matters, followed by a grey box listing what you need to know first, with references back to earlier chapters.
2. **The core material.** Definitions, explanations, results and proofs, developed in prose. Each important idea is explained four ways: what it says (the definition), what it means (intuition), why it matters (its role in the module and in economics), and how to use it (worked examples).
3. **Worked examples.** Every example from the lectures is reproduced and explained in full, usually with more steps shown than on the slides. Further examples have been added wherever an idea needed more practice.
4. **Chapter review.** A summary in prose, the key definitions and key equations collected in one place, connections to other chapters, the common mistakes, and an exam checklist.
5. **Practice questions with full solutions.** Questions are labelled by type: *basic* (direct use of a definition), *conceptual* (explain why), *mathematical* (calculation or proof), *application* (an economic or applied setting), *multi-step* (several ideas combined) and *exam-style* (written in the format of the past ECN115 papers). Every question uses only material covered up to that point. Full solutions follow the questions.

The boxes

Different kinds of content are set in different boxes so that you can see at a glance what you are reading.

Definition

A precise statement of what a term means. Definitions must be learned exactly: past ECN115 papers award marks for stating them.

Theorem / Proposition

A statement that has been, or will be, proved from the definitions and axioms.

Worked Example

A problem solved step by step, with the reasoning behind each step.

CORE CONCEPT

An idea on which a large part of the module rests. If you are short of time, make sure you understand every box with this label.

Key Formula

An important formula, given the full treatment: what each symbol means, where it comes from, when to use it, an example, common mistakes, rearrangements, what happens when the inputs change, and how it appears in questions.

Common mistake

An error that students frequently make, and how to avoid it.

Recognising this in an exam

How a type of question is usually worded and what it is really asking.

Note on the source material

A place where the slides or the study guide contain an error, a typographical slip, or an ambiguity. The correct version is given, and nothing is silently changed.

Beyond the course materials

Knowledge that is not in the ECN115 materials but helps you understand them. You are not expected to reproduce it unless your lecturer says otherwise.

Note

A remark, a qualification or a piece of context.

Source tags

Small grey tags such as [L1, slide 17] at the right-hand edge of a box tell you where the material comes from: *L1* and *L2* are the two lecture slide sets, *SG* is the Topic 1 study and revision guide, and *Jan 2024* and similar labels are past examination papers. Use these tags to find the original when you want to compare.

A note on examinability

This book never claims that a particular item will be examined. Where the course materials themselves say something about assessment it is reported: for example, the slides describe the Least Upper Bound property as “extra material, non-examinable”. Where past papers show that a type of question has been set before, the book says so, because that is a matter of record; it is not a prediction.

Suggested way to study

On a first reading, work through each chapter in order and attempt every worked example yourself before reading its solution. On a second reading, concentrate on the CORE CONCEPT boxes, the key formulas and the chapter reviews, and attempt the practice questions with the solutions covered. Before an assessment, use the *Final Master Review* at the end of the book: it contains a summary of the whole module, a concept map, a master formula sheet, a glossary, a checklist of everything you should be able to do, and a final set of mixed practice questions with solutions.

Module Overview

The module

ECN115 *Mathematical Methods in Economics and Finance* is taught at Queen Mary University of London in 2026–27 by Dr Evgenii Safonov (e-mail e.safonov@qmul.ac.uk, office GC332, office hours Tuesdays 15:00–17:00). Lectures take place on Wednesdays 09:00–11:00 (Arts Two lecture theatre) and Thursdays 09:00–11:00 (Bancroft Mason lecture theatre); classes are listed on MyTimetable, and undergraduate support is through AskQM. [L1, slides 2–6]

Assessment consists of a one-hour closed-book midterm test worth 30% (on 11 November 2026) and a two-hour closed-book final examination worth 70% (in January). Calculators were not permitted on any of the three past papers summarised in the study guide. Weekly homework is not graded but is described as an important component: each week you attempt the homework, the most challenging problems are analysed in class, and then solutions are posted. As the slides put it, “it is almost impossible to learn mathematics without practice”.

No textbook is required. The slides suggest three for supplementary reading: Sydsæter, Hammond and others, *Essential Mathematics for Economic Analysis*; Simon and Blume, *Mathematics for Economists*; and, for more advanced study, Sydsæter, Hammond, Seierstad and Strøm, *Further Mathematics for Economic Analysis*.

The six topics of the module

- Topic 1. Introduction.** Mathematical arguments. Sets, numbers, vectors and sequences. Absolute value, inequalities, mathematical induction, the binomial formula.
- Topic 2. Functions.** The concept of a function. Elementary numerical functions. Basic properties of functions. Relationships between functions.
- Topic 3. Limits and smooth functions.** Limits of sequences. Continuity and differentiability of functions. Properties of the derivative. The chain rule and other rules of differentiation. Higher-order derivatives.
- Topic 4. Univariate optimisation.** The existence of maxima and minima. The first-order condition for optimality. Corner solutions. Dependence of the solution on the parameter.
- Topic 5. Linear algebra.** Systems of linear equations. Matrix algebra. Matrix inversion.
- Topic 6. Multivariate analysis and optimisation.** Multivariate functions. Partial derivatives. Unconstrained and constrained optimisation: existence, first-order conditions.

What this book covers

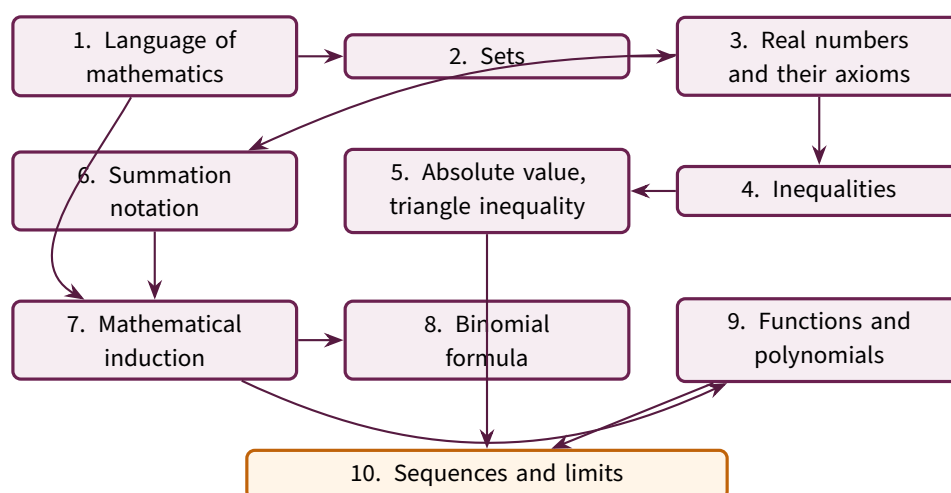
The materials provided for this book are the slides of Lectures 1 and 2, a study and revision guide for Topic 1 (which also summarises the January 2024, 2025 and 2026 papers), and nothing else. The book therefore covers exactly the content of those two lectures: all of Topic 1 except vectors, the concept of a function and polynomials from Topic 2, and the definition of the limit of a sequence from Topic 3. The slides list the plan for the following two lectures (properties of limits, calculating limits, infinite sums and an introduction to derivatives in Lecture 3; more functions and derivatives in Lecture 4), but no material from those lectures was provided, so it is not covered here.

Coverage

The Topic 1 syllabus lists *vectors*, but vectors do not appear in the slides of Lectures 1 and 2. They are therefore not taught in this book. If vectors are covered later in the module, treat that material as an addition to Chapters 2 and 3.

How the chapters fit together

The chapters build on one another as shown in the map below. The first three chapters provide the language (logic and sets) and the objects (real numbers and their properties). Chapters 4 and 5 use the properties of real numbers to solve inequalities, which is the single most frequently used skill in the module. Chapters 6–8 develop sums and proof by induction, and use them to prove the binomial formula. Chapter 9 defines functions and proves that a polynomial can be factorised through each of its roots. Chapter 10 brings everything together in the definition of the limit of a sequence, which uses quantifiers, absolute values, inequalities and the triangle inequality all at once, and which is the foundation of Topic 3.



Map of the book. How the chapters depend on one another. An arrow from one chapter to another means that the second uses ideas from the first. The limit of a sequence (Chapter 10) draws on almost everything before it.

Where the lectures appear in this book

Lecture material	Slides	Chapter
Elementary logic: propositions, proof, axiom, theorem, $\Rightarrow$, $\Leftrightarrow$, negation	L1, 8	1
“There exists” and “for all”	L2, 21	1
Sets, subsets, intersection, union, set-builder notation, set difference	L1, 10–14	2
Number systems, properties O1–O2, A1–A5, M1–M6, Least Upper Bound, intervals	L1, 16–20	3
Basic and quadratic inequalities (Examples 1–3)	L1, 22–25	4
Absolute value, triangle inequality	L2, 10–11	5
Summation operator and its properties	L1, 27–30	6
Mathematical induction	L1, 32–36; L2, 4–8	7
The binomial formula, factorials, binomial coefficients	L2, 13–14	8
Functions, polynomials, Theorem L2.1	L2, 16–19	9
Sequences, numerical sequences, limits	L2, 22–30	10

Why mathematics in an economics degree?

Economics makes claims about how people, firms and governments behave and about the consequences of that behaviour. Many of these claims are precise enough to be written down mathematically: a consumer chooses the bundle that maximises utility subject to a budget, a firm chooses output to maximise profit, an interest rate compounds over time, a sequence of quarterly output figures approaches a long-run level. Writing such claims mathematically has two great advantages. First, it forces precision: a statement such as “demand falls when the price rises” becomes an inequality that can be checked. Second, it allows *proof*: once the assumptions are written down, their consequences can be derived with certainty rather than argued about. The material in this book is the toolkit for both. Inequalities describe constraints and comparisons; sets describe the options available to a decision-maker; sums describe totals over time or over people; induction proves statements about every period at once; functions describe relationships such as cost as a function of output; and limits describe long-run behaviour.

The Language of Mathematics

Mathematics is written in a small, precise language. A handful of words — proposition, proof, axiom, theorem, implies, if and only if, not, for all, there exists — carry almost all of the logical weight in every argument you will meet in this module. This chapter teaches that language. Everything later in the book, from solving an inequality to proving that a sequence has a limit, is written in it.

Before you start: prerequisites

None. This chapter starts from the beginning. It helps to be comfortable with elementary algebra (rearranging an equation such as $2x + 1 = 0$) because the examples use it.

1.1 Propositions

The basic unit of a mathematical argument is a sentence that makes a claim which is either correct or incorrect.

Definition 1.1 (Proposition)

A **proposition** is a statement that may be either true or false.

[L1, slide 8]

The definition has two parts. A proposition is a *statement* — a complete declarative sentence, not a question, a command or a fragment — and it must be possible to say that it is *true* or *false*. It does not matter whether we know which: “every even number greater than 2 is the sum of two primes” is a proposition even though nobody currently knows whether it is true.

Worked Example 1.1 – Recognising propositions

Decide which of the following are propositions.

- (a) $2 + 3 = 5$.
- (b) $2 + 3 = 6$.
- (c) $2 + 3$.
- (d) Is 7 greater than 4?
- (e) The unemployment rate in the United Kingdom in March 2026 was below 5%.

(f) $x > 3$.

Solution. (a) is a proposition and it is true. (b) is a proposition and it is false; being false does not stop a statement from being a proposition. (c) is not a statement at all: it is the name of a number, with no verb and no claim. (d) is a question, so it is neither true nor false. (e) is a proposition: it is either true or false as a matter of fact, whether or not you know the figure. (f) needs care. On its own, " $x > 3$ " is neither true nor false until we know what x is: it is true when $x = 5$ and false when $x = 1$. Such a sentence is a *proposition depending on x* , sometimes called an *open sentence* or a *property of x* . It becomes a proposition once a value is substituted for x , or once it is turned into a claim about many values of x with a quantifier (Section 1.6).

Propositions depending on a variable

Much of this module works with statements such as " $x > 3$ " or " $P_n: 1+2+\dots+n = n(n+1)/2$ ", which depend on a variable. We write $P(x)$ or P_n to show the dependence. Solving an inequality means finding the set of all x for which $P(x)$ is true (Chapter 4); proving a statement by induction means showing that P_n is true for every natural number n (Chapter 7).

1.2 Proofs, axioms and theorems

Definition 1.2 (Proof, axiom, theorem)

- A **proof** is a demonstration that one proposition follows from one or more other propositions.
- An **axiom** is a proposition that is accepted without proof.
- A **theorem** is a proposition that has been proved on the basis of axioms and previously established theorems.

[L1, slide 8]

These three words describe how mathematical knowledge is organised. Any argument has to start somewhere: we cannot prove everything from nothing, because every proof needs something to start from. The starting points are the *axioms*. From them we derive *theorems*, by *proofs* in which each step follows logically from earlier steps. Once a theorem has been proved it may be used in later proofs exactly as if it were an axiom.

Intuition

Think of the axioms as the rules of a game and the theorems as positions that can be reached by legal moves. A proof is the record of the moves. Nobody asks you to justify the rules of chess; they ask you to show that a position can be reached by playing by the rules.

In ECN115 the axioms that matter most are the properties of the real numbers listed in Chapter 3 (labelled O1–O2, A1–A5 and M1–M6). They describe how addition, multiplication and the order " $>$ " behave. Everything we do with inequalities is a sequence of steps, each justified by one of

these properties. The slides make the point explicitly when they solve $2x + 1 > 0$ by naming, at each step, which property is being used (Section 4.2).

An axiom is not “a fact too obvious to prove”

An axiom is a starting assumption, chosen because we have to start somewhere. Some very obvious facts are *not* axioms in this module. For example, $0 \cdot a = 0$ for every real number a is not on the list of properties O1–M6; it is a theorem that can be derived from them (Proposition 3.2). Whether something is an axiom depends on which list of assumptions has been chosen, not on how obvious it is.

1.3 Implication

Definition 1.3 (Implication)

Given propositions P and Q , we write $P \Rightarrow Q$ for the proposition that P **implies** Q , that is, “if P , then Q ”.

[L1, slide 8]

The proposition P is called the *hypothesis* (or antecedent) and Q the *conclusion* (or consequent). The claim $P \Rightarrow Q$ is a promise: *whenever P is true, Q is true as well*. It says nothing about what happens when P is false.

When is “ $P \Rightarrow Q$ ” true?

The slides define implication in words. In formal logic, the truth of $P \Rightarrow Q$ is fixed by the following table, where T means true and F means false.

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

The only way to break the promise “if P then Q ” is for P to be true and Q to be false. If P is false the promise has not been tested, so the implication counts as true.

Worked Example 1.2 – Implications in algebra

Consider, for real numbers x , the propositions P : “ $x > 2$ ” and Q : “ $x^2 > 4$ ”.

- (a) Is $P \Rightarrow Q$ true for every real x ?
- (b) Is $Q \Rightarrow P$ true for every real x ?

Solution. (a) Yes. If $x > 2$ then x is positive and $x \cdot x > 2 \cdot x > 2 \cdot 2 = 4$ (multiplying an inequality by a positive number keeps its direction, property M6 in Chapter 3). So $x^2 > 4$.

(b) No. Take $x = -3$. Then $x^2 = 9 > 4$, so Q is true, but $x > 2$ is false. One such value, called

a **counterexample**, is enough to show that " $Q \Rightarrow P$ for every x " is false.

Ways of saying " $P \Rightarrow Q$ ". The same implication is expressed in many ways in English, and you need to recognise all of them: "if P then Q ", " P implies Q ", " Q if P ", " Q follows from P ", " P only if Q ", " P is a *sufficient* condition for Q ", and " Q is a *necessary* condition for P ".

The words *necessary* and *sufficient* are used constantly in economics (for example, "a first-order condition is necessary for an interior maximum"). In $P \Rightarrow Q$, knowing P is *sufficient* to know Q ; and Q is *necessary* for P , because P cannot hold without Q .

"Only if" points the same way as "implies"

" P only if Q " means $P \Rightarrow Q$, not $Q \Rightarrow P$. "A firm makes a profit only if price exceeds average cost" means "profit $\Rightarrow$ price $>$ average cost".

1.3.1 Converse and contrapositive

Converse and contrapositive

From an implication $P \Rightarrow Q$ we can build two related implications.

- The **converse** is $Q \Rightarrow P$. It is a *different* claim: the example above shows that $x > 2 \Rightarrow x^2 > 4$ is true while its converse is false.
- The **contrapositive** is $(\text{not } Q) \Rightarrow (\text{not } P)$. It is *equivalent* to the original: an implication and its contrapositive are always both true or both false. For example, "if $x > 2$ then $x^2 > 4$ " has contrapositive "if $x^2 \leq 4$ then $x \leq 2$ ".

Confusing an implication with its converse is one of the most common errors in all of mathematics.

1.4 Equivalence

Definition 1.4 (Equivalence)

When $P \Rightarrow Q$ and $Q \Rightarrow P$, we write $P \Leftrightarrow Q$ and say that P and Q are **equivalent**. [L1, slide 8]

" $P \Leftrightarrow Q$ " is read " P if and only if Q ", often shortened to " P iff Q ". The "if" half is $Q \Rightarrow P$ and the "only if" half is $P \Rightarrow Q$. Equivalent propositions are true in exactly the same circumstances.

CORE CONCEPT | Why equivalence matters when solving

When you solve an equation or inequality, you transform it step by step. If each step is an *equivalence*, the final, simple statement is true for exactly the same values of x as the original, so it describes the solution set exactly. If some step is only a one-way implication, the final statement may be true for values that do not satisfy the original, and the "answer" may be wrong.

Worked Example 1.3 – A one-way step that introduces a false solution

Solve $\sqrt{x+2} = x$ for real $x \geq -2$.

Solution. Squaring both sides gives $x+2 = x^2$, that is, $x^2 - x - 2 = 0$, so $(x-2)(x+1) = 0$ and $x = 2$ or $x = -1$. But squaring is only a one-way step: $\sqrt{x+2} = x \Rightarrow x+2 = x^2$ is true, while the converse is not (because x could be negative). Checking in the original equation: $x = 2$ gives $\sqrt{4} = 2$, which is correct; $x = -1$ gives $\sqrt{1} = 1 \neq -1$, which is wrong. The solution is $x = 2$ only. The false “solution” $x = -1$ appeared because one step was an implication rather than an equivalence.

The slides illustrate the same principle with inequalities: in solving $2x + 1 > 0$ they show not only $2x + 1 > 0 \Rightarrow 2x > -1$ but also the reverse, $2x > -1 \Rightarrow 2x + 1 > 0$, and only then conclude $2x + 1 > 0 \Leftrightarrow 2x > -1$ (Worked Example 4.1).

1.5 Negation

Definition 1.5 (Negation)

The proposition “**not** P ” is true when the proposition P is false, and “not P ” is false when P is true.

[L1, slide 8]

Negation reverses the truth value. Forming the negation of a mathematical statement correctly is a skill in its own right, needed for proofs by contradiction and for showing that something does *not* happen (for example, that a sequence does not have a limit, Worked Example 10.7).

Negating an inequality. For real numbers, exactly one of $a > b$, $a = b$ and $a < b$ is true (property O1, Chapter 3). Therefore

$$\text{not}(x > 0) \iff x \leq 0, \quad \text{not}(x \leq 3) \iff x > 3.$$

The negation of “ $x > 0$ ” is *not* “ $x < 0$ ”: it must also include $x = 0$.

Negating “and” and “or”: De Morgan’s laws

The study guide adds the two rules for negating compound statements, known as De Morgan’s laws:

$$\text{not}(P \text{ and } Q) \iff (\text{not } P) \text{ or } (\text{not } Q), \quad \text{not}(P \text{ or } Q) \iff (\text{not } P) \text{ and } (\text{not } Q).$$

For example, the negation of “ $x > 1$ and $y > 1$ ” is “ $x \leq 1$ or $y \leq 1$ ”. In mathematics “or” is always *inclusive*: “ P or Q ” is true when at least one of them is true, including when both are. Finally, the negation of an implication is $\text{not}(P \Rightarrow Q) \iff P \text{ and } (\text{not } Q)$: to deny a promise you must show that the condition held and the promised outcome did not.

1.6 Quantifiers: “for all” and “there exists”

Statements about *many* objects at once need two further symbols.

Definition 1.6 (Quantifiers)

Let A be a set and P a property (a proposition depending on an element x).

- “ $\exists x \in A : P$ ”, or “ $\exists x \in A$ such that P ”, means *there exists at least one element x in the set A such that the property P holds for x* . The symbol $\exists$ is the **existential quantifier**.
- “ $\forall x \in A, P$ ” means *for all elements x of the set A , the property P is true*. The symbol $\forall$ is the **universal quantifier**.

[L2, slide 21]

Worked Example 1.4 – Quantifiers on a finite set

Let $A = \{2, 4, 6, 8, 10, 12\}$. Show that $\exists x \in A : x \geq 10$ and that $\forall x \in A, x < 20$.

Solution. For the first statement it is enough to exhibit one element: $10 \in A$ and $10 \geq 10$ (so is 12). For the second we must check every element: 2, 4, 6, 8, 10, 12 are all less than 20. Both statements are therefore true.

[L2, slide 21]

CORE CONCEPT | How to prove and disprove quantified statements

- To prove $\exists x \in A : P(x)$, exhibit one particular $x \in A$ and check that $P(x)$ holds.
- To prove $\forall x \in A, P(x)$, take an *arbitrary* $x \in A$ — one about which you assume nothing except that it belongs to A — and show that $P(x)$ holds. Checking examples is not enough unless A is finite and you check every element.
- To disprove $\forall x \in A, P(x)$, find one counterexample: an $x \in A$ for which $P(x)$ is false.
- To disprove $\exists x \in A : P(x)$, show that $P(x)$ is false for every $x \in A$.

1.6.1 Negating quantified statements

Negation swaps the quantifiers

$$\text{not}(\forall x \in A, P(x)) \Leftrightarrow \exists x \in A : \text{not } P(x), \quad \text{not}(\exists x \in A : P(x)) \Leftrightarrow \forall x \in A, \text{not } P(x).$$

“Not every student passed” means “some student did not pass”; “there is no student who passed” means “every student failed”. When negating, the set A after the quantifier does not change: the negation of a claim about real numbers is still a claim about real numbers.

1.6.2 The order of quantifiers

When a statement contains several quantifiers, their order matters.

Worked Example 1.5 – Order of quantifiers

Compare

$$(i) \forall n \in \mathbb{Z}, \exists m \in \mathbb{Z} : m > n \quad \text{and} \quad (ii) \exists m \in \mathbb{Z} : \forall n \in \mathbb{Z}, m > n.$$

Solution. Statement (i) says that for every integer n there is some integer larger than it. This is true: given n , take $m = n + 1$. The choice of m is allowed to depend on n .

Statement (ii) says that there is a single integer m that is larger than every integer. This is false: whatever m is proposed, the integer $n = m$ is not smaller than it. Here m must be chosen first and then work for every n simultaneously.

The definition of the limit of a sequence (Chapter 10) has the shape “ $\forall \varepsilon > 0, \exists N \in \mathbb{N} : \forall n > N, \dots$ ”. Reading it correctly depends entirely on this point: N comes after ε , so N may depend on ε .

1.7 How proofs are built

The proofs in this book use a small number of patterns. They are introduced in the chapters where they are first needed, and are listed here so that you can recognise them.

Direct proof (chain of implications)

Start from what is given and reach the conclusion by steps each of which follows from earlier ones. Used for almost every computation, for example solving an inequality (Chapter 4).

Proof of an equivalence

Prove $P \Rightarrow Q$ and $Q \Rightarrow P$ separately, or build a chain of steps each of which is reversible.

Proof by cases

If one of several cases must hold, and the conclusion follows in each case, the conclusion holds. Used for the sign analysis of a product (Worked Example 4.6) and for absolute values (Chapter 5).

Proof by contradiction

To prove P , assume “not P ” and derive something impossible (for example $1 < \frac{1}{2}$). Then “not P ” cannot hold, so P does. Used to show that the sequence $1, 0, 1, 0, \dots$ has no limit (Worked Example 10.7).

Proof by mathematical induction

To prove P_n for every $n \in \mathbb{N}$, prove P_1 and prove that $P_n \Rightarrow P_{n+1}$ for every n (Chapter 7).

1.8 Chapter review

Summary

A proposition is a statement that is either true or false. Mathematics organises propositions into axioms (accepted without proof) and theorems (proved from axioms and earlier theorems), connected by proofs. Propositions are combined with “implies” ($\Rightarrow$), “if and only if” ($\Leftrightarrow$) and

“not”. An implication is a one-way promise; an equivalence is a two-way one, and only a chain of equivalences is guaranteed to preserve a solution set exactly. Quantifiers turn properties of an element into claims about whole sets: $\exists$ is proved by an example, $\forall$ by an argument about an arbitrary element, and a false $\forall$ is disproved by one counterexample. Negation swaps $\forall$ and $\exists$, and the order of mixed quantifiers changes the meaning.

Key definitions

- **Proposition:** a statement that may be either true or false.
- **Proof:** a demonstration that one proposition follows from one or more others.
- **Axiom:** a proposition accepted without proof.
- **Theorem:** a proposition proved from axioms and previously established theorems.
- $P \Rightarrow Q$: if P then Q . $P \Leftrightarrow Q$: $P \Rightarrow Q$ and $Q \Rightarrow P$. **not** P : true exactly when P is false.
- $\exists x \in A : P$: at least one $x \in A$ has property P . $\forall x \in A, P$: every $x \in A$ has property P .

Key logical equivalences

$$P \Leftrightarrow Q \text{ means } (P \Rightarrow Q) \text{ and } (Q \Rightarrow P); \quad \text{not}(x > a) \Leftrightarrow x \leq a;$$

$$\text{not}(\forall x \in A, P) \Leftrightarrow \exists x \in A : \text{not } P; \quad \text{not}(\exists x \in A : P) \Leftrightarrow \forall x \in A, \text{not } P.$$

Connections

The vocabulary of this chapter is used throughout. Set equality is defined by a two-way condition (Chapter 2); the properties of the real numbers are axioms (Chapter 3); solving inequalities is the construction of a chain of equivalences (Chapter 4); induction is a way of proving a $\forall$ statement about $\mathbb{N}$ (Chapter 7); and the definition of a limit is a statement with three nested quantifiers (Chapter 10).

Common mistakes

- Treating $P \Rightarrow Q$ as if it meant $Q \Rightarrow P$ (confusing an implication with its converse).
- Reading “ P only if Q ” as $Q \Rightarrow P$.
- Negating $x > 0$ as $x < 0$ instead of $x \leq 0$.
- “Proving” a $\forall$ statement by checking a few examples.
- Swapping the order of $\forall$ and $\exists$.
- Using a one-way step (such as squaring) when solving and not checking the answers.

Exam checklist

- ☐ Define proposition, proof, axiom and theorem in the words of the course.
- ☐ Explain the difference between $\Rightarrow$ and $\Leftrightarrow$, with an example.
- ☐ Form the negation of an inequality and of a quantified statement.

- Prove an $\exists$ statement with an example and disprove a $\forall$ statement with a counterexample.
- Explain why the order of quantifiers matters, using Worked Example 1.5.

1.9 Practice questions

Question 1.1 [basic]

Which of the following are propositions? For each proposition, say whether it is true or false. (a) 7 is an even number. (b) $3x + 2$. (c) $\sqrt{16} = 4$. (d) Calculate 5×6 . (e) Every natural number is greater than 0.

Question 1.2 [basic]

State the converse and the contrapositive of: “If a firm’s revenue exceeds its costs, then the firm makes a profit.”

Question 1.3 [conceptual]

Explain why checking that a statement holds for $n = 1, 2, \dots, 1000$ does not prove that it holds for every natural number n .

Question 1.4 [mathematical]

Write the negation of each statement, simplifying as far as possible. (a) $x \geq 5$. (b) $\forall x \in \mathbb{R}, x^2 \geq 0$. (c) $\exists n \in \mathbb{N} : n^2 = 2$. (d) $x > 1$ and $x < 4$.

Question 1.5 [mathematical]

Let $A = \{1, 3, 5, 7, 9\}$. Decide whether each statement is true, with a reason. (a) $\forall x \in A, x$ is odd. (b) $\exists x \in A : x > 8$. (c) $\forall x \in A, x < 9$. (d) $\exists x \in A : x^2 = 25$.

Question 1.6 [application]

An economist claims: “For every price $p > 0$ there is a quantity $q > 0$ that the consumer is willing to buy.” A second economist claims: “There is a quantity $q > 0$ that the consumer is willing to buy at every price $p > 0$.” Write both claims with quantifiers and explain why they say different things.

Question 1.7 [multi-step]

Consider the claim “for all real x , if $x^2 = 9$ then $x = 3$ ”. (a) Is it true? Justify. (b) State its converse and decide whether the converse is true. (c) Replace the conclusion so that the resulting implication is true and its converse is also true, and write it as an equivalence.

Question 1.8 [exam-style]

(a) Explain what is meant by a *proposition*, an *axiom* and a *theorem*. (b) Let $A = \{x \in \mathbb{R} : 0 < x < 1\}$. Determine whether each of the following is true, giving a proof or a counterexample: (i) $\forall x \in A, x^2 < x$; (ii) $\exists x \in A : x^2 = x$.

1.10 Solutions to practice questions

Solution to Question 1.1

(a) A proposition; false. (b) Not a proposition: it is an expression (a number once x is known), not a statement. (c) A proposition; true. (d) Not a proposition: it is an instruction. (e) A proposition; true ($\mathbb{N} = \{1, 2, 3, \dots\}$, Chapter 3).

Solution to Question 1.2

Let P be “revenue exceeds costs” and Q be “the firm makes a profit”. The converse is “If the firm makes a profit, then its revenue exceeds its costs.” The contrapositive is “If the firm does not make a profit, then its revenue does not exceed its costs.” (With profit defined as revenue minus costs, all three statements are true here, but in general only the contrapositive is guaranteed to have the same truth value as the original.)

Solution to Question 1.3

The statement is a claim about infinitely many propositions $P_1, P_2, P_3, \dots$. Checking 1000 of them establishes those 1000 and says nothing about P_{1001} or any later one; however many cases are checked, infinitely many remain. A proof must cover an *arbitrary* natural number, for example by mathematical induction (Chapter 7). There are famous statements that hold for very many initial cases and then fail.

Solution to Question 1.4

(a) $x < 5$. (b) $\exists x \in \mathbb{R} : x^2 < 0$ (a false statement, as it should be, since the original is true). (c) $\forall n \in \mathbb{N}, n^2 \neq 2$. (d) By De Morgan’s law, “ $x \leq 1$ or $x \geq 4$ ”.

Solution to Question 1.5

(a) True: 1, 3, 5, 7, 9 are all odd. (b) True: $9 \in A$ and $9 > 8$. (c) False: $9 \in A$ but $9 < 9$ is false; $x = 9$ is a counterexample. (d) True: $x = 5$.

Solution to Question 1.6

The first claim is $\forall p > 0, \exists q > 0$: the consumer buys q at price p . The quantity may depend on the price, which is how demand normally works. The second is $\exists q > 0 : \forall p > 0$, the consumer buys q at price p : one fixed quantity would be bought however high the price, which is a far stronger (and economically implausible) claim. The quantifiers are the same; only their order differs, and that changes the meaning (Worked Example 1.5).

Solution to Question 1.7

(a) False: $x = -3$ satisfies $x^2 = 9$ but $x \neq 3$. (b) The converse is “if $x = 3$ then $x^2 = 9$ ”, which is true. (c) “If $x^2 = 9$ then $x = 3$ or $x = -3$ ” is true (since $x^2 - 9 = (x - 3)(x + 3) = 0$ forces one factor to be zero), and its converse is true as well, so $x^2 = 9 \Leftrightarrow (x = 3 \text{ or } x = -3)$.

Solution to Question 1.8

(a) A proposition is a statement that may be either true or false. An axiom is a proposition accepted without proof. A theorem is a proposition that has been proved on the basis of axioms and previously established theorems. (b)(i) True. Take an arbitrary $x \in A$, so $0 < x < 1$. Multiplying $x < 1$ by the positive number x keeps the direction (M6): $x \cdot x < 1 \cdot x$, that is, $x^2 < x$. Since x was arbitrary, the statement holds for all $x \in A$. (ii) False. If $x^2 = x$ then $x(x - 1) = 0$, so $x = 0$ or $x = 1$; neither lies in A . Hence no $x \in A$ satisfies $x^2 = x$. (Alternatively, (ii) contradicts (i).)

Sets

A set is a collection of objects. Sets are the containers of mathematics: the solutions of an inequality form a set, the options open to a consumer form a set, the natural numbers form a set. This chapter introduces the notation for sets, the relations between them (membership, subset, equality) and the operations that build new sets from old ones (intersection, union and difference).

Before you start: prerequisites

The language of Chapter 1: propositions, implication and equivalence (Sections 1.3 and 1.4), and the meaning of “for all” and “there exists” (Section 1.6).

2.1 Sets and elements

Definition 2.1 (Set, element)

A **set** is a collection of objects. If the set A consists of the objects $x, y, z, \dots$, we write $A = \{x, y, z, \dots\}$. If an object x is in the set A , we write $x \in A$ and call x an **element** of A . If not, we write $x \notin A$.

[L1, slide 10]

! “What is a collection?”

The slide defines a set as “a collection (what is a collection?) of objects”. The parenthesis is deliberate: “collection” is no more precise than “set”, so the definition is not really a definition. In this module “set” is treated as a basic, undefined notion, as it is in most of mathematics, and you will not be asked to define it from anything simpler. What matters is how sets are *used*: an object is either an element of a given set or it is not.

Two features of sets follow from the fact that a set is determined only by *which* objects belong to it.

Order and repetition do not matter

$\{1, 2, 3\}$, $\{3, 1, 2\}$ and $\{1, 1, 2, 3, 3\}$ are the same set: each contains exactly the elements 1, 2 and 3. Listing an element twice does not put it in the set twice, and the order of the list has no meaning. (This follows from the definition of equality in Section 2.2.)

Worked Example 2.1 – Membership

Let $A = \{2, 4, 6, 8\}$ and $B = \{\text{apples}, \text{oranges}\}$. Then $4 \in A$, $5 \notin A$, $\text{apples} \in B$ and $\text{bananas} \notin B$. Elements of a set can be any objects at all, including other sets: the set $C = \{\{1, 2\}, \{3\}\}$ has two elements, the set $\{1, 2\}$ and the set $\{3\}$. Note that $1 \notin C$: the number 1 is an element of an element of C , not of C itself.

Definition 2.2 (Empty set)

A set that contains no objects is called the **empty set**. It is denoted by the symbol $\emptyset$; sometimes it is written $\{\}$.

[L1, slide 10]

The empty set is the set analogue of the number zero. It appears naturally as the answer to questions whose answer is “nothing”: the set of real numbers x with $x^2 < 0$ is $\emptyset$, and so is the set of solutions of the inequality $0 > 1$.

$\emptyset$ is not the same as $\{\emptyset\}$

$\emptyset$ has no elements. $\{\emptyset\}$ is a set with exactly one element, namely the empty set. A box containing an empty box is not an empty box.

2.1.1 Not everything is a set

Russell's paradox

The slides remark that not everything can be described as a set: for example, there is no “set of all sets” (nor certain similar collections), because assuming that there is leads to a contradiction known as *Russell's paradox*.

[L1, slide 10]

What goes wrong

Suppose any description at all could define a set. Consider $R = \{x : x \notin x\}$, the set of all sets that are not elements of themselves, and ask whether $R \in R$. If $R \in R$, then R satisfies the condition defining R , so $R \notin R$. If $R \notin R$, then R satisfies the condition, so $R \in R$. Either answer contradicts itself. The way out is to restrict which collections count as sets; in particular, the collection of all sets is not one. Nothing in ECN115 comes close to these limits, which is why “set” can safely be used informally.

2.2 Equality of sets

Definition 2.3 (Equal sets)

Two sets A and B are **equal**, written $A = B$, if every element of A is also an element of B , and every element of B is an element of A . [L1, slide 10]

In the language of Chapter 1, $A = B$ means: for every object x , $x \in A \Leftrightarrow x \in B$. Sets are equal when they have exactly the same elements, however they are described. For example, $\{x \in \mathbb{R} : x^2 = 1\} = \{-1, 1\}$, and the set of even prime numbers equals $\{2\}$.

2.3 Subsets

Definition 2.4 (Subset, superset)

If each element of A is also an element of B , then we write $A \subseteq B$ and call A a **subset** of B , and B a **superset** of A . [L1, slide 11]

With quantifiers: $A \subseteq B$ means $\forall x \in A, x \in B$. To prove $A \subseteq B$ you take an arbitrary element of A and show it lies in B . To disprove it you find one element of A that is not in B .

! Notation: $\subseteq$ and $\subset$

The slides note that some textbooks write $A \subset B$ for “ A is a subset of B ”. Be aware that many other texts use $A \subset B$ to mean a *proper* subset (a subset with $A \neq B$). This book always writes $\subseteq$ for “subset”, which is unambiguous. When reading another source, check which convention it uses.

Proposition 2.1 (Every set is a subset of itself)

For every set A , $A \subseteq A$. [L1, slide 12]

Proof. Every element of A is an element of A . □

CORE CONCEPT | Equality is two subset relations

$$A = B \iff (A \subseteq B \text{ and } B \subseteq A).$$

To prove that two sets are equal, prove that each is a subset of the other. [L1, slide 11]

Proposition 2.2 (Equality via subsets)

For sets A and B , $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

Proof. By Definition 2.3, $A = B$ means “every element of A is an element of B ” and “every element of B is an element of A ”. By Definition 2.4, the first part says exactly $A \subseteq B$ and the second says exactly $B \subseteq A$. So the two conditions are the same statement written in different words, and each implies the other. $\square$

The empty set is a subset of every set

For every set A , $\emptyset \subseteq A$. To disprove it we would need an element of $\emptyset$ that is not in A , and $\emptyset$ has no elements at all. A statement of the form “every element of $\emptyset$ has property P ” is therefore true whatever P is; such statements are called *vacuously true*.

2.4 Intersection and union

Definition 2.5 (Intersection and union)

- The collection of objects in both set A and set B is written $A \cap B$ and called the **intersection** of the two sets.
- The collection of objects in either set A , or set B , or in both sets A and B is written $A \cup B$ and called the **union** of the two sets.

[L1, slide 11]

In symbols, $x \in A \cap B \Leftrightarrow (x \in A \text{ and } x \in B)$, and $x \in A \cup B \Leftrightarrow (x \in A \text{ or } x \in B)$, where “or” is inclusive. Intersection corresponds to “and”, union to “or”. Because “ P and Q ” means the same as “ Q and P ”, and similarly for “or”, both operations are *commutative*:

$$B \cap A = A \cap B, \quad B \cup A = A \cup B.$$

[L1, slide 11]

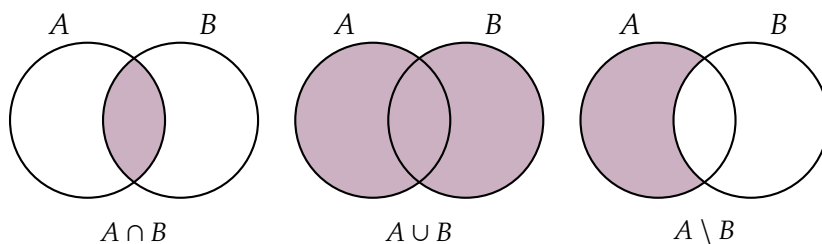


Figure 2.1: Venn diagrams. Each circle represents a set; the shaded region is the set named below it. Left: the intersection $A \cap B$ (in both). Centre: the union $A \cup B$ (in at least one). Right: the set difference $A \setminus B$ (in A but not in B), defined in Section 2.6.

A picture is not a proof

Venn diagrams such as Figure 2.1 are excellent for seeing what a statement says and for guessing whether it is true. They are not proofs: a proof must argue from the definitions, for an arbitrary element.

Worked Example 2.2 – Preferences for fruit

Emma, Bob, Hannah and Sam like the following fruit:

$$\begin{aligned} E &= \{\text{oranges, strawberries}\}, & B &= \{\text{oranges, apples, bananas}\}, \\ H &= \{\text{bananas, blueberries, apples}\}, & S &= \{\text{bananas, oranges, apples, blueberries}\}. \end{aligned}$$

Find $B \cap H$, $B \cup H$, $S \cap E$, $B \cap E$ and $E \cap H$, and decide whether $B \subseteq S$ and $H \subseteq S$. [L1, slide 12]

Solution. $B \cap H$ contains the fruit that both Bob and Hannah like: $B \cap H = \{\text{apples, bananas}\}$.
 $B \cup H$ contains the fruit liked by at least one of them:

$$B \cup H = \{\text{oranges, apples, bananas, blueberries}\}.$$

This is exactly S : as sets (order irrelevant) the two lists contain the same four fruit, so $B \cup H = S$.

$S \cap E = \{\text{oranges}\}$ and $B \cap E = \{\text{oranges}\}$, so $S \cap E = B \cap E$.

$E \cap H = \emptyset$: Emma and Hannah have no fruit in common.

Every fruit Bob likes is liked by Sam, so $B \subseteq S$; likewise $H \subseteq S$. This also follows from $B \cup H = S$, since every set is a subset of its union with another set. Finally, every set is a subset of itself: $E \subseteq E$, $B \subseteq B$ and so on.

Further algebra of $\cap$ and $\cup$

Because $\cap$ and $\cup$ mirror “and” and “or”, they obey the same laws:

- associativity: $(A \cap B) \cap C = A \cap (B \cap C)$ and $(A \cup B) \cup C = A \cup (B \cup C)$;
- distributivity: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$;
- $A \cap B \subseteq A \subseteq A \cup B$.

Each is proved by taking an arbitrary x and translating membership into “and” and “or”.

2.5 Set-builder notation

Listing elements works for small sets but not for large or infinite ones. Usually a set is described by a property that its elements share.

Definition 2.6 (Set-builder notation)

Suppose that a set A might contain some elements that have property P . Then

$$\{x \in A \mid x \text{ has property } P\}$$

denotes the subset of A that consists of all elements of the set A that have property P . [L1, slide 13]

The vertical bar is read “such that”. The set A in front of the bar says where the elements are taken from; the condition after the bar says which of them are kept. The result is always a subset of A .

Worked Example 2.3 – Reading set-builder notation

- (a) If A is the set of students at QMUL and P is the property of being born after 2005, then $\{x \in A \mid x \text{ was born after 2005}\}$ is the set of students at QMUL who were born after 2005. [L1, slide 13]
- (b) $\{n \in \mathbb{N} \mid n \text{ is even}\} = \{2, 4, 6, \dots\}$.
- (c) $\{x \in \mathbb{R} \mid x^2 < 4\}$ is the set of real numbers strictly between -2 and 2 ; in the interval notation of Section 3.5 it is $(-2, 2)$.
- (d) $\{x \in \mathbb{R} \mid x^2 = -1\} = \emptyset$, since no real number squares to -1 .

Solving is describing a set

“Find all x that satisfy $2x + 1 > 0$ ” is the same request as “describe the set $\{x \in \mathbb{R} \mid 2x + 1 > 0\}$ more simply”. The answer $(-\frac{1}{2}, \infty)$ in Worked Example 4.1 is a simpler name for the same set.

2.6 Set difference

Definition 2.7 (Set difference)

Given two sets A and B , the set

$$A \setminus B = \{x \in A \mid x \notin B\}$$

is called the **difference** between sets A and B ; we also say that $A \setminus B$ is the result of **subtraction** of the set B from the set A . Thus $A \setminus B$ is the subset of elements of A that are not elements of B . [L1, slide 14]

Worked Example 2.4 – Set differences

Compute

- (a) $\{\text{apples, oranges, blueberries}\} \setminus \{\text{bananas, oranges}\};$
- (b) $\{2, 4, 6, 8, 10, 12, 14, \dots\} \setminus \{3, 6, 9, 12, 15, \dots\}.$

[L1, slide 14]

Solution. (a) Keep the elements of the first set that are not in the second. Apples and blueberries are not in the second set; oranges are. So the difference is $\{\text{apples, blueberries}\}$. Bananas play no role: removing something that was never there changes nothing.

(b) From the even natural numbers remove the multiples of 3. The even numbers that are multiples of 3 are the multiples of 6, so we remove $6, 12, 18, \dots$ and are left with $\{2, 4, 8, 10, 14, 16, 20, \dots\}$, the even numbers not divisible by 3.

Proposition 2.3 (Subset and difference)

For sets A and B : $A \subseteq B$ implies $A \setminus B = \emptyset$.
 $A \subseteq B \Leftrightarrow A \setminus B = \emptyset$.

[L1, slide 14] The converse is also true, so

Proof. ($\Rightarrow$) Suppose $A \subseteq B$. An element of $A \setminus B$ would be an element of A that is not in B ; but every element of A is in B . So $A \setminus B$ has no elements: $A \setminus B = \emptyset$. ($\Leftarrow$) Suppose $A \setminus B = \emptyset$ and take any $x \in A$. If $x \notin B$ we would have $x \in A \setminus B$, which is impossible; hence $x \in B$. Since x was arbitrary, $A \subseteq B$. $\square$

Set difference is not commutative

Unlike $\cap$ and $\cup$, the order matters: in general $A \setminus B \neq B \setminus A$. For example, with $A = \{1, 2\}$ and $B = \{2, 3\}$ we have $A \setminus B = \{1\}$ but $B \setminus A = \{3\}$.

Two useful facts about $\setminus$

For any sets A and B :

- $A \setminus B \subseteq A$, and $A \setminus B$ has no element in common with B .
- $A \setminus B = A$ if and only if $A \cap B = \emptyset$ (removing B removes nothing exactly when A and B share no element).

2.7 Sets in examination questions

The January 2024 ECN115 paper opened with a sixteen-mark question on sets, consisting of a definition, two “does such a thing exist?” parts and one “find all” part (as summarised in the study guide). The following worked example reproduces that structure.

Worked Example 2.5 – Questions on set difference

- Explain the notion of set difference $A \setminus B$.
- Are there non-empty sets of natural numbers A and B with $A \setminus B = A$? If yes, give an example; if no, argue why.
- Are there non-empty sets of natural numbers A and B with $A \setminus B = B$? If yes, give an example; if no, argue why.
- Find all sets $D \subseteq \mathbb{N}$ such that $\{1, 2\} \cup D = \{1, 2, 3\}$.

[SG: Jan 2024 Q1]

Solution. (a) For sets A and B , $A \setminus B = \{x \in A \mid x \notin B\}$: the set of elements of A that are not elements of B .

(b) Yes. Take $A = \{1\}$ and $B = \{2\}$. Both are non-empty sets of natural numbers, $1 \notin B$, so $A \setminus B = \{1\} = A$. (Any two non-empty sets with no common element work.)

(c) No. Suppose $A \setminus B = B$ with B non-empty, and take some $b \in B$. Then $b \in A \setminus B$, so by the definition of set difference $b \notin B$. This contradicts $b \in B$. Hence there are no such sets. (The

argument shows more: $A \setminus B = B$ forces $B = \emptyset$.)

(d) Since $3 \in \{1, 2, 3\} = \{1, 2\} \cup D$ and $3 \notin \{1, 2\}$, we need $3 \in D$. Since every element of D lies in $\{1, 2\} \cup D = \{1, 2, 3\}$, we need $D \subseteq \{1, 2, 3\}$. The elements 1 and 2 may or may not be in D , because they are in the union anyway. Hence

$$D \in \{\{3\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\},$$

four sets in all, and each of them does satisfy the condition.

“If yes, give an example; if no, argue why”

This phrasing appears on several past ECN115 papers. A “yes” answer needs one explicit example, checked against every condition (including “non-empty”). A “no” answer needs a general argument, usually by contradiction: assume such objects exist and derive something impossible. Decide which it is before writing; a quick experiment with small sets usually tells you.

2.8 Sets in economics

Worked Example 2.6 – A budget set

A consumer with income $m = 100$ buys quantities $x \geq 0$ and $y \geq 0$ of two goods priced $p_x = 5$ and $p_y = 10$. Describe the bundles she can afford as a set, and decide whether the bundles $(10, 5)$ and $(12, 5)$ are affordable.

Solution. A bundle is a pair (x, y) of real numbers. The affordable bundles form the *budget set*

$$B = \{(x, y) \in \mathbb{R}^2 \mid x \geq 0, y \geq 0, 5x + 10y \leq 100\},$$

where $\mathbb{R}^2$ denotes the set of all pairs of real numbers. The bundle $(10, 5)$ costs $5 \cdot 10 + 10 \cdot 5 = 100$, so $(10, 5) \in B$ (it is on the budget line). The bundle $(12, 5)$ costs $110 > 100$, so $(12, 5) \notin B$.

If a rationing scheme also limits the first good to at most 8 units, the bundles that are both affordable and permitted form the intersection $B \cap \{(x, y) \in \mathbb{R}^2 \mid x \leq 8\}$. Each extra constraint on a decision-maker corresponds to intersecting with one more set.

2.9 Chapter review

Summary

A set is determined by its elements; order and repetition are irrelevant, and the empty set $\emptyset$ has no elements. $A \subseteq B$ means every element of A is in B ; every set is a subset of itself; and two sets are equal exactly when each is a subset of the other. Intersection ($\cap$, “and”), union ($\cup$, “or”) and difference ($\setminus$, “and not”) build new sets; the first two are commutative, the third is not. Set-builder notation $\{x \in A \mid P\}$ describes the elements of A with property P , and solving an equation or inequality amounts to describing such a set in simpler terms.

Key definitions

- $x \in A$: x is an element of A ; $x \notin A$: it is not.
- $A = B$: every element of A is in B and every element of B is in A .
- $A \subseteq B$: every element of A is in B (A a subset, B a superset).
- $A \cap B$: objects in both; $A \cup B$: objects in A , in B or in both.
- $\{x \in A \mid P\}$: elements of A with property P . $A \setminus B = \{x \in A \mid x \notin B\}$.

Key facts

$$A = B \Leftrightarrow (A \subseteq B \text{ and } B \subseteq A), \quad A \subseteq A, \quad A \cap B = B \cap A, \quad A \cup B = B \cup A, \quad A \subseteq B \Rightarrow A \setminus B = \emptyset.$$

Connections

Membership statements are propositions (Chapter 1), and $\cap$, $\cup$ correspond to “and” and “or”. The number systems $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ are sets, one inside the next (Chapter 3); intervals are subsets of $\mathbb{R}$ described in set-builder notation (Section 3.5); and the solution of every inequality in Chapters 4 and 5 is a set, often a union of intervals.

Common mistakes

- Confusing $\in$ (element of) with $\subseteq$ (subset of): $1 \in \{1, 2\}$ but $\{1\} \subseteq \{1, 2\}$.
- Assuming $A \setminus B = B \setminus A$.
- Forgetting that “or” in $A \cup B$ includes elements in both sets.
- Treating a Venn diagram as a proof.
- In “find all sets” questions, listing only one set when several satisfy the conditions.

Exam checklist

- ☐ Define subset and set equality, and state $A = B \Leftrightarrow (A \subseteq B \text{ and } B \subseteq A)$.
- ☐ Define set difference in set-builder notation.
- ☐ Compute intersections, unions and differences of given sets.
- ☐ Answer “does such a set exist?” questions with an example or an argument by contradiction.
- ☐ Find all sets satisfying given union, intersection and subset conditions.

2.10 Practice questions

Question 2.1 [basic]

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6\}$. Find $A \cap B$, $A \cup B$, $A \setminus B$ and $B \setminus A$.

Question 2.2 [basic]

List the elements of (a) $\{n \in \mathbb{N} \mid n < 5\}$; (b) $\{n \in \mathbb{N} \mid n^2 \leq 30\}$; (c) $\{x \in \mathbb{R} \mid x^2 = 4\}$; (d) $\{x \in \mathbb{R} \mid x^2 + 1 = 0\}$.

Question 2.3 [conceptual]

Let $C = \{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$. Which of the following are true? (a) $\{1, 2\} \in C$; (b) $1 \in C$; (c) $\{2, 1\} \in C$; (d) $\{\{1, 2\}\} \subseteq C$; (e) $\{1, 2\} \subseteq C$.

Question 2.4 [mathematical]

Prove that for any sets A and B , $A \cap B \subseteq A$ and $A \subseteq A \cup B$.

Question 2.5 [mathematical]

Prove that if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Question 2.6 [application]

A firm can produce any whole number of units from 0 to 50 inclusive. A regulator requires output of at least 20 units, and capacity constraints in a second plant exclude the outputs 30 to 35 inclusive. Describe the set of permitted outputs using set-builder notation and set operations, and state how many outputs are permitted.

Question 2.7 [multi-step]

Are there non-empty sets of natural numbers A and B such that $A \cup B = A \cap B$? If yes, give an example and describe all such pairs; if no, argue why.

Question 2.8 [exam-style]

(a) Explain what it means for a set A to be a subset of a set B , and state what it means for two sets to be equal in terms of subsets. (b) Find all sets D of natural numbers that satisfy both $\{2, 4\} \cap D = \{2\}$ and $D \subseteq \{1, 2, 3, 4\}$. (c) Let $A = \{x \in \mathbb{N} \mid x \leq 10\}$ and $B = \{x \in \mathbb{N} \mid x \text{ is even}\}$. List the elements of $A \setminus B$ and decide whether $A \setminus B = B \setminus A$.

2.11 Solutions to practice questions

Solution to Question 2.1

$A \cap B = \{4, 5\}$; $A \cup B = \{1, 2, 3, 4, 5, 6\}$; $A \setminus B = \{1, 2, 3\}$; $B \setminus A = \{6\}$. Note that $A \setminus B \neq B \setminus A$.

Solution to Question 2.2

(a) $\{1, 2, 3, 4\}$ (recall $0 \notin \mathbb{N}$). (b) $n^2 \leq 30$ holds for $n = 1, 2, 3, 4, 5$ ($5^2 = 25$) but not $n = 6$ (36); so $\{1, 2, 3, 4, 5\}$. (c) $\{-2, 2\}$. (d) $\emptyset$.

Solution to Question 2.3

(a) True. (b) False: the elements of C are sets, not numbers. (c) True: $\{2, 1\} = \{1, 2\}$, since order does not matter. (d) True: the only element of $\{\{1, 2\}\}$ is $\{1, 2\}$, which is in C . (e) False: 1 is an element of $\{1, 2\}$ but $1 \notin C$.

Solution to Question 2.4

Let $x \in A \cap B$ be arbitrary. Then $x \in A$ and $x \in B$; in particular $x \in A$. So $A \cap B \subseteq A$. Now let $x \in A$ be arbitrary. Then " $x \in A$ or $x \in B$ " is true, so $x \in A \cup B$. So $A \subseteq A \cup B$.

Solution to Question 2.5

Let $x \in A$ be arbitrary. Since $A \subseteq B$, $x \in B$. Since $B \subseteq C$, $x \in C$. As x was an arbitrary element of A , $A \subseteq C$.

Solution to Question 2.6

Let $X = \{n \in \mathbb{Z} \mid 0 \leq n \leq 50\}$ be the feasible outputs. The permitted outputs are

$$\{n \in X \mid n \geq 20\} \setminus \{n \in \mathbb{Z} \mid 30 \leq n \leq 35\} = \{20, 21, \dots, 29\} \cup \{36, 37, \dots, 50\}.$$

There are $10 + 15 = 25$ permitted outputs.

Solution to Question 2.7

Yes: $A = B = \{1\}$ gives $A \cup B = \{1\} = A \cap B$. In fact this happens exactly when $A = B$. Always $A \cap B \subseteq A \subseteq A \cup B$ (Question 2.4). If $A \cup B = A \cap B$, the outer two sets coincide, so A is squeezed between equal sets and $A = A \cap B = A \cup B$; the same argument gives $B = A \cup B$; hence $A = B$. Conversely, if $A = B$ then $A \cup B = A = A \cap B$.

Solution to Question 2.8

(a) $A \subseteq B$ means that every element of A is also an element of B . Two sets are equal, $A = B$, exactly when $A \subseteq B$ and $B \subseteq A$. (b) $\{2, 4\} \cap D = \{2\}$ says $2 \in D$ and $4 \notin D$. $D \subseteq \{1, 2, 3, 4\}$ says D has no other elements. So 1 and 3 are each free to be in or out: $D \in \{\{2\}, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}\}$. (c) $A = \{1, \dots, 10\}$, so $A \setminus B = \{1, 3, 5, 7, 9\}$. But $B \setminus A = \{12, 14, 16, \dots\}$, the even numbers greater than 10. For example $12 \in B \setminus A$ while $12 \notin A \setminus B$, so $A \setminus B \neq B \setminus A$.

Number Systems and the Properties of Real Numbers

Economics measures prices, quantities, incomes and rates, and all of these are numbers. This chapter introduces the four number systems used in the module — the natural numbers, the integers, the rational numbers and the real numbers — and then lists the basic properties of the real numbers from which every calculation in this book can be justified. These properties are the axioms of the module: when we solve an inequality, every step is an application of one of them.

Before you start: prerequisites

Sets, subsets and set-builder notation (Chapter 2); the meaning of axiom and theorem (Section 1.2); implication and equivalence (Sections 1.3 and 1.4).

3.1 The four number systems

Definition 3.1 (Number systems)

- The **natural numbers** are the set $\mathbb{N} = \{1, 2, 3, \dots\}$.
- The **integers** are the set $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$.
- The **rational numbers** are the set $\mathbb{Q}$ consisting of all numbers that can be written as p/q with $p \in \mathbb{Z}$ and $q \in \mathbb{N}$.
- The **real numbers** are (informally) the set $\mathbb{R}$ of points on the number line.

[L1, slide 16]

CORE CONCEPT | Successor property of $\mathbb{N}$

For any $n \in \mathbb{N}$, we have $n + 1 \in \mathbb{N}$.

[L1, slide 16]

Starting from 1 and repeatedly adding 1 reaches every natural number and never leaves $\mathbb{N}$. This simple fact is what makes proof by induction work (Chapter 7).

0 is not a natural number in this module

In ECN115, $\mathbb{N} = \{1, 2, 3, \dots\}$ starts at 1. Some books include 0. The difference matters for index ranges in sums and for the starting point of an induction. When 0 is wanted it is added explicitly, as in $\{0\} \cup \mathbb{N}$ in the binomial formula (Chapter 8).

Why $q \in \mathbb{N}$ in the definition of $\mathbb{Q}$. Requiring the denominator to be a natural number does two jobs at once. It rules out $q = 0$, so the fraction p/q always makes sense, without a separate clause. And it puts the sign in the numerator: $-\frac{3}{4}$ is written $\frac{-3}{4}$ rather than $\frac{3}{-4}$. Every rational number can be written in many ways ($\frac{1}{2} = \frac{2}{4} = \frac{50}{100}$), but this does not matter: $\mathbb{Q}$ is the set of numbers that can be written in this form at least once.

Proposition 3.1 (The number systems are nested)

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}.$$

[L1, slide 16]

Every natural number is an integer; every integer p is the rational number $p/1$; and every rational number is a point on the number line. Equality and the inequalities $\geq, \leq, >, <$ are defined on all four sets as usual, and they agree with one another: $2 < 3$ means the same whether 2 and 3 are thought of as natural numbers or as real numbers.

3.1.1 Why the real numbers are needed

The slides note that the rational numbers are adequate for *arithmetic*: adding, subtracting, multiplying or dividing (by a non-zero number) two rational numbers always gives a rational number. The real numbers are needed for *algebra* (for example, solving $x^2 = 2$), *geometry* (for example, the ratio of the circumference of a circle to its diameter, π) and *limits*.

[L1, slide 16]

$\sqrt{2}$ is not rational

The equation $x^2 = 2$ has no solution in $\mathbb{Q}$. Suppose it did: $x = p/q$ with $p \in \mathbb{Z}$, $q \in \mathbb{N}$, and the fraction in lowest terms (so p and q are not both even). Then $p^2 = 2q^2$, so p^2 is even, and therefore p is even (the square of an odd number is odd). Write $p = 2k$; then $4k^2 = 2q^2$, so $q^2 = 2k^2$ is even and q is even. So p and q are both even, contradicting lowest terms. Hence no rational number squares to 2. On the number line there is nevertheless a point at distance $\sqrt{2}$ from 0 (the diagonal of a unit square), so the rational numbers have “gaps” that the real numbers fill. Section 3.4 describes the property of $\mathbb{R}$ that captures the absence of gaps.

3.2 The properties of the real numbers

The slides list thirteen properties of the real numbers in three groups: two for the order relation, five for addition and six for multiplication. In this module they are the axioms: we do not prove them, and every other fact about real numbers is derived from them.

3.2.1 The order relation

Definition 3.2 (Order: O1–O2)

- O1** For any real numbers a, b , exactly one of the following statements is true: either $a > b$, or $a = b$, or $a < b$.
- $a < b$ is the same as $b > a$.
 - We write $a \geq b$ if $a > b$ or $a = b$. We write $a \leq b$ if $a < b$ or $a = b$.
- O2** *Transitivity*: if $a > b$ and $b > c$, then $a > c$.

[L1, slide 17]

O1 is called the **trichotomy** property (“division into three”). It has two halves: *at least* one of the three possibilities holds, so any two real numbers can be compared; and *at most* one holds, so the possibilities exclude each other. The first half is what makes case analyses complete: when a factor in a product is not zero, it must be positive or negative, and there is no fourth option (Worked Example 4.6). The second half is what makes negation simple: “not ($a > b$)” is the same as $a \leq b$. Note that $\geq$ and $\leq$ are *defined* from $>$ and $=$; they are not separate primitive notions.

3.2.2 Addition

Addition assigns to any numbers a, b a unique number $a + b$.

Definition 3.3 (Addition: A1–A5)

For all real numbers a, b, c :

- A1** *Commutativity*: $a + b = b + a$.
- A2** *Associativity*: $a + (b + c) = (a + b) + c$.
- A3** *Property of the number zero*: $a + 0 = a$.
- A4** *Existence of the additive inverse*: $a + (-a) = 0$.
- A5** *Additivity of the order*: $a > b \Rightarrow a + c > b + c$.

[L1, slide 17]

A1 and A2 say that, in a sum, neither the order of the terms nor the way they are grouped matters, which is why we may write $a + b + c$ without brackets. A3 says that 0 changes nothing when added. A4 says that every number a has a partner $-a$ that cancels it; subtraction is *defined* by $a - b = a + (-b)$. A5 connects addition to order: adding the same number to both sides of a strict inequality preserves it.

A5 is an implication, not an equivalence

As stated, A5 says only $a > b \Rightarrow a + c > b + c$. The reverse direction is true but needs an argument: from $a + c > b + c$, apply A5 again with $-c$ to get $(a + c) + (-c) > (b + c) + (-c)$, and simplify with A2–A4 to $a > b$. So

$$a > b \iff a + c > b + c \quad \text{for every real } c.$$

This is a small theorem, not the axiom itself. The slides use exactly this two-step argument in Example 1 of Chapter 4.

3.2.3 Multiplication

Multiplication assigns to any numbers a, b a unique number $a \cdot b$ (also written ab).

Definition 3.4 (Multiplication: M1–M6)

For all real numbers a, b, c :

- M1** *Commutativity:* $a \cdot b = b \cdot a$.
- M2** *Associativity:* $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
- M3** *Properties of the number one:* $1 \neq 0$, and $a \cdot 1 = a$.
- M4** *Existence of the inverse of a non-zero number:* $a \cdot (1/a) = 1$ for $a \neq 0$.
- M5** *Distributive law:* $a \cdot (b + c) = a \cdot b + a \cdot c$.
- M6** *Multiplicative properties of the order relation:*
 - if $a > b$ and $c > 0$, then $a \cdot c > b \cdot c$;
 - if $a > b$ and $c < 0$, then $a \cdot c < b \cdot c$.

[L1, slide 18]

M1–M4 mirror A1–A4. Division is defined by $a/b = a \cdot (1/b)$ for $b \neq 0$; there is no $1/0$, which is why M4 carries the condition $a \neq 0$. M5 is the only property that links addition and multiplication: it is what allows brackets to be multiplied out and common factors to be taken out.

CORE CONCEPT | M6: the sign of the multiplier decides the direction

Multiplying both sides of an inequality by a *positive* number keeps its direction; multiplying by a *negative* number reverses it. Multiplying by 0 destroys it (both sides become 0). Consequently:

- before multiplying or dividing an inequality by anything, establish its sign;
- if the sign is unknown (a variable or a parameter), split into the cases positive, zero and negative.

This one habit accounts for a large share of the marks in inequality and parameter questions on past ECN115 papers.

The rational numbers satisfy all thirteen properties

Every one of O1–O2, A1–A5 and M1–M6 is also true in $\mathbb{Q}$: sums, products, negatives and reciprocals of rational numbers are rational, and the order behaves in the same way. [L1, slide 19]
So these properties cannot, on their own, tell $\mathbb{R}$ apart from $\mathbb{Q}$. Whatever distinguishes the real numbers — the existence of $\sqrt{2}$, the good behaviour of limits — must come from an additional property (Section 3.4).

3.3 Consequences of the properties

Many familiar rules of algebra are not on the list. They are theorems, derived from the list. The slides use some of them without proof (the study guide points out that Example 2 of Chapter 4 relies on $0 \cdot a = 0$); this section proves the ones needed in the book. The proofs show how theorems are built from axioms, and you may use the results freely afterwards.

Proposition 3.2 (Multiplying by zero)

For every real number a , $0 \cdot a = 0$.

Proof. By A3, $0 + 0 = 0$. Multiply by a and use M1 and M5: $0 \cdot a = (0 + 0) \cdot a = 0 \cdot a + 0 \cdot a$. Now add $-(0 \cdot a)$ to both sides and use A2, A4 and A3: $0 = 0 \cdot a$. $\square$

Proposition 3.3 (Zero products)

For real numbers a, b : if $ab = 0$ then $a = 0$ or $b = 0$. Equivalently, if $a \neq 0$ and $b \neq 0$ then $ab \neq 0$.

Proof. Suppose $ab = 0$ and $a \neq 0$. By M4, $1/a$ exists. Then, using M2, M4, M3 and Proposition 3.2, $b = 1 \cdot b = ((1/a) \cdot a) \cdot b = (1/a) \cdot (ab) = (1/a) \cdot 0 = 0$. So if $a \neq 0$ then $b = 0$; hence $a = 0$ or $b = 0$. $\square$

Proposition 3.4 (Signs)

For real numbers a, b :

- (i) $(-1) \cdot a = -a$, and $-(-a) = a$;
- (ii) $a > 0 \Leftrightarrow -a < 0$;
- (iii) $a > b \Leftrightarrow a - b > 0$;
- (iv) if $a \neq 0$ then $a^2 > 0$; hence $a^2 \geq 0$ for every a , and in particular $1 = 1^2 > 0$;
- (v) if $a > 0$ then $1/a > 0$; if $a < 0$ then $1/a < 0$;
- (vi) the product of two numbers with the same sign is positive; the product of two numbers with opposite signs is negative.

Proof. (i) $a + (-1)a = 1 \cdot a + (-1)a = (1 + (-1))a = 0 \cdot a = 0$ by M3, M5, A4 and Proposition 3.2; so $(-1)a$ is an additive inverse of a . Additive inverses are unique (if $a + b = 0$ and $a + c = 0$ then $b = b + (a + c) = (b + a) + c = 0 + c = c$), so $(-1)a = -a$. Also $-a + a = 0$ shows that a is the inverse of $-a$, i.e. $-(-a) = a$.

(ii) If $a > 0$, add $-a$ to both sides (A5): $0 > -a$. Conversely, if $-a < 0$, add a : $0 < a$.

(iii) Add $-b$ to both sides of $a > b$ (A5, A4): $a - b > 0$. Conversely add b .

(iv) If $a > 0$, M6 with $c = a > 0$ applied to $a > 0$ gives $a \cdot a > 0 \cdot a = 0$. If $a < 0$, apply M6 to the inequality $0 > a$ with $c = a < 0$: the direction reverses, giving $0 \cdot a < a \cdot a$, that is, $a \cdot a > 0$. Either way $a^2 > 0$.

(v) Let $a > 0$. By M4, $a \cdot (1/a) = 1 > 0$. If $1/a$ were 0 the product would be 0; if $1/a$ were negative, M6 applied to $a > 0$ with $c = 1/a < 0$ would give $a \cdot (1/a) < 0$. By O1, $1/a > 0$. The case $a < 0$ is similar.

(vi) If $a > 0$ and $b > 0$, M6 gives $ab > 0 \cdot b = 0$. If $a < 0$ and $b < 0$, M6 with $c = b < 0$ applied to $0 > a$ gives $0 \cdot b < ab$, i.e. $ab > 0$. If $a > 0$ and $b < 0$, M6 with $c = b < 0$ applied to $a > 0$ gives $ab < 0$. $\square$

Note

You will not be asked to reproduce these derivations in ECN115: the slides say explicitly that, when solving problems, “it suffices just to do correct calculations”. They are included so that you can see that every rule you use rests on the thirteen properties, and so that the reasoning behind sign analysis (Section 4.3) is complete.

3.4 The Least Upper Bound property

Non-examinable

The slides present this property as “extra material, non-examinable”. It is included because it explains *why* the real numbers, rather than the rational numbers, are the right setting for limits.

Definition 3.5 (Upper bound and least upper bound)

Let $A \subseteq \mathbb{R}$ be non-empty. A number $y \in \mathbb{R}$ is an **upper bound** for A if $x \leq y$ for all $x \in A$. A number $z \in \mathbb{R}$ is a **least upper bound** for A if

- (i) z is an upper bound: $x \leq z$ for all $x \in A$; and
- (ii) no smaller number is an upper bound: for every $w < z$ there is $x \in A$ such that $x > w$.

[L1, slide 19]

Principle 3.5 (Least Upper Bound property)

If a non-empty subset $A \subseteq \mathbb{R}$ has an upper bound, then A has a least upper bound. [L1, slide 19]

The rational numbers do *not* have this property. The set $A = \{q \in \mathbb{Q} \mid q > 0, q^2 < 2\}$ is non-empty ($1 \in A$) and bounded above (by 2), but there is no smallest *rational* upper bound: the least upper bound would have to be $\sqrt{2}$, which is not rational. In $\mathbb{R}$ the number $\sqrt{2}$ exists, and it is the least upper bound. The Least Upper Bound property is precisely the statement that the real number line has no gaps; it is what guarantees, in Topic 3, that limits exist when we expect them to.

3.5 Intervals

Subsets of $\mathbb{R}$ consisting of all numbers between two endpoints occur so often that they have their own notation.

Definition 3.6 (Intervals)

For real numbers $a \leq b$:

$$\begin{aligned}\text{closed interval: } [a, b] &= \{x \in \mathbb{R} \mid a \leq x \leq b\}, \\ \text{open interval: } (a, b) &= \{x \in \mathbb{R} \mid a < x < b\}, \\ \text{half-open intervals: } (a, b] &= \{x \in \mathbb{R} \mid a < x \leq b\}, & [a, b) &= \{x \in \mathbb{R} \mid a \leq x < b\}, \\ \text{infinite intervals: } (a, \infty) &= \{x \in \mathbb{R} \mid a < x\}, & (-\infty, b) &= \{x \in \mathbb{R} \mid x < b\}, \\ \text{half-infinite intervals: } [a, \infty) &= \{x \in \mathbb{R} \mid a \leq x\}, & (-\infty, b] &= \{x \in \mathbb{R} \mid x \leq b\}, \\ & & (-\infty, \infty) &= \mathbb{R}.\end{aligned}$$

[L1, slide 20]

A square bracket means the endpoint is *included* ($\leq$); a round bracket means it is *excluded* ($<$). The symbols ∞ and $-\infty$ are not real numbers; they only indicate that the interval has no upper or lower end. They can therefore never be included, and always take a round bracket.

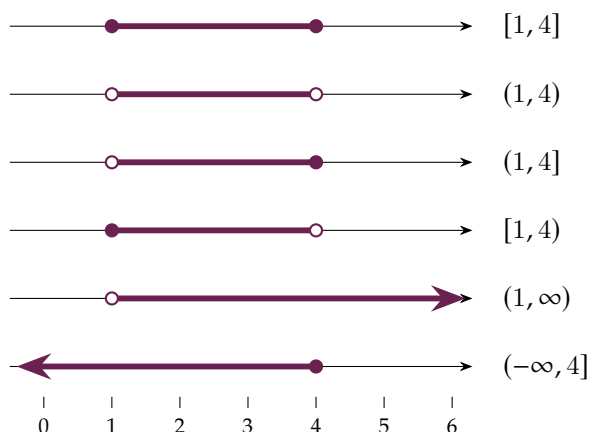


Figure 3.1: Intervals on the number line. A filled dot marks an included endpoint (square bracket), an open dot an excluded endpoint (round bracket). An arrow shows that the interval continues without end.

Worked Example 3.1 – Writing sets as intervals

Write each set as an interval or a union of intervals: (a) $\{x \in \mathbb{R} \mid -2 \leq x < 7\}$; (b) $\{x \in \mathbb{R} \mid x > 3\}$; (c) $\{x \in \mathbb{R} \mid x \leq 0 \text{ or } x > 5\}$; (d) $\{x \in \mathbb{R} \mid x \neq 1\}$.

Solution. (a) $[-2, 7)$. (b) $(3, \infty)$. (c) $(-\infty, 0] \cup (5, \infty)$. (d) $(-\infty, 1) \cup (1, \infty)$, which can also be written $\mathbb{R} \setminus \{1\}$.

Why the type of endpoint matters

$[0, 1)$ has a smallest element (0) but no largest: for any $x < 1$ in the interval, $(x + 1)/2$ is a larger element of the interval. $(0, 1]$ has a largest element (1) but no smallest. In optimisation (Topic 4) this decides whether a function defined on the interval can attain a maximum or a minimum, which is why a question may be posed on a half-open interval such as $(0, c]$.

3.6 Chapter review

Summary

$\mathbb{N} = \{1, 2, 3, \dots\}$, with the successor property; $\mathbb{Z}$ adds zero and the negatives; $\mathbb{Q}$ consists of fractions p/q with $p \in \mathbb{Z}$, $q \in \mathbb{N}$; and $\mathbb{R}$ is, informally, the number line. They are nested: $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$. The rationals suffice for arithmetic, but solving $x^2 = 2$, measuring a circle and taking limits need $\mathbb{R}$. The real numbers satisfy the order properties O1 (trichotomy) and O2 (transitivity), the addition properties A1–A5 and the multiplication properties M1–M6; in ECN115 these are the axioms. M6 — a positive multiplier keeps an inequality, a negative one reverses it — is the most important for calculation. $\mathbb{Q}$ satisfies all thirteen properties too; the additional Least Upper Bound property (non-examinable) is what distinguishes $\mathbb{R}$. Intervals name the subsets of $\mathbb{R}$ between two endpoints.

Key definitions and properties

- $\mathbb{N} = \{1, 2, 3, \dots\}$; $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$; $\mathbb{Q} = \{p/q : p \in \mathbb{Z}, q \in \mathbb{N}\}$; $\mathbb{R}$: the number line.
- O1: exactly one of $a > b$, $a = b$, $a < b$. O2: $a > b, b > c \Rightarrow a > c$.
- A1–A4: $a + b = b + a$; $a + (b + c) = (a + b) + c$; $a + 0 = a$; $a + (-a) = 0$. A5: $a > b \Rightarrow a + c > b + c$.
- M1–M5: $ab = ba$; $a(bc) = (ab)c$; $1 \neq 0$, $a \cdot 1 = a$; $a \cdot (1/a) = 1$ ($a \neq 0$); $a(b + c) = ab + ac$.
- M6: $a > b, c > 0 \Rightarrow ac > bc$; $a > b, c < 0 \Rightarrow ac < bc$.
- Intervals: $[a, b]$, (a, b) , $(a, b]$, $[a, b)$, (a, ∞) , $[a, \infty)$, $(-\infty, b)$, $(-\infty, b]$, $\mathbb{R}$.

Connections

The axioms justify every step of the inequality solving in Chapter 4; the sign rules of Proposition 3.4 drive the sign analysis there. The successor property underlies induction (Chapter 7). Intervals are the natural form of answers to inequalities and the domains of functions (Chapter 9). The Least Upper Bound property is the reason limits (Chapter 10) are studied in $\mathbb{R}$ rather than $\mathbb{Q}$.

Common mistakes

- Including 0 in $\mathbb{N}$.
- Forgetting to reverse an inequality after multiplying or dividing by a negative number.
- Dividing by a quantity that might be zero.
- Putting a square bracket next to ∞ , or the wrong bracket for a strict inequality.
- Assuming A5 gives an equivalence without the second step.

Exam checklist

- ☐ Write down $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and the inclusions between them; state the successor property.
- ☐ Explain why $\mathbb{R}$ is needed in addition to $\mathbb{Q}$.
- ☐ State O1, O2, A1–A5 and M1–M6, and say which property justifies a given step.
- ☐ State both clauses of M6 exactly.

- Write any interval in set-builder notation and vice versa.

3.7 Practice questions

Question 3.1 [basic]

Decide whether each number belongs to $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ (list all that apply): 5 , -3 , 0 , $\frac{-7}{2}$, 0.25 , $\sqrt{9}$, π .

Question 3.2 [basic]

Write in set-builder notation: (a) $(-3, 5]$; (b) $[2, \infty)$; (c) $(-\infty, 0) \cup [1, 2]$.

Question 3.3 [conceptual]

Explain why the definition of $\mathbb{Q}$ uses $q \in \mathbb{N}$ rather than $q \in \mathbb{Z}$.

Question 3.4 [mathematical]

Starting from $a > b$, and naming the property used at each step, show that $-a < -b$.

Question 3.5 [mathematical]

Using the properties of the real numbers, prove that if $a > b$ and $c > d$ then $a + c > b + d$.

Question 3.6 [conceptual]

Which of the thirteen properties O1–M6 fail for the natural numbers $\mathbb{N}$ (with the usual addition and multiplication)? Give a reason for each.

Question 3.7 [multi-step]

Let a, b be real numbers with $0 < a < b$. Prove that $a^2 < b^2$ and that $1/a > 1/b$.

Question 3.8 [exam-style]

(a) State both clauses of the multiplicative property of the order relation (M6). (b) A student argues: “ $x > y$, so multiplying both sides by x gives $x^2 > xy$.” Give conditions on x under which the conclusion is guaranteed, and give an example with $x > y$ for which $x^2 > xy$ is false.

3.8 Solutions to practice questions

Solution to Question 3.1

5 : $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$. -3 : $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$. 0 : $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ (not $\mathbb{N}$). $\frac{-7}{2}$: $\mathbb{Q}, \mathbb{R}$. $0.25 = \frac{1}{4}$: $\mathbb{Q}, \mathbb{R}$. $\sqrt{9} = 3$: $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$. π : $\mathbb{R}$ only.

Solution to Question 3.2

(a) $\{x \in \mathbb{R} \mid -3 < x \leq 5\}$. (b) $\{x \in \mathbb{R} \mid x \geq 2\}$. (c) $\{x \in \mathbb{R} \mid x < 0 \text{ or } 1 \leq x \leq 2\}$.

Solution to Question 3.3

Since $0 \notin \mathbb{N}$, the condition $q \in \mathbb{N}$ excludes a zero denominator automatically, so p/q is always defined. It also fixes the sign in the numerator. With $q \in \mathbb{Z}$ a separate condition $q \neq 0$ would be needed.

Solution to Question 3.4

Add $(-a) + (-b)$ to both sides of $a > b$ (A5): $a + ((-a) + (-b)) > b + ((-a) + (-b))$. Rearranging with A1 and A2, the left side is $(a + (-a)) + (-b) = 0 + (-b) = -b$ (A4, A1, A3) and the right side is $(b + (-b)) + (-a) = -a$. Hence $-b > -a$, i.e. $-a < -b$. (Alternatively: multiply by $-1 < 0$ using M6 and Proposition 3.4(i).)

Solution to Question 3.5

From $a > b$, A5 with c gives $a + c > b + c$. From $c > d$, A5 with b gives $c + b > d + b$, i.e. $b + c > b + d$ (A1). By transitivity (O2), $a + c > b + d$.

Solution to Question 3.6

A3 fails: there is no 0 in $\mathbb{N}$, so " $a + 0 = a$ " has no meaning inside $\mathbb{N}$. A4 fails: $-1 \notin \mathbb{N}$, so 1 has no additive inverse in $\mathbb{N}$. M4 fails: $2 \in \mathbb{N}$ but $1/2 \notin \mathbb{N}$. The order properties O1, O2, A5 and M6 (with $c \in \mathbb{N}$; the second clause never applies since no natural number is negative), and A1, A2, M1–M3, M5 continue to hold.

Solution to Question 3.7

Since $b > a$ and $a > 0$, M6 gives $b \cdot a > a \cdot a$; since $b > a$ and $b > 0$, M6 gives $b \cdot b > a \cdot b$. By M1 and O2, $b^2 > ab > a^2$. For the reciprocals: $a > 0$ and $b > 0$ give $ab > 0$, so $1/(ab) > 0$ by Proposition 3.4(v). Multiply $b > a$ by $1/(ab) > 0$ (M6): $b/(ab) > a/(ab)$, i.e. $1/a > 1/b$.

Solution to Question 3.8

(a) If $a > b$ and $c > 0$, then $a \cdot c > b \cdot c$. If $a > b$ and $c < 0$, then $a \cdot c < b \cdot c$. (b) The step is M6 with multiplier x , so it is guaranteed when $x > 0$. If $x < 0$ the direction reverses and $x^2 < xy$; if $x = 0$ both sides are 0. Example: $x = -1$, $y = -2$ gives $x > y$, but $x^2 = 1$ and $xy = 2$, so $x^2 > xy$ is false.

Solving Inequalities

Inequalities are everywhere in economics: a budget must not be exceeded, output cannot be negative, a project is worth undertaking if its benefits exceed its costs. Solving an inequality means finding every value of the unknown for which it is true. This chapter develops the method in stages — linear inequalities, products of factors, quadratics and fractions — and shows how each step rests on the properties of the real numbers from Chapter 3.

Before you start: prerequisites

Equivalence and why it matters for solving (Section 1.4); set-builder notation, unions and intersections (Chapter 2); the properties O1, A1–A5 and M1–M6 and the sign rules of Proposition 3.4 (Chapter 3); interval notation (Section 3.5).

4.1 What it means to solve an inequality

An inequality in an unknown x , such as $2x + 1 > 0$, is a proposition depending on x . Its **solution set** is the set of all real x for which it is true, $\{x \in \mathbb{R} \mid 2x + 1 > 0\}$. To *solve* the inequality is to describe this set as simply as possible, usually as an interval or a union of intervals.

CORE CONCEPT | Solve by a chain of equivalences

Transform the inequality step by step, making sure that each step is an *equivalence* (Section 1.4). Then the final, simple inequality has exactly the same solution set as the original, and its solution set is the answer. The two tools are:

- **adding** the same number to both sides, which is always reversible (A5, applied twice);
- **multiplying** both sides by the same number, which is reversible if the number is non-zero, and which keeps the direction if the number is positive and reverses it if the number is negative (M6).

How much justification to write

The slides justify each step by naming the property used, in order to show that the methods rest on the properties of the real numbers. They also say: “When solving problems, it suffices just to do correct calculations.” [L1, slide 22] In your own answers you do not need to cite A5 or

M4 at every line; but you must get the direction right, and when you multiply or divide by something whose sign matters you should say what that sign is.

4.2 Linear inequalities

Worked Example 4.1 – The first inequality, step by step

Find all x that satisfy $2x + 1 > 0$.

[L1, slides 22–23]

Solution. *Step 1: remove the constant.* Using A5 and A4, add the number -1 to both sides; then use A2–A4 to simplify the left-hand side, and A1, A3 to find $0 + (-1) = -1$:

$$2x + 1 > 0 \implies 2x + 1 + (-1) > 0 + (-1) \implies 2x > -1.$$

The reverse implication works as well: add 1 to both sides of $2x > -1$,

$$2x > -1 \implies 2x + 1 > -1 + 1 \implies 2x + 1 > 0.$$

Therefore the two inequalities are equivalent: $2x + 1 > 0 \Leftrightarrow 2x > -1$.

Step 2: remove the coefficient. By M4 there is an inverse $1/2$ of the non-zero number 2, and $1/2 > 0$ (Proposition 3.4(v)). Using M6, multiply both sides of $2x > -1$ by $1/2$; the direction is kept:

$$2x > -1 \implies \frac{1}{2} \cdot 2x > \frac{1}{2} \cdot (-1) \implies x > -\frac{1}{2}.$$

Similarly the reverse implication holds, multiplying by $2 > 0$: $x > -\frac{1}{2} \implies 2x > -1$.

Conclusion. All three inequalities are equivalent,

$$2x + 1 > 0 \iff 2x > -1 \iff x > -\frac{1}{2},$$

and the solution of $2x + 1 > 0$ is $x \in (-\frac{1}{2}, \infty)$.

The example shows why both directions are checked. If we had shown only $2x + 1 > 0 \implies x > -\frac{1}{2}$, we would know that every solution lies in $(-\frac{1}{2}, \infty)$, but not that every number in $(-\frac{1}{2}, \infty)$ is a solution.

4.2.1 The general linear inequality

Proposition 4.1 (Solving $ax + b > 0$)

Let $a, b \in \mathbb{R}$. The solution set of $ax + b > 0$ is

$$\begin{cases} \left(-\frac{b}{a}, \infty\right) & \text{if } a > 0, \\ \left(-\infty, -\frac{b}{a}\right) & \text{if } a < 0, \\ \mathbb{R} \text{ if } b > 0, \text{ and } \emptyset \text{ if } b \leq 0, & \text{if } a = 0. \end{cases}$$

Proof. Adding $-b$ to both sides gives the equivalent inequality $ax > -b$. If $a > 0$, multiply by $1/a > 0$, which keeps the direction: $x > -b/a$. If $a < 0$, multiply by $1/a < 0$, which reverses it: $x < -b/a$. Both steps are reversible (multiply back by a). If $a = 0$ the inequality reads $0 > -b$, i.e. $b > 0$, which does not involve x at all: it is true for every x if $b > 0$ and for no x if $b \leq 0$. $\square$

Worked Example 4.2 – Dividing by a negative number

Solve $5 - 3x > 11$.

Solution. Add -5 to both sides: $-3x > 6$. Now divide by -3 , i.e. multiply by $-\frac{1}{3}$. Since $-\frac{1}{3} < 0$, the second clause of M6 applies and the direction *reverses*: $x < -2$. The solution is $x \in (-\infty, -2)$.

Check. $x = -3$ gives $5 + 9 = 14 > 11$, true; $x = 0$ gives $5 > 11$, false. Both agree with the answer.

Worked Example 4.3 – Collecting terms first

Find all x satisfying $\frac{1}{2}x - 3 > 2x + 1$.

Solution. Add $-2x$ and $+3$ to both sides (A5, twice): $\frac{1}{2}x - 2x > 1 + 3$, that is, $-\frac{3}{2}x > 4$. Multiply by $-\frac{2}{3} < 0$; the direction reverses (M6): $x < -\frac{8}{3}$. The solution is $x \in (-\infty, -\frac{8}{3})$. **Check:** $x = -3$ gives $-4.5 > -5$, true; $x = 0$ gives $-3 > 1$, false. [SG, mock Q2(b)]

4.2.2 Inequalities with a parameter

A **parameter** is a letter standing for a fixed but unspecified number, such as a price or a tax rate. When an inequality involves a parameter, the answer usually depends on it, and the dependence arises exactly at the step where we multiply or divide by an expression containing it.

Worked Example 4.4 – Solving $ax > 1$

Let $a \in \mathbb{R}$ be a parameter. Find all x with $ax > 1$.

Solution. The step we would like to take is “divide by a ”. Whether this is allowed, and what it does to the direction, depends on the sign of a , so we split into three exhaustive cases (O1).

- $a > 0$: $1/a$ exists (M4) and is positive, so the direction is kept: $x > 1/a$, i.e. $x \in (1/a, \infty)$.
- $a < 0$: $1/a$ exists and is negative, so the direction reverses: $x < 1/a$, i.e. $x \in (-\infty, 1/a)$.
- $a = 0$: there is no $1/0$, so we cannot divide. The inequality reads $0 > 1$, false for every x : the solution set is $\emptyset$.

Worked Example 4.5 – A parameter multiplying a bracket

Let $a \in \mathbb{R}$. Find all x satisfying $a(x - 1) > 0$. [SG, mock Q2(d)]

Solution.

- $a > 0$: multiply by $1/a > 0$: $x - 1 > 0$, so $x \in (1, \infty)$.
- $a < 0$: multiply by $1/a < 0$, reversing: $x - 1 < 0$, so $x \in (-\infty, 1)$.

- $a = 0$: the left side is 0 for every x , and $0 > 0$ is false, so the solution set is $\emptyset$.

“Your answer should depend on the parameter”

Parameter questions appeared on all three past ECN115 papers summarised in the study guide, usually with a hint such as “consider three cases: $a > 0$, $a = 0$, $a < 0$ ”. The marks are for carrying out each case correctly and for not forgetting the case $a = 0$, which can never be folded into the other two because $1/0$ does not exist.

4.3 Products: sign analysis

Worked Example 4.6 – A product of two linear factors

Find all x that satisfy $(2x + 1)(x - 1) > 0$.

[L1, slide 24]

Solution. Neither $2x + 1$ nor $x - 1$ can be equal to zero: otherwise $(2x + 1)(x - 1) = 0$ (Proposition 3.2), and $0 > 0$ is false. So by trichotomy (O1) each factor is either positive or negative, and it suffices to consider four cases (Proposition 3.4(vi)):

$2x + 1$	$x - 1$	$(2x + 1)(x - 1)$
> 0	> 0	> 0
> 0	< 0	< 0
< 0	> 0	< 0
< 0	< 0	> 0

Therefore $(2x + 1)(x - 1) > 0$ if and only if

$$[2x + 1 > 0 \text{ and } x - 1 > 0] \quad \text{or} \quad [2x + 1 < 0 \text{ and } x - 1 < 0].$$

Since $2x + 1 > 0$ is equivalent to $x > -\frac{1}{2}$, $x - 1 > 0$ is equivalent to $x > 1$, and $x > 1 \Rightarrow x > -\frac{1}{2}$, we have $[2x + 1 > 0 \text{ and } x - 1 > 0] \Leftrightarrow x > 1$. Similarly $[2x + 1 < 0 \text{ and } x - 1 < 0] \Leftrightarrow x < -\frac{1}{2}$.

Thus the solution is

$$x \in (-\infty, -\frac{1}{2}) \cup (1, \infty).$$

Two small but important points of logic appear in this example. Inside each case the two conditions are joined by “and”, so their solution sets are *intersected*; and a pair of conditions such as “ $x > 1$ and $x > -\frac{1}{2}$ ” collapses to the stronger one, $x > 1$. The cases themselves are joined by “or”, so their solution sets are combined by *union*.

4.3.1 The sign chart

The four-case table can be organised on the number line. The **critical points** are the values where a factor is zero, here $x = -\frac{1}{2}$ and $x = 1$. They split the line into intervals, and on each interval every factor keeps one sign. Record the signs in a chart and multiply.

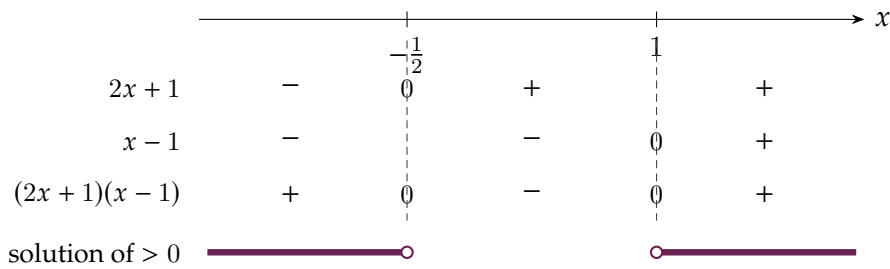


Figure 4.1: Sign chart for $(2x + 1)(x - 1)$. The critical points $-\frac{1}{2}$ and 1 divide the line into three intervals; on each, each factor has a constant sign. The product is positive on the two outer intervals, so the solution of $(2x + 1)(x - 1) > 0$ is $(-\infty, -\frac{1}{2}) \cup (1, \infty)$. Open dots show that the critical points are excluded, because the inequality is strict.

Why the signs are constant between critical points

A linear factor $mx + c$ with $m \neq 0$ changes sign only at its root $-c/m$: by Proposition 4.1 it is positive on one side of the root and negative on the other. So between consecutive critical points no factor changes sign, and neither does the product. This is why testing one convenient point in each interval is enough.

Worked Example 4.7 – A product that must be negative

Solve $(x + 3)(2x - 7) < 0$.

Solution. The critical points are $x = -3$ and $x = \frac{7}{2}$. A product is negative exactly when the factors have opposite signs:

- $x + 3 > 0$ and $2x - 7 < 0$: $x > -3$ and $x < \frac{7}{2}$, i.e. $-3 < x < \frac{7}{2}$;
- $x + 3 < 0$ and $2x - 7 > 0$: $x < -3$ and $x > \frac{7}{2}$, which is impossible.

The solution is $x \in (-3, \frac{7}{2})$.

Worked Example 4.8 – A non-strict inequality

Find all x satisfying $(3 - x)(x + 2) \geq 0$.

[SG, mock Q2(c)]

Solution. Now the product may also be zero, which happens at $x = 3$ and $x = -2$; both satisfy ≥ 0 , so they are included. Away from these points each factor is positive or negative, and the product is positive when the signs agree:

- $3 - x > 0$ and $x + 2 > 0$: $x < 3$ and $x > -2$, i.e. $-2 < x < 3$;
- $3 - x < 0$ and $x + 2 < 0$: $x > 3$ and $x < -2$, impossible.

Adding the two points where the product is zero gives $x \in [-2, 3]$. Note the *square* brackets.

Strict and non-strict inequalities

For > 0 or < 0 the critical points are excluded (round brackets); for ≥ 0 or ≤ 0 they are included (square brackets). Writing $(-2, 3)$ instead of $[-2, 3]$ in Worked Example 4.8 omits two solutions.

4.4 Quadratic inequalities

A quadratic inequality, such as $3x^2 - 5x + 2 > 0$, can be solved by rewriting the quadratic as a product of two linear factors and then using sign analysis.

Worked Example 4.9 – Factorising through the roots

Find all x that satisfy $3x^2 - 5x + 2 > 0$.

[L1, slide 25]

Solution. *Idea.* Represent $3x^2 - 5x + 2$ as a product of two linear terms (if possible), $3x^2 - 5x + 2 = e \cdot (x + f) \cdot (x + g)$ for some real numbers e, f, g .

Finding e . Expanding, $e(x + f)(x + g) = ex^2 + e(f + g)x + efg$. Comparing the coefficients of x^2 gives $e = 3$. Comparing the others gives only $f + g = -\frac{5}{3}$ and $fg = \frac{2}{3}$, which do not immediately give f and g .

Observation. Substitute $x = -f$: then $e(x + f)(x + g) = e \cdot 0 \cdot (g - f) = 0$. Thus $x = -f$ is a solution (root) of the equation $3x^2 - 5x + 2 = 0$, and similarly $x = -g$ is a root. So f and g can be found by solving the equation.

Solving the equation. The discriminant is $D = (-5)^2 - 4 \cdot 3 \cdot 2 = 25 - 24 = 1$, so the roots are

$$x_{1,2} = \frac{-(-5) \pm \sqrt{1}}{2 \cdot 3} = \frac{5 \pm 1}{6}, \quad x_1 = \frac{2}{3}, \quad x_2 = 1.$$

Hence $3x^2 - 5x + 2 = 3(x - \frac{2}{3})(x - 1)$.

Sign analysis. Solving $3(x - \frac{2}{3})(x - 1) > 0$ as in Worked Example 4.6 (the factor 3 is positive and does not affect the sign), the inequality $3x^2 - 5x + 2 > 0$ is equivalent to

$$x \in (-\infty, \frac{2}{3}) \cup (1, \infty).$$

4.4.1 The quadratic formula

Key Formula 4.1 – Roots of a quadratic

$$x_{1,2} = \frac{-b \pm \sqrt{D}}{2a}, \quad D = b^2 - 4ac$$

Symbols

a, b, c are the real coefficients of $ax^2 + bx + c$, with $a \neq 0$; D is the *discriminant*; x_1, x_2 are the roots, the values of x where $ax^2 + bx + c = 0$; the symbol $\pm$ means that one root uses $+$ and the other $-$.

What it means

The formula gives every real solution of $ax^2 + bx + c = 0$. The discriminant tells you how many there are: two distinct real roots if $D > 0$, one repeated root $x = -b/(2a)$ if $D = 0$, and no real roots if $D < 0$ (the square root of a negative number is not a real number).

Where it comes from

Completing the square: $ax^2 + bx + c = a[(x + \frac{b}{2a})^2 - \frac{D}{4a^2}]$ (Key Formula 4.2). Setting this to zero gives $(x + \frac{b}{2a})^2 = \frac{D}{4a^2}$, and taking square roots (possible only when $D \geq 0$) gives the formula.

When to use it

Whenever a quadratic must be factorised and the factors are not obvious; in particular, as the first step in solving any quadratic inequality.

Step-by-step example

For $3x^2 - 5x + 2$: $a = 3$, $b = -5$, $c = 2$. Then $D = (-5)^2 - 4(3)(2) = 1$, $\sqrt{D} = 1$, and $x = \frac{5 \pm 1}{6}$, giving $\frac{2}{3}$ and 1. Check: $3(\frac{2}{3})^2 - 5(\frac{2}{3}) + 2 = \frac{4}{3} - \frac{10}{3} + 2 = 0$.

Common mistakes

Writing $-b$ incorrectly when b is negative (here $-b = +5$); dividing only $\sqrt{D}$ by $2a$ instead of the whole numerator; forgetting that $D < 0$ means there are no real roots, not that the formula “fails”.

Rearrangements

Once the roots are known, the quadratic factorises as $ax^2 + bx + c = a(x - x_1)(x - x_2)$. Comparing coefficients gives $x_1 + x_2 = -b/a$ and $x_1x_2 = c/a$, a quick check on the roots.

Effect of changing the inputs

Increasing c (with $a > 0$) lowers $D = b^2 - 4ac$: the two roots move towards each other, merge when $D = 0$, and disappear when $D < 0$. Geometrically the parabola is lifted upwards until it no longer meets the x -axis.

How it appears in questions

Usually as an unmarked first step (“factorise”, “find the roots”). A parameter in c leads to a question such as “for which values of c does the equation have two distinct roots?”, answered by solving $D > 0$.

4.4.2 The sign of a quadratic

Once $ax^2 + bx + c = a(x - x_1)(x - x_2)$ with $x_1 < x_2$, sign analysis gives a rule worth remembering.

Proposition 4.2 (Sign of a quadratic with two roots)

Let $a > 0$ and $x_1 < x_2$. Then $a(x - x_1)(x - x_2)$ is positive for $x < x_1$ and for $x > x_2$, zero at x_1 and x_2 , and negative for $x_1 < x < x_2$. If $a < 0$ all the signs are reversed.

Proof. For $x < x_1$ both factors are negative; for $x > x_2$ both are positive; between the roots $x - x_1 > 0$ and $x - x_2 < 0$. Apply Proposition 3.4(vi), then multiply by a using M6. $\square$

Answering between the roots

For a quadratic with positive leading coefficient, “ > 0 ” is solved *outside* the roots and “ < 0 ” *between* them. A quick test of one point (for example $x = 0$, if it is not a root) catches the error: in Worked Example 4.9, $x = 0$ gives $2 > 0$, and 0 is indeed outside $[\frac{2}{3}, 1]$.

4.4.3 When the discriminant is zero or negative

The lecture example has $D > 0$. The other two cases are handled by completing the square.

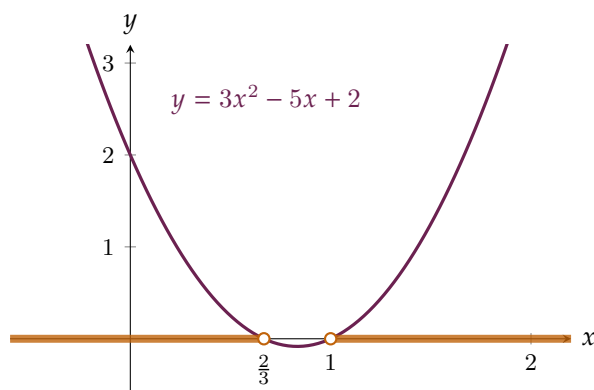


Figure 4.2: The parabola $y = 3x^2 - 5x + 2$ opens upwards because the leading coefficient 3 is positive. It crosses the x -axis at the roots $\frac{2}{3}$ and 1, lies below the axis between them and above it outside them. The thick segments show the solution set of $3x^2 - 5x + 2 > 0$.

Key Formula 4.2 – Completing the square

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 - \frac{D}{4a}, \quad D = b^2 - 4ac$$

Symbols

$a \neq 0$, b , c real coefficients; D the discriminant.

What it means

Every quadratic is a multiple of a perfect square plus a constant. Since a square is never negative, the sign of the quadratic can be read off directly.

Where it comes from

Expand the right-hand side: $a \left(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} \right) - \frac{b^2 - 4ac}{4a} = ax^2 + bx + \frac{b^2}{4a} - \frac{b^2}{4a} + c = ax^2 + bx + c$.

When to use it

When $D = 0$ or $D < 0$, or when you need the minimum (or maximum) value of a quadratic.

Step-by-step example

$x^2 - 5x + \frac{25}{4} = (x - \frac{5}{2})^2$ ($D = 0$), and $x^2 - 5x + 7 = (x - \frac{5}{2})^2 + \frac{3}{4}$ ($D = -3$).

Common mistakes

Forgetting to factor out a before completing the square; sign errors in $\frac{b}{2a}$.

Effect of changing the inputs

If $a > 0$ the quadratic is smallest at $x = -\frac{b}{2a}$, where it equals $-\frac{D}{4a}$. If $D < 0$ this minimum is positive, so the quadratic is positive everywhere; if $D = 0$ the minimum is 0, attained only at $x = -\frac{b}{2a}$.

How it appears in questions

“Show that the expression is positive for all x ”, with the hint “write it as a square plus a constant”.

Proposition 4.3 (Sign of a quadratic when $D \leq 0$)

Let $a > 0$. If $D = 0$, then $ax^2 + bx + c = a(x - x_0)^2$ with $x_0 = -\frac{b}{2a}$, which is positive for every $x \neq x_0$ and zero at x_0 . If $D < 0$, then $ax^2 + bx + c > 0$ for every real x .

Proof. By Key Formula 4.2, $ax^2 + bx + c = a(x - x_0)^2 - \frac{D}{4a}$. A square is ≥ 0 , and is > 0 unless what is squared is zero (Proposition 3.4(iv)). If $D = 0$ the constant vanishes; if $D < 0$ the constant $-\frac{D}{4a}$ is positive, so the sum is positive. $\square$

Worked Example 4.10 – A quadratic with a parameter

Let $c \in \mathbb{R}$ and consider $x^2 - 5x + c$.

[SG, mock Q3]

- (a) For which c does $x^2 - 5x + c = 0$ have two distinct real roots?
- (b) Solve $x^2 - 5x + 6 > 0$.
- (c) Solve $x^2 - 5x + \frac{25}{4} > 0$.
- (d) Show that if $c > \frac{25}{4}$ then $x^2 - 5x + c > 0$ for every $x \in \mathbb{R}$.

Solution. (a) $D = (-5)^2 - 4 \cdot 1 \cdot c = 25 - 4c$. Two distinct roots exactly when $D > 0$: $25 - 4c > 0 \Leftrightarrow c < \frac{25}{4}$ (dividing by $4 > 0$ keeps the direction). So $c \in (-\infty, \frac{25}{4})$.

(b) $D = 1$, roots $\frac{5 \pm 1}{2} = 2, 3$, so $x^2 - 5x + 6 = (x - 2)(x - 3)$. The leading coefficient is positive, so the expression is positive outside the roots: $x \in (-\infty, 2) \cup (3, \infty)$.

(c) $D = 0$ and $x^2 - 5x + \frac{25}{4} = (x - \frac{5}{2})^2$, which is positive except at $x = \frac{5}{2}$. So $x \in (-\infty, \frac{5}{2}) \cup (\frac{5}{2}, \infty) = \mathbb{R} \setminus \{\frac{5}{2}\}$. In (b) a whole interval was excluded; here the two roots have merged and only one point is excluded.

(d) Completing the square, $x^2 - 5x + c = (x - \frac{5}{2})^2 + (c - \frac{25}{4})$. The square is ≥ 0 and, when $c > \frac{25}{4}$, the constant is > 0 ; so the sum is positive for every x . (This is the case $D < 0$.)

Worked Example 4.11 – A non-strict quadratic inequality

Solve $2x^2 - 7x + 3 \geq 0$.

Solution. $D = 49 - 24 = 25$, roots $\frac{7 \pm 5}{4}$, i.e. $\frac{1}{2}$ and 3 . So $2x^2 - 7x + 3 = 2(x - \frac{1}{2})(x - 3)$. With leading coefficient $2 > 0$, the expression is ≥ 0 outside the roots and at the roots themselves: $x \in (-\infty, \frac{1}{2}] \cup [3, \infty)$.

4.5 Inequalities involving fractions

Worked Example 4.12 – An incorrect argument

A student solves $\frac{1}{x} > 2$ as follows: “Multiply both sides by x to obtain $1 > 2x$. Dividing by 2 gives $x < \frac{1}{2}$. Therefore the solution is $x \in (-\infty, \frac{1}{2})$.” Explain the error and find the correct solution.

[SG, mock Q6]

Solution. *The error.* Multiplying an inequality by x is governed by M6, which keeps the direction only if $x > 0$ and reverses it if $x < 0$. The sign of x is unknown, so the step is not justified. Moreover $x = 0$ is not allowed at all, since $1/0$ is undefined.

A value wrongly included. $x = -1$ lies in the student’s answer, but $1/(-1) = -1 > 2$ is false.

Correct solution. Split by the sign of x .

- $x > 0$: multiplying by $x > 0$ keeps the direction, $1 > 2x$, so $x < \frac{1}{2}$. With $x > 0$: $0 < x < \frac{1}{2}$.

- $x < 0$: then $1/x < 0 < 2$ (Proposition 3.4(v)), so the inequality fails for every negative x .
- $x = 0$: excluded, as $1/x$ is undefined.

The solution set is $(0, \frac{1}{2})$.

The safest general method avoids multiplying by an expression of unknown sign altogether.

CORE CONCEPT | Method for inequalities with fractions

1. Move everything to one side, so that the other side is 0.
2. Combine into a single fraction $\frac{N(x)}{M(x)}$ and factorise numerator and denominator.
3. Apply sign analysis to the factors of N and M : a quotient has the same sign as the corresponding product, because $N/M = N \cdot M \cdot (1/M^2)$ and $1/M^2 > 0$.
4. Exclude the zeros of M (the fraction is undefined there); include the zeros of N only for $\geq$ or $\leq$.

Worked Example 4.13 – Using the method

Solve $\frac{1}{x} > 2$ and $\frac{x+1}{x-2} \geq 0$.

Solution. For the first, $\frac{1}{x} - 2 > 0 \Leftrightarrow \frac{1-2x}{x} > 0$. Critical points $x = 0$ and $x = \frac{1}{2}$. Signs: for $x < 0$, numerator +, denominator –, quotient –; for $0 < x < \frac{1}{2}$, both +, quotient +; for $x > \frac{1}{2}$, numerator –, denominator +, quotient –. Solution $(0, \frac{1}{2})$, as before.

For the second, critical points $x = -1$ (numerator zero) and $x = 2$ (denominator zero). For $x < -1$ both factors are negative, quotient positive; for $-1 < x < 2$ the quotient is negative; for $x > 2$ it is positive. Include $x = -1$ (where the quotient is 0) but not $x = 2$ (undefined): $x \in (-\infty, -1] \cup (2, \infty)$.

4.6 Inequalities in economics

Worked Example 4.14 – Break-even output

A firm sells its product at price $p = 12$ per unit. Its total cost of producing q units is $C(q) = 200 + 7q$ (a fixed cost of 200 and a cost of 7 per unit). For which outputs $q \geq 0$ does the firm make a strictly positive profit?

Solution. Profit is revenue minus cost: $\pi(q) = 12q - (200 + 7q) = 5q - 200$. We need $5q - 200 > 0 \Leftrightarrow 5q > 200 \Leftrightarrow q > 40$ (multiplying by $\frac{1}{5} > 0$). The firm makes a profit exactly when $q \in (40, \infty)$. The value $q = 40$ is the *break-even output*.

Worked Example 4.15 – A quadratic profit function

Suppose instead that profit is $\pi(q) = -q^2 + 10q - 16$. For which q is profit positive?

Solution. $-q^2 + 10q - 16 > 0 \Leftrightarrow q^2 - 10q + 16 < 0$ (multiplying by -1 reverses the direction).

The roots of $q^2 - 10q + 16$ are $\frac{10 \pm \sqrt{100 - 64}}{2} = \frac{10 \pm 6}{2}$, i.e. 2 and 8, so $q^2 - 10q + 16 = (q - 2)(q - 8)$, which is negative between the roots. Profit is positive exactly for $q \in (2, 8)$: producing too little fails to cover fixed costs, and producing too much drives costs up faster than revenue.

4.7 Chapter review

Summary

Solving an inequality means describing its solution set exactly, which requires a chain of equivalences. Adding a number to both sides is always reversible; multiplying by a positive number keeps the direction, by a negative number reverses it, and by zero destroys it (M6). A linear inequality $ax + b > 0$ is solved by one addition and one multiplication, with three cases when a is a parameter. A product is positive when its factors share a sign and negative when they have opposite signs; organising this on the number line around the critical points gives the sign chart. A quadratic is factorised through its roots, found with the discriminant; with positive leading coefficient it is positive outside its roots and negative between them, and completing the square handles the cases $D = 0$ and $D < 0$. Fractions are handled by moving everything to one side and analysing the signs of numerator and denominator, never by multiplying by an expression of unknown sign.

Key definitions

- **Solution set** of an inequality: the set of all values of the unknown for which it is true.
- **Critical points**: the values at which a factor (of a numerator or denominator) is zero.
- **Discriminant** of $ax^2 + bx + c$: $D = b^2 - 4ac$.

Key equations

$$D = b^2 - 4ac, \quad x_{1,2} = \frac{-b \pm \sqrt{D}}{2a},$$

$$ax^2 + bx + c = a(x - x_1)(x - x_2) \quad (D \geq 0), \quad ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 - \frac{D}{4a}.$$

Connections

Every step uses the properties of Chapter 3. Answers are sets, written with the interval and union notation of Chapter 2. Absolute-value inequalities (Chapter 5) reduce to the methods here, and the definition of a limit (Chapter 10) requires solving inequalities such as $2^{-n} < \varepsilon$ for n . In Topic 4, the sign of a derivative written as a fraction $P(x)/Q(x)$ is found by exactly this sign analysis.

Common mistakes

- Not reversing the inequality when multiplying or dividing by a negative number.

- Dividing by a parameter or variable without considering its sign, or forgetting the zero case.
- Multiplying both sides by x in an inequality with $1/x$.
- Using $\cap$ to combine the cases of a sign analysis (cases are combined with $\cup$).
- Solving “ > 0 ” for a quadratic with positive leading coefficient by taking the interval between the roots.
- Wrong brackets: round for strict inequalities and for excluded points, square for included endpoints.

Exam checklist

- ☐ Solve a linear inequality, stating where the direction changes and why.
- ☐ Solve $ax > b$ or $a(x - k) > 0$ for all values of a parameter a .
- ☐ Carry out a sign analysis for a product of linear factors, including non-strict cases.
- ☐ Find roots with the discriminant and factorise a quadratic.
- ☐ Solve quadratic inequalities for $D > 0$, $D = 0$ and $D < 0$.
- ☐ Solve inequalities involving fractions without multiplying by an expression of unknown sign.

4.8 Practice questions

Question 4.1 [basic]

Solve: (a) $3x - 4 \leq 8$; (b) $7 - 2x > 1$; (c) $\frac{x}{3} + 1 \geq \frac{x}{2}$.

Question 4.2 [basic]

Solve $(x - 4)(x + 1) > 0$ and $(x - 4)(x + 1) \leq 0$.

Question 4.3 [mathematical]

Solve $x^2 - x - 6 < 0$ and $4x^2 + 4x + 1 > 0$.

Question 4.4 [mathematical]

Show that $x^2 + 2x + 5 > 0$ for every real x .

Question 4.5 [conceptual]

Explain why, in a sign analysis, the cases are combined with $\cup$ while the conditions within a case are combined with $\cap$.

Question 4.6 [mathematical]

Solve $\frac{2x - 1}{x + 3} < 1$.

Question 4.7 [application]

A consumer has £60 to spend on two goods priced £4 and £6 per unit. She wants to buy x units of the first good and exactly 6 units of the second. For which $x \geq 0$ can she afford this?

Question 4.8 [multi-step]

Let $b \in \mathbb{R}$ be a parameter. Find all x with $bx + 3 > x$. Your answer should depend on b .

Question 4.9 [multi-step]

For which values of the parameter k is $x^2 + kx + 4 > 0$ for every real x ?

Question 4.10 [exam-style]

Let $a \in \mathbb{R}$ be a parameter. (a) Find all x satisfying $(x - a)(x - 1) < 0$. Your answer should depend on a . Hint: consider the cases $a > 1$, $a = 1$ and $a < 1$. (b) A student claims that $\frac{x-1}{x} \geq 1$ is equivalent to $x - 1 \geq x$ and hence has no solutions. Explain why the argument is not correct and find the solution set.

4.9 Solutions to practice questions

Solution to Question 4.1

(a) $3x \leq 12$, so $x \leq 4$: $x \in (-\infty, 4]$. (b) $-2x > -6$; dividing by $-2 < 0$ reverses: $x < 3$, so $x \in (-\infty, 3)$. (c) Multiply by $6 > 0$: $2x + 6 \geq 3x$, so $6 \geq x$: $x \in (-\infty, 6]$.

Solution to Question 4.2

Critical points -1 and 4 . The product is positive when both factors have the same sign: $x < -1$ or $x > 4$. So $(x - 4)(x + 1) > 0 \Leftrightarrow x \in (-\infty, -1) \cup (4, \infty)$, and $(x - 4)(x + 1) \leq 0 \Leftrightarrow x \in [-1, 4]$ (between the roots, roots included).

Solution to Question 4.3

$x^2 - x - 6 = (x - 3)(x + 2)$, negative between the roots: $x \in (-2, 3)$. $4x^2 + 4x + 1 = (2x + 1)^2$ ($D = 0$), positive except at $x = -\frac{1}{2}$: $x \in \mathbb{R} \setminus \{-\frac{1}{2}\}$.

Solution to Question 4.4

$x^2 + 2x + 5 = (x + 1)^2 + 4$. A square is ≥ 0 , so the sum is $\geq 4 > 0$ for every x . (Equivalently $D = 4 - 20 < 0$ with a positive leading coefficient.)

Solution to Question 4.5

Within a case, *both* conditions must hold (for instance “ $2x + 1 > 0$ and $x - 1 > 0$ ”), and the set of x satisfying two conditions simultaneously is the intersection of their solution sets. A number solves the original inequality if it falls into *some* case (“case 1 or case 4”), and the set of x satisfying at least one of several conditions is the union of their solution sets. Intersection corresponds to “and” and union to “or” (Section 2.4).

Solution to Question 4.6

$\frac{2x-1}{x+3} - 1 < 0 \Leftrightarrow \frac{2x-1-(x+3)}{x+3} < 0 \Leftrightarrow \frac{x-4}{x+3} < 0$. Critical points -3 and 4 ; the quotient is negative when the signs differ, which happens for $-3 < x < 4$. Solution $(-3, 4)$.

Solution to Question 4.7

Cost $4x + 6 \cdot 6 = 4x + 36 \leq 60 \Leftrightarrow 4x \leq 24 \Leftrightarrow x \leq 6$. With $x \geq 0$: $x \in [0, 6]$.

Solution to Question 4.8

$bx + 3 > x \Leftrightarrow (b - 1)x > -3$. If $b > 1$: $x > -\frac{3}{b-1}$, i.e. $x \in (-\frac{3}{b-1}, \infty)$. If $b < 1$: dividing by $b - 1 < 0$ reverses, $x < -\frac{3}{b-1} = \frac{3}{1-b}$, i.e. $x \in (-\infty, \frac{3}{1-b})$. If $b = 1$: $0 > -3$, true for all x : the solution set is $\mathbb{R}$.

Solution to Question 4.9

With leading coefficient $1 > 0$, $x^2 + kx + 4 > 0$ for all x exactly when there are no real roots, $D < 0$ (if $D = 0$ the quadratic is zero at one point; if $D > 0$ it is negative between the roots). $D = k^2 - 16 < 0 \Leftrightarrow (k - 4)(k + 4) < 0 \Leftrightarrow k \in (-4, 4)$.

Solution to Question 4.10

(a) The critical points are a and 1 ; the product is negative strictly between them. If $a > 1$: $x \in (1, a)$. If $a < 1$: $x \in (a, 1)$. If $a = 1$: $(x - 1)^2 < 0$ has no solution, so the solution set is $\emptyset$. (b) Multiplying by x is valid only for $x > 0$; for $x < 0$ the direction reverses, and $x = 0$ is excluded. Correct method: $\frac{x-1}{x} - 1 \geq 0 \Leftrightarrow \frac{-1}{x} \geq 0$. The numerator is negative, so the quotient is ≥ 0 exactly when $x < 0$ (and it is never 0). Solution set $(-\infty, 0)$. For example $x = -1$ gives $\frac{-2}{-1} = 2 \geq 1$, a solution the student's argument missed.

Absolute Value and the Triangle Inequality

The absolute value of a number measures its size regardless of sign; the absolute value of a difference measures the distance between two numbers. Distance is the key idea behind limits: a sequence approaches a number when the distance between them becomes as small as we please. This chapter defines the absolute value, proves its basic properties, establishes the triangle inequality, and shows how to solve inequalities involving absolute values.

Before you start: prerequisites

The properties of the real numbers and the sign rules (Chapter 3, especially Proposition 3.4); proof by cases (Section 1.7); solving linear and product inequalities (Chapter 4); intervals (Section 3.5).

5.1 Definition

Definition 5.1 (Absolute value)

The **absolute value** $|x|$ of a real number x is defined as follows:

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

Equivalently, $|x| = \max\{x, -x\}$, the larger of the two numbers x and $-x$.

[L2, slide 10]

For a non-negative number the absolute value changes nothing; for a negative number it removes the minus sign. The two descriptions agree: if $x \geq 0$ then $x \geq -x$ (since $-x \leq 0 \leq x$), so $\max\{x, -x\} = x$; if $x < 0$ then $-x > 0 > x$, so $\max\{x, -x\} = -x$.

Worked Example 5.1 – Computing absolute values

$$|2| = 2, \quad |-5| = 5, \quad \left|\frac{7}{3}\right| = \frac{7}{3}, \quad |-\sqrt{11}| = \sqrt{11}, \quad |0| = 0.$$

[L2, slide 10]

$|-x|$ is not always x

It is tempting to think “the absolute value removes the minus sign”, so that $|-x| = x$. That

is true only if $x \geq 0$. For $x = -3$, $|-x| = |3| = 3 = -x$. The correct statement is $|-x| = |x|$ (Proposition 5.1).

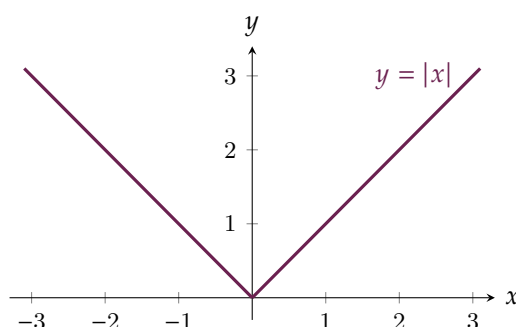


Figure 5.1: The graph of $y = |x|$. To the right of 0 it coincides with $y = x$; to the left it coincides with $y = -x$. The graph has a corner at the origin.

5.2 Properties

Proposition 5.1 (Properties of the absolute value)

For all real numbers a and b :

[L2, slide 10]

- (i) $|a| \geq a$;
- (ii) $|a| \geq -a$;
- (iii) $|-a| = |a|$;
- (iv) $|a \cdot b| = |a| \cdot |b|$;
- (v) if $b \neq 0$, then $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$ (this follows from the previous property).

In addition, $|a| \geq 0$, with $|a| = 0$ if and only if $a = 0$.

Proof. (i) and (ii) are immediate from $|a| = \max\{a, -a\}$: the maximum of two numbers is at least as large as each of them. Combining them with Proposition 3.4 gives the useful double inequality

$$-|a| \leq a \leq |a|. \quad (5.1)$$

(iii) $|-a| = \max\{-a, -(-a)\} = \max\{-a, a\} = |a|$.

(iv) Consider the signs. If $a \geq 0$ and $b \geq 0$ then $ab \geq 0$ and $|ab| = ab = |a||b|$. If $a \geq 0$ and $b < 0$ then $ab \leq 0$, so $|ab| = -ab = a \cdot (-b) = |a||b|$. The case $a < 0$, $b \geq 0$ is symmetric. If $a < 0$ and $b < 0$ then $ab > 0$, so $|ab| = ab = (-a)(-b) = |a||b|$.

(v) Apply (iv) to $a = \frac{a}{b} \cdot b$: $|a| = \left|\frac{a}{b}\right| \cdot |b|$. Since $b \neq 0$, $|b| > 0$, and dividing by $|b|$ gives the result.

Finally, if $a > 0$ then $|a| = a > 0$; if $a < 0$ then $|a| = -a > 0$; and $|0| = 0$. $\square$

Absolute value and square roots

For every real a , $|a|^2 = a^2$ and $\sqrt{a^2} = |a|$ (where $\sqrt{}$ denotes the non-negative square root).

The second identity is a frequent source of errors: $\sqrt{(-3)^2} = \sqrt{9} = 3$, not -3 .

5.3 Absolute value as distance

CORE CONCEPT | Distance on the number line

For real numbers x and y , $|x - y|$ is the **distance** between x and y on the number line. In particular $|x| = |x - 0|$ is the distance from x to 0.

[L2, slide 11]

Distance does not depend on direction: $|x - y| = |y - x|$ by Proposition 5.1(iii). For example, the distance between -2 and 5 is $|-2 - 5| = 7 = |5 - (-2)|$.

Statements about distance translate directly into absolute values, and vice versa:

In words	In symbols
x is within 2 of 3	$ x - 3 < 2$
x is at least 4 away from -1	$ x + 1 \geq 4$
a_n is within ε of x	$ a_n - x < \varepsilon$

The last line is exactly the condition that appears in the definition of a limit (Chapter 10).

5.4 The triangle inequality

Theorem 5.2 (Triangle inequality)

For any real numbers x, y, z ,

$$|x - z| \leq |x - y| + |y - z|.$$

[L2, slide 11; HW-1 Problem B1]

Intuition

$|x - z|$ is the distance from x to z ; $|x - y|$ is the distance from x to y ; $|y - z|$ is the distance from y to z . Going from x to z directly should not be longer than going first from x to y and then from y to z . A detour cannot be shorter than the direct route. The name comes from geometry: in a triangle, one side is never longer than the sum of the other two.

⚠ The proof

The slides pose the triangle inequality as Problem B1 of the first homework and refer to “notes” for the argument, but no proof appears in the materials provided. The proof below is supplied for completeness.

The cleanest route is to prove a two-number version first.

Theorem 5.3 (Triangle inequality, two-number form)

For any real numbers a and b , $|a + b| \leq |a| + |b|$.

Proof. By (5.1), $-|a| \leq a \leq |a|$ and $-|b| \leq b \leq |b|$. Adding these (Chapter 3, Question 3.5 applied to $\leq$),

$$-(|a| + |b|) \leq a + b \leq |a| + |b|.$$

Now $|a + b|$ equals either $a + b$ or $-(a + b)$. The right-hand inequality gives $a + b \leq |a| + |b|$; the left-hand inequality, multiplied by -1 (reversing it), gives $-(a + b) \leq |a| + |b|$. In both cases $|a + b| \leq |a| + |b|$. $\square$

Proof of Theorem 5.2. Put $a = x - y$ and $b = y - z$. Then $a + b = x - z$, and Theorem 5.3 gives $|x - z| \leq |x - y| + |y - z|$. $\square$

The slides ask how the three-number form can be used to prove two further inequalities. Both follow by choosing the three numbers cleverly.

Worked Example 5.2 – Applying the triangle inequality

Use Theorem 5.2 to show that for any real a, b : (i) $|a + b| \leq |a| + |b|$; (ii) $|a| \leq |a - b| + |b|$. [L2, slide 11]

Solution. (i) Take $x = a$, $y = 0$, $z = -b$. Then $|x - z| = |a - (-b)| = |a + b|$, $|x - y| = |a - 0| = |a|$ and $|y - z| = |0 - (-b)| = |b|$. So $|a + b| \leq |a| + |b|$.

(ii) Take $x = a$, $y = b$, $z = 0$. Then $|x - z| = |a|$, $|x - y| = |a - b|$ and $|y - z| = |b|$. So $|a| \leq |a - b| + |b|$.

Reverse triangle inequality

Rearranging (ii) gives $|a| - |b| \leq |a - b|$; exchanging a and b gives $|b| - |a| \leq |a - b|$. Together: $||a| - |b|| \leq |a - b|$. The sizes of two numbers differ by no more than the distance between them.

Key Formula 5.1 – The triangle inequality

$$|x - z| \leq |x - y| + |y - z| \quad \text{equivalently} \quad |a + b| \leq |a| + |b|$$

Symbols

x, y, z, a, b are any real numbers; $|\cdot|$ is the absolute value.

What it means

The distance between two points is at most the length of any route between them that passes through a third point. In the second form: the size of a sum is at most the sum of the sizes.

Where it comes from

From $-|a| \leq a \leq |a|$ and $-|b| \leq b \leq |b|$ by adding (Theorem 5.3); the three-number form follows by substituting $a = x - y$, $b = y - z$.

When to use it

To bound the distance between two numbers by passing through a convenient intermediate

number; to bound the size of a sum; above all in proofs about limits (Worked Example 10.7).

Step-by-step example

With $x = 1, y = x_0, z = 0$: $1 = |1 - 0| \leq |1 - x_0| + |x_0 - 0|$. If both distances on the right were less than $\frac{1}{4}$, we would get $1 < \frac{1}{2}$, which is false. This is exactly the argument that $1, 0, 1, 0, \dots$ has no limit.

Common mistakes

Writing $|a + b| = |a| + |b|$ (equality holds only when a and b do not have opposite signs: $|3 + (-5)| = 2$ but $|3| + |-5| = 8$); writing $|a - b| \leq |a| - |b|$ (false: $a = 0, b = 1$ gives $1 \leq -1$).

Rearrangements

$|a| - |b| \leq |a - b|$ and $||a| - |b|| \leq |a - b|$ (reverse triangle inequality); $|a - b| \leq |a| + |b|$ (apply the two-number form to a and $-b$).

Effect of changing the inputs

Equality $|a + b| = |a| + |b|$ holds exactly when $ab \geq 0$. In the three-number form, equality holds exactly when y lies between x and z (inclusive): moving y outside that interval makes the detour strictly longer.

How it appears in questions

As a step in a proof (“by the triangle inequality, ...”), particularly in arguments by contradiction about limits; or as a short “show that” question.

5.5 Solving inequalities with absolute values

Proposition 5.4 (Within a distance)

Let $a \in \mathbb{R}$ and $r > 0$. Then

$$\begin{aligned} |x - a| < r &\iff a - r < x < a + r \iff x \in (a - r, a + r), \\ |x - a| > r &\iff x < a - r \text{ or } x > a + r. \end{aligned}$$

The same holds with $\leq$ and $\geq$, with the endpoints included.

Proof. Split by the sign of $x - a$. If $x \geq a$, then $|x - a| = x - a$, and $|x - a| < r \iff x < a + r$; combined with $x \geq a$ this gives $a \leq x < a + r$. If $x < a$, then $|x - a| = a - x$, and $|x - a| < r \iff x > a - r$; combined with $x < a$ this gives $a - r < x < a$. The union of the two cases is $a - r < x < a + r$. The second statement is the negation of the first with $\leq$ (Section 1.5): $\text{not}(a - r \leq x \leq a + r) \iff x < a - r$ or $x > a + r$. $\square$

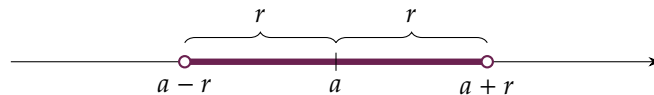


Figure 5.2: The set $\{x \in \mathbb{R} \mid |x - a| < r\}$ is the open interval of radius r centred at a : all points whose distance from a is less than r .

Worked Example 5.3 – Inequalities with absolute values

Solve (a) $|x - 3| < 2$; (b) $|2x + 1| \leq 5$; (c) $|x| > 3$; (d) $|x - 1| < |x + 2|$.

Solution. (a) By Proposition 5.4 with $a = 3$, $r = 2$: $1 < x < 5$, so $x \in (1, 5)$. In words: the numbers within distance 2 of 3.

(b) $|2x + 1| \leq 5 \Leftrightarrow -5 \leq 2x + 1 \leq 5 \Leftrightarrow -6 \leq 2x \leq 4 \Leftrightarrow -3 \leq x \leq 2$, so $x \in [-3, 2]$. (Alternatively write $|2x + 1| = 2|x + \frac{1}{2}|$ and use distance from $-\frac{1}{2}$.)

(c) $x < -3$ or $x > 3$: $x \in (-\infty, -3) \cup (3, \infty)$.

(d) Both sides are non-negative, and for non-negative numbers $u < v \Leftrightarrow u^2 < v^2$ (by Chapter 3, Question 3.7 and its converse). So the inequality is equivalent to $(x - 1)^2 < (x + 2)^2$, i.e. $x^2 - 2x + 1 < x^2 + 4x + 4$, i.e. $-6x < 3$, i.e. $x > -\frac{1}{2}$. In words: x is closer to 1 than to -2 exactly when it lies to the right of the midpoint $-\frac{1}{2}$.

Splitting $|x - a| < r$ wrongly

$|x - 3| < 2$ does *not* mean “ $x - 3 < 2$ ” alone; that drops the lower bound. Nor does $|x| > 3$ mean $-3 > x > 3$, which is impossible. Inequalities of the form $|\cdot| < r$ give a single interval (“and”); those of the form $|\cdot| > r$ give two pieces (“or”).

Worked Example 5.4 – An economic tolerance band

A central bank targets an inflation rate of 2% and regards outcomes within one percentage point of the target as acceptable. Write the acceptable range as an absolute-value inequality and as an interval.

Solution. If π denotes inflation in per cent, the condition is $|\pi - 2| \leq 1$, i.e. $1 \leq \pi \leq 3$, or $\pi \in [1, 3]$. Inflation outside this band, $|\pi - 2| > 1$, means $\pi < 1$ or $\pi > 3$.

5.6 Chapter review

Summary

$|x|$ equals x for $x \geq 0$ and $-x$ for $x < 0$; equivalently $|x| = \max\{x, -x\}$. It is never negative and is zero only at 0. It satisfies $|a| \geq a$, $|a| \geq -a$, $|-a| = |a|$, $|ab| = |a||b|$ and $|a/b| = |a|/|b|$. The quantity $|x - y|$ is the distance between x and y . The triangle inequality, $|x - z| \leq |x - y| + |y - z|$, says a detour is never shorter than the direct route; its two-number form is $|a + b| \leq |a| + |b|$. Inequalities $|x - a| < r$ describe intervals of radius r about a , and $|x - a| > r$ describes the two pieces outside.

Key definitions

- $|x| = x$ if $x \geq 0$, $|x| = -x$ if $x < 0$; equivalently $|x| = \max\{x, -x\}$.
- $|x - y|$: the distance between x and y .

Key equations

$$\begin{aligned} -|a| \leq a \leq |a|, \quad & |-a| = |a|, \quad & |ab| = |a||b|, \quad & \left| \frac{a}{b} \right| = \frac{|a|}{|b|} \quad (b \neq 0), \\ |x - z| \leq |x - y| + |y - z|, \quad & |a + b| \leq |a| + |b|, \quad & |a| \leq |a - b| + |b|, \\ & |x - a| < r \Leftrightarrow a - r < x < a + r. \end{aligned}$$

Connections

The proofs use cases (Section 1.7) and the sign rules of Proposition 3.4. Absolute-value inequalities reduce to the linear inequalities of Chapter 4. The distance interpretation and the triangle inequality are the main tools of Chapter 10, where they are used to prove that a sequence converges or that it does not.

Common mistakes

- Writing $|-x| = x$ or $\sqrt{x^2} = x$ without the condition $x \geq 0$.
- Writing $|a + b| = |a| + |b|$ or $|a - b| = |a| - |b|$.
- Solving $|x - a| < r$ as $x - a < r$ only.
- Squaring both sides of an inequality whose sides might be negative.

Exam checklist

- ☐ State the definition of $|x|$ in both forms.
- ☐ State and prove the properties $|a| \geq a$, $|-a| = |a|$, $|ab| = |a||b|$.
- ☐ State the triangle inequality and explain it as a statement about distances.
- ☐ Derive $|a + b| \leq |a| + |b|$ and $|a| \leq |a - b| + |b|$ from the three-number form.
- ☐ Solve inequalities of the forms $|x - a| < r$, $|x - a| \geq r$ and $|\cdot| < |\cdot|$.

5.7 Practice questions

Question 5.1 [basic]

Evaluate $|-7|$, $|3 - 8|$, $|2 - \sqrt{5}|$, $|\frac{4}{9}|$ and $|x^2 + 1|$ (for real x).

Question 5.2 [basic]

Solve (a) $|x + 4| < 1$; (b) $|3x - 6| \geq 9$; (c) $|5 - x| \leq 0$.

Question 5.3 [conceptual]

Give an example of real numbers a, b with $|a + b| < |a| + |b|$, and one with $|a + b| = |a| + |b|$. What distinguishes the two situations?

Question 5.4 [mathematical]

Prove that for all real a, b , $|a - b| \leq |a| + |b|$.

Question 5.5 [mathematical]

Prove that $||a| - |b|| \leq |a - b|$ for all real a, b .

Question 5.6 [mathematical]

Solve $|x - 2| > |x|$ and interpret your answer as a statement about distances.

Question 5.7 [application]

A firm's production line is calibrated to fill bags with 500 grams of flour. A bag is rejected if its weight w differs from 500 grams by more than 2% of 500 grams. Write the condition for acceptance as an absolute-value inequality and solve it.

Question 5.8 [multi-step]

Show that if $|x - 3| < 1$ then $|x^2 - 9| < 7$. (Hint: write $x^2 - 9 = (x - 3)(x + 3)$ and bound $|x + 3|$.)

Question 5.9 [exam-style]

(a) State the triangle inequality for three real numbers x, y, z and explain its meaning in terms of distances on the number line. (b) Suppose that the real number c satisfies both $|c - 1| < \frac{1}{4}$ and $|c| < \frac{1}{4}$. Use the triangle inequality to derive a contradiction, and explain what this shows about a number that is simultaneously close to 0 and to 1.

5.8 Solutions to practice questions

Solution to Question 5.1

$|-7| = 7$; $|3 - 8| = |-5| = 5$; since $\sqrt{5} > 2$, $|2 - \sqrt{5}| = \sqrt{5} - 2$; $|- \frac{4}{9}| = \frac{4}{9}$; $x^2 + 1 \geq 1 > 0$, so $|x^2 + 1| = x^2 + 1$.

Solution to Question 5.2

(a) $-1 < x + 4 < 1 \Leftrightarrow -5 < x < -3$: $x \in (-5, -3)$. (b) $|3x - 6| = 3|x - 2| \geq 9 \Leftrightarrow |x - 2| \geq 3 \Leftrightarrow x \leq -1$ or $x \geq 5$: $x \in (-\infty, -1] \cup [5, \infty)$. (c) An absolute value is never negative, so $|5 - x| \leq 0 \Leftrightarrow |5 - x| = 0 \Leftrightarrow x = 5$. Solution $\{5\}$.

Solution to Question 5.3

$a = 3, b = -5$: $|a + b| = 2 < 8 = |a| + |b|$. $a = 3, b = 5$: $|a + b| = 8 = |a| + |b|$. Equality holds exactly when a and b do not have opposite signs ($ab \geq 0$); when they have opposite signs, part of one cancels part of the other.

Solution to Question 5.4

Apply $|u + v| \leq |u| + |v|$ with $u = a, v = -b$: $|a - b| \leq |a| + |-b| = |a| + |b|$.

Solution to Question 5.5

From $|a| \leq |a - b| + |b|$ (Worked Example 5.2), $|a| - |b| \leq |a - b|$. Exchanging the roles of a and b : $|b| - |a| \leq |b - a| = |a - b|$. So both $|a| - |b|$ and its negative are at most $|a - b|$; since $||a| - |b||$ equals one of them, $||a| - |b|| \leq |a - b|$.

Solution to Question 5.6

Both sides are non-negative, so square: $(x - 2)^2 > x^2 \Leftrightarrow x^2 - 4x + 4 > x^2 \Leftrightarrow -4x > -4 \Leftrightarrow x < 1$. Solution $(-\infty, 1)$: the points closer to 0 than to 2 are those to the left of the midpoint 1.

Solution to Question 5.7

2% of 500 is 10. Acceptance: $|w - 500| \leq 10 \Leftrightarrow 490 \leq w \leq 510$, i.e. $w \in [490, 510]$ grams.

Solution to Question 5.8

If $|x - 3| < 1$ then $2 < x < 4$, so $5 < x + 3 < 7$ and $|x + 3| < 7$. Then $|x^2 - 9| = |x - 3| |x + 3| < 1 \cdot 7 = 7$. (Both factors are non-negative, so the strict inequalities may be multiplied: if $0 \leq u < 1$ and $0 \leq v < 7$ then $uv \leq 1 \cdot v < 7$ when $v > 0$, and $uv = 0 < 7$ when $v = 0$.)

Solution to Question 5.9

(a) For any real x, y, z , $|x - z| \leq |x - y| + |y - z|$. Since $|u - v|$ is the distance between u and v , it says that the distance from x to z is at most the distance from x to y plus the distance from y to z : going via y is never shorter than going directly. (b) Take $x = 1$, $y = c$, $z = 0$: $1 = |1 - 0| \leq |1 - c| + |c - 0| < \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$, so $1 < \frac{1}{2}$, a contradiction. Hence no number can be within $\frac{1}{4}$ of both 0 and 1. This is the key step in showing that $1, 0, 1, 0, \dots$ has no limit (Worked Example 10.7): a limit would have to be close to both 0 and 1 at once.

Summation Notation

Totals are central to economics: total revenue over twelve months, total consumption over a lifetime, the sum of the incomes of everyone in a country. Writing out such sums term by term is impractical, and the summation operator $\sum$ provides a compact and precise notation for them. This chapter defines the notation, fixes the conventions for unusual cases, and establishes the three properties that allow sums to be manipulated.

Before you start: prerequisites

The properties of addition and multiplication A1–A4, M1–M5 (Chapter 3); the natural numbers and integers (Section 3.1).

6.1 The summation operator

We can add many numbers, such as $3 + 5 + 2 + 9 + 5$. When the numbers are labelled by an index that runs through consecutive integers, there is a shorthand.

Definition 6.1 (Summation operator)

Let $a_k, a_{k+1}, \dots, a_n$ be numbers, indexed by an integer j that takes the consecutive values $k, k+1, \dots$ up to n . Their sum is written

$$\sum_{j=k}^n a_j = a_k + a_{k+1} + \dots + a_{n-1} + a_n.$$

We read this as follows: as the index j changes from k to n , we sum up all the corresponding numbers a_j . In running text the same sum is often written $\sum_{j=k}^n a_j$. [L1, slide 27]

The integer j is the **index of summation**, k is the **lower limit** and n the **upper limit**. The index is a *dummy variable*: it has no meaning outside the sum, and any other letter can be used without changing the value, $\sum_{j=1}^4 j^2 = \sum_{i=1}^4 i^2 = 30$. The limits k and n may be any integers, including negative ones and zero.

Worked Example 6.1 – Evaluating sums

Evaluate, for $a_j = j$ and $b_j = j^2$,

$$\sum_{j=3}^7 a_j, \quad \sum_{j=-2}^3 a_j, \quad \sum_{j=3}^7 b_j, \quad \sum_{j=-2}^3 b_j.$$

[L1, slide 28]

Solution. Write out each term and add:

$$\sum_{j=3}^7 j = 3 + 4 + 5 + 6 + 7 = 25,$$

$$\sum_{j=-2}^3 j = -2 - 1 + 0 + 1 + 2 + 3 = 3,$$

$$\sum_{j=3}^7 j^2 = 3^2 + 4^2 + 5^2 + 6^2 + 7^2 = 9 + 16 + 25 + 36 + 49 = 135,$$

$$\sum_{j=-2}^3 j^2 = (-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2 + 3^2 = 4 + 1 + 0 + 1 + 4 + 9 = 19.$$

Note that the sums starting at $j = -2$ include the terms for $j = -2, -1$ and 0 .

6.2 Counting terms, one-term sums and empty sums

CORE CONCEPT | The number of terms

The sum $\sum_{j=k}^n a_j$ (with $n \geq k$) has $n - k + 1$ terms, because both the lower and the upper limit are included. For example, $\sum_{j=3}^7$ has $7 - 3 + 1 = 5$ terms ($j = 3, 4, 5, 6, 7$), not 4.

Two boundary cases need to be fixed.

Definition 6.2 (One term and zero terms)

- When $n = k$, the sum consists of one term, with index $j = n = k$: $\sum_{j=k}^k a_j = a_k$.
- By analogy, when $n = k - 1$, we define the value of the summation operator as a sum of zero terms, which we take to be zero: $\sum_{j=k}^{k-1} a_j = 0$.

[L1, slide 29]

The first is a consequence of the definition; the second is a *convention*, a choice made so that the notation behaves well. Why zero? Adding a sum of no numbers should not change a total, and

zero is the number whose addition changes nothing (A3). Proposition 6.2 below shows that zero is in fact the only value consistent with the splitting property P3.

Key Formula 6.1 – Sum of a constant

$$\sum_{j=k}^n a = \underbrace{a + a + \cdots + a}_{n-k+1 \text{ terms}} = (n - k + 1) a$$

Symbols

a is a fixed real number (the same for every j); $k \leq n$ are integers; $n - k + 1$ is the number of terms.

What it means

If every term equals a , the sum is the number of terms times a .

[L1, slide 29]

Where it comes from

Repeated addition is multiplication: the sum has $n - k + 1$ terms, each equal to a , and $a + a + \cdots + a = (1 + 1 + \cdots + 1)a$ by M5.

When to use it

Whenever a sum contains a constant part, for example after splitting $\sum(4j + 3)$ into $4 \sum j + \sum 3$.

Step-by-step example

$\sum_{j=-5}^9 4$: the count is $9 - (-5) + 1 = 15$, so the sum is $15 \times 4 = 60$.

Common mistakes

Using $n - k$ instead of $n - k + 1$ (which would give 56 above); with a negative lower limit, computing $9 - 5$ instead of $9 - (-5)$.

Effect of changing the inputs

Raising the upper limit by one adds one more copy of a ; for $n = k - 1$ the formula gives $0 \cdot a = 0$, in agreement with the empty-sum convention.

How it appears in questions

As a short computation ("evaluate $\sum_{j=-4}^{11} 7$ "), or as a step in finding a closed form.

6.3 Properties of the summation operator

Proposition 6.1 (Properties P1–P3)

For numbers a_j, b_j and a constant c :

$$(P1) \quad \sum_{j=k}^n (a_j + b_j) = \sum_{j=k}^n a_j + \sum_{j=k}^n b_j,$$

$$(P2) \quad \sum_{j=k}^n c \cdot a_j = c \cdot \sum_{j=k}^n a_j,$$

$$(P3) \quad \sum_{j=k}^n a_j + \sum_{j=n+1}^m a_j = \sum_{j=k}^m a_j \quad \text{for } k \leq n \leq m.$$

P1 and P2 follow from the properties of the real numbers; P3 comes from the definition of the operator.

[L1, slide 30]

Proof. (P1) The left side is $(a_k + b_k) + (a_{k+1} + b_{k+1}) + \cdots + (a_n + b_n)$. By commutativity and associativity of addition (A1, A2), the terms may be reordered and regrouped as $(a_k + \cdots + a_n) + (b_k + \cdots + b_n)$.

(P2) $ca_k + ca_{k+1} + \cdots + ca_n = c(a_k + a_{k+1} + \cdots + a_n)$ by the distributive law (M5, applied repeatedly).

(P3) The first sum adds the terms with indices $k, \dots, n$, the second those with indices $n+1, \dots, m$; together they add each term with index from k to m exactly once. (When $n = m$ the second sum is empty and equals 0.) $\square$

The difference between the two kinds of justification is worth noticing. P1 and P2 are about the numbers being added, and they hold because of how addition and multiplication behave. P3 is about the notation: splitting the range of the index and re-joining it gives back the same list of terms.

Proposition 6.2 (The empty sum must be zero)

If P3 is to hold also when $n = k - 1$, then $\sum_{j=k}^{k-1} a_j$ must equal 0.

Proof. Put $n = k - 1$ in P3 with any $m \geq k$: $\sum_{j=k}^{k-1} a_j + \sum_{j=k}^m a_j = \sum_{j=k}^m a_j$. Adding $-\sum_{j=k}^m a_j$ to both sides (A5, A4) gives $\sum_{j=k}^{k-1} a_j = 0$. $\square$

What the properties do not say

There is no rule for products or powers inside a sum:

$$\sum_j a_j b_j \neq \left(\sum_j a_j \right) \left(\sum_j b_j \right), \quad \sum_j a_j^2 \neq \left(\sum_j a_j \right)^2 \quad \text{in general.}$$

For example, $\sum_{j=1}^2 j^2 = 1 + 4 = 5$, while $\left(\sum_{j=1}^2 j \right)^2 = 3^2 = 9$. Also, P2 allows only a factor c that does not depend on the index to be taken outside.

Shifting the index

Replacing j by $i + 1$ throughout shifts the limits down by one: $\sum_{j=k}^n a_j = \sum_{i=k-1}^{n-1} a_{i+1}$. Both sides list the same terms. For example $\sum_{j=1}^n j = \sum_{i=0}^{n-1} (i + 1)$. Index shifts are needed to combine sums with different limits.

6.4 Computing sums with the properties

Worked Example 6.2 – A sum with a negative lower limit

Evaluate $\sum_{j=-1}^4 (j^2 - j)$, showing the individual terms.

[SG, mock Q4(a)]

Solution. For $j = -1, 0, 1, 2, 3, 4$ the terms $j^2 - j$ are $(1+1)+(0-0)+(1-1)+(4-2)+(9-3)+(16-4) = 2 + 0 + 0 + 2 + 6 + 12 = 22$.

Worked Example 6.3 – Two methods for one sum

Compute $\sum_{j=2}^6 (2j - 1)$ directly and by the properties.

Solution. Directly: $3 + 5 + 7 + 9 + 11 = 35$. By the properties, with $\sum_{j=2}^6 j = 2 + 3 + 4 + 5 + 6 = 20$ and $6 - 2 + 1 = 5$ terms:

$$\sum_{j=2}^6 (2j - 1) \stackrel{P1}{=} \sum_{j=2}^6 2j + \sum_{j=2}^6 (-1) \stackrel{P2}{=} 2 \sum_{j=2}^6 j + 5 \cdot (-1) = 40 - 5 = 35.$$

Worked Example 6.4 – A closed form

Given that $\sum_{j=1}^n j = \frac{n(n+1)}{2}$ (proved in Chapter 7), find a closed-form expression for $\sum_{j=1}^n (4j+3)$, naming each property used.

[SG, mock Q4(b)]

Solution.

$$\sum_{j=1}^n (4j+3) \stackrel{P1}{=} \sum_{j=1}^n 4j + \sum_{j=1}^n 3 \stackrel{P2}{=} 4 \sum_{j=1}^n j + \sum_{j=1}^n 3 = 4 \cdot \frac{n(n+1)}{2} + (n-1+1) \cdot 3 = 2n^2 + 2n + 3n = 2n^2 + 5n,$$

using the constant-sum rule (Key Formula 6.1) for $\sum 3$. Check with $n = 2$: $7 + 11 = 18$ and $2 \cdot 4 + 5 \cdot 2 = 18$.

Worked Example 6.5 – Further evaluations

Evaluate (a) $\sum_{j=-3}^2 j^3$; (b) $\sum_{j=-4}^{11} 7$; (c) $\sum_{j=7}^6 aj$.

Solution. (a) $(-27) + (-8) + (-1) + 0 + 1 + 8 = -27$. (The terms for $j = \pm 1$ and $j = \pm 2$ cancel in pairs.) (b) $11 - (-4) + 1 = 16$ terms, so $16 \times 7 = 112$. (c) The upper limit is one below the lower limit, so this is an empty sum, equal to 0 by convention.

Recalling $\sum_{j=1}^n j$ by pairing

Write the sum forwards and backwards and add: each of the n columns of $1 + 2 + \cdots + n$ and $n + (n-1) + \cdots + 1$ adds to $n+1$, so twice the sum is $n(n+1)$. This is a way to *recall* the formula; in the course it is *proved* by induction (Worked Example 7.1).

6.5 Sums in economics

Worked Example 6.6 – Total and average revenue

A shop's revenue in month t is R_t for $t = 1, \dots, 12$. Write total annual revenue and average monthly revenue in summation notation. If $R_t = 1000 + 50t$, compute both.

Solution. Total revenue is $\sum_{t=1}^{12} R_t$ and the average is $\frac{1}{12} \sum_{t=1}^{12} R_t$. With $R_t = 1000 + 50t$,

$$\sum_{t=1}^{12} (1000 + 50t) = 12 \cdot 1000 + 50 \sum_{t=1}^{12} t = 12000 + 50 \cdot \frac{12 \cdot 13}{2} = 12000 + 3900 = 15900,$$

and the average is $15900/12 = 1325$.

Worked Example 6.7 – Discounting a stream of payments

An asset pays £100 at the end of each of the next three years, and the interest rate is $r = 5\%$. Its present value is $\sum_{t=1}^3 \frac{100}{(1.05)^t}$. Evaluate it to the nearest penny.

Solution. $\frac{100}{1.05} + \frac{100}{1.05^2} + \frac{100}{1.05^3} = 95.238 + 90.703 + 86.384 = 272.32$ (to the nearest penny). A closed form for sums of this geometric type is proved by induction in Chapter 7.

6.6 Chapter review

Summary

$\sum_{j=k}^n a_j$ denotes $a_k + \dots + a_n$; the index is a dummy variable and the sum has $n - k + 1$ terms. A sum with $n = k$ has one term; a sum with $n = k - 1$ has none and is defined to be 0, the only value compatible with P3. A sum of a constant is the number of terms times the constant. P1 and P2 — sums split over addition and constants come outside — follow from the properties of the real numbers; P3 — adjacent ranges can be joined — follows from the definition. Products and squares cannot be taken through a sum.

Key definitions

- $\sum_{j=k}^n a_j = a_k + a_{k+1} + \dots + a_n$; index j , limits k and n .
- $\sum_{j=k}^k a_j = a_k$; $\sum_{j=k}^{k-1} a_j = 0$ (empty sum, by definition).

Key equations

$$\sum_{j=k}^n a = (n - k + 1)a, \quad \sum (a_j + b_j) = \sum a_j + \sum b_j, \quad \sum c a_j = c \sum a_j,$$
$$\sum_{j=k}^n a_j + \sum_{j=n+1}^m a_j = \sum_{j=k}^m a_j \quad (k \leq n \leq m).$$

Connections

Sums are the natural setting for induction (Chapter 7): a statement about $\sum_{j=1}^n$ for every n is proved by peeling off the last term with P3. The binomial formula (Chapter 8) and polynomials (Chapter 9) are written with $\sum$. Infinite sums, the subject of Lecture 3, are limits (Chapter 10) of finite sums.

Common mistakes

- Counting $n - k$ terms instead of $n - k + 1$.
- Starting at $j = 0$ or $j = 1$ when the lower limit is negative, and losing terms.
- Taking a factor depending on j outside the sum.
- Writing $\sum a_j b_j = \sum a_j \sum b_j$ or $\sum a_j^2 = (\sum a_j)^2$.

Exam checklist

- ☐ Expand and evaluate a sum with any integer limits.
- ☐ State the one-term and empty-sum conventions and justify the empty-sum value.
- ☐ State P1, P2 and P3 and say which come from the properties of $\mathbb{R}$ and which from the definition.
- ☐ Use the properties and the constant-sum rule to find closed forms.

6.7 Practice questions

Question 6.1 [basic]

Evaluate $\sum_{j=1}^5 (3j - 2)$, $\sum_{j=0}^4 2^j$ and $\sum_{j=-2}^2 (j^2 + 1)$.

Question 6.2 [basic]

Write in summation notation: (a) $1 + 4 + 9 + 16 + 25 + 36$; (b) $2 + 4 + 6 + \cdots + 100$; (c) $1 - 1 + 1 - 1 + 1 - 1$.

Question 6.3 [conceptual]

How many terms does $\sum_{j=-3}^{10} a_j$ have? Explain why the formula for the number of terms has a “+1”.

Question 6.4 [mathematical]

Using $\sum_{j=1}^n j = \frac{n(n+1)}{2}$ and the properties of $\sum$, find closed forms for $\sum_{j=1}^n (6j - 1)$ and $\sum_{j=11}^{20} j$.

Question 6.5 [mathematical]

Show that $\sum_{j=1}^n (a_{j+1} - a_j) = a_{n+1} - a_1$ for any numbers $a_1, \dots, a_{n+1}$. Use this to evaluate $\sum_{j=1}^n \left(\frac{1}{j} - \frac{1}{j+1} \right)$.

Question 6.6 [application]

A firm's cost of producing the j -th unit is $c_j = 20 + 0.5j$ (so the first unit costs 20.5). Write the total cost of producing n units as a sum and find a closed form. How much does it cost to produce 40 units?

Question 6.7 [multi-step]

Show by a counterexample that $\sum_{j=1}^n a_j b_j = \left(\sum_{j=1}^n a_j \right) \left(\sum_{j=1}^n b_j \right)$ is false in general, and find a condition on n under which it is always true.

Question 6.8 [exam-style]

(a) State the value of $\sum_{j=7}^6 a_j$ and explain why this value is adopted. (b) Suppose $a_j = c$ for every j . Explain why $\sum_{j=k}^n a_j = (n - k + 1)c$ and evaluate $\sum_{j=-5}^9 4$. (c) Find a closed form for $\sum_{j=1}^n (4j + 3)$, naming each property of the summation operator used.

6.8 Solutions to practice questions

Solution to Question 6.1

$1 + 4 + 7 + 10 + 13 = 35$; $1 + 2 + 4 + 8 + 16 = 31$; $5 + 2 + 1 + 2 + 5 = 15$.

Solution to Question 6.2

(a) $\sum_{j=1}^6 j^2$. (b) $\sum_{j=1}^{50} 2j$. (c) $\sum_{j=1}^6 (-1)^{j+1}$ (or $\sum_{j=0}^5 (-1)^j$).

Solution to Question 6.3

$10 - (-3) + 1 = 14$ terms. The indices run $-3, -2, \dots, 10$. The difference $n - k = 13$ counts the steps between the first and last index; the number of indices is one more than the number of steps, because both endpoints are included.

Solution to Question 6.4

$\sum_{j=1}^n (6j - 1) = 6 \cdot \frac{n(n+1)}{2} - n = 3n^2 + 3n - n = 3n^2 + 2n$. By P3, $\sum_{j=11}^{20} j = \sum_{j=1}^{20} j - \sum_{j=1}^{10} j = 210 - 55 = 155$.

Solution to Question 6.5

Write out the sum: $(a_2 - a_1) + (a_3 - a_2) + \dots + (a_{n+1} - a_n)$. Each a_j with $2 \leq j \leq n$ appears once with $+$ and once with $-$, so everything cancels except $a_{n+1} - a_1$. (Formally: by P1 and P2 the sum is $\sum_{j=1}^n a_{j+1} - \sum_{j=1}^n a_j = \sum_{i=2}^{n+1} a_i - \sum_{j=1}^n a_j = a_{n+1} - a_1$, using an index shift and P3.) With $a_j = -\frac{1}{j}$: $\frac{1}{j} - \frac{1}{j+1} = a_{j+1} - a_j$, so the sum is $a_{n+1} - a_1 = 1 - \frac{1}{n+1} = \frac{n}{n+1}$.

Solution to Question 6.6

$C(n) = \sum_{j=1}^n (20 + 0.5j) = 20n + 0.5 \cdot \frac{n(n+1)}{2} = 20n + \frac{n(n+1)}{4}$. $C(40) = 800 + \frac{40 \cdot 41}{4} = 800 + 410 = 1210$.

Solution to Question 6.7

$n = 2$, $a_1 = a_2 = b_1 = b_2 = 1$: the left side is 2, the right side is $2 \cdot 2 = 4$. When $n = 1$ both sides equal $a_1 b_1$, so the identity holds for every choice of numbers exactly when $n = 1$ (for $n \geq 2$ the example above, padded with zeros, is a counterexample).

Solution to Question 6.8

(a) 0. The upper limit is one less than the lower limit, so the sum has no terms, and a sum of zero terms is defined to be 0. The choice keeps P3 valid in the boundary case (Proposition 6.2), and zero is the number whose addition changes nothing. (b) The sum is $c + c + \cdots + c$ with one term for each index $k, k + 1, \dots, n$, i.e. $n - k + 1$ terms, so it equals $(n - k + 1)c$. Here $9 - (-5) + 1 = 15$ terms, so the sum is 60. (c) See Worked Example 6.4: $2n^2 + 5n$.

Mathematical Induction

Many statements in mathematics and economics are claims about every natural number at once: a formula for the sum of the first n numbers, a bound on a sequence that holds in every period, a property of a polynomial of every degree. Checking such a claim for a few values of n proves nothing about the infinitely many values that remain. Mathematical induction is the technique that proves all of them together, by proving just two things.

Before you start: prerequisites

Propositions and implication (Chapter 1); the natural numbers and the successor property (Section 3.1); summation notation and property P3 (Chapter 6).

7.1 The principle of induction

Induction applies when there is a sequence of similar propositions $P_1, P_2, \dots, P_n, \dots$ that depend on a parameter $n \in \mathbb{N}$.

[L1, slide 32; L2, slide 4]

Principle 7.1 (Mathematical induction)

Instead of proving each of the propositions $P_1, P_2, P_3, \dots$ separately, we prove only two propositions:

- the **induction base**: the first proposition, P_1 ;
- the **induction step**: the proposition that for any index n , $P_n \Rightarrow P_{n+1}$.

When both the induction base and the induction step are proven, all the propositions $P_1, P_2, P_3, \dots$ are proven. When proving the induction step $P_n \Rightarrow P_{n+1}$, the statement that P_n is true is called the **induction hypothesis**.

[L2, slide 4]

Dominoes

Imagine an infinite row of dominoes, one for each natural number. The base knocks over the first domino. The step guarantees that whenever a domino falls, it knocks over the next one. Then every domino falls:

$$P_1 \Rightarrow P_2 \Rightarrow P_3 \Rightarrow P_4 \Rightarrow P_5 \Rightarrow \dots$$

Both parts are essential. A perfect chain of dominoes achieves nothing if nobody pushes the first; and pushing the first achieves nothing if there is a gap somewhere in the row.

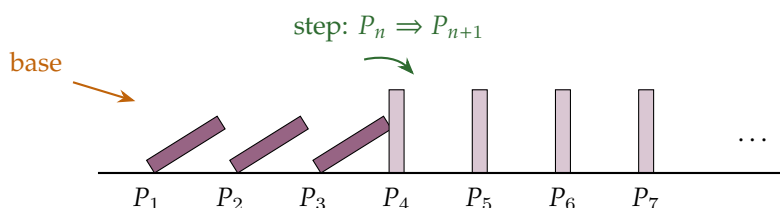


Figure 7.1: Induction as a row of dominoes. Proving the base P_1 knocks over the first; proving the step shows that each falling domino knocks over its successor. Together they guarantee that every P_n is true.

Why induction is valid

Suppose the base and the step have been proved, but some P_n is false. Among the natural numbers n for which P_n is false there is a smallest one, say m . It is not 1, because P_1 is true; so $m - 1 \in \mathbb{N}$, and P_{m-1} is true because m was the smallest failure. But the step gives $P_{m-1} \Rightarrow P_m$, so P_m is true, a contradiction. (The argument uses the fact that every non-empty set of natural numbers has a smallest element, which is equivalent to the principle of induction itself. In this module induction is taken as a basic property of $\mathbb{N}$, resting on the successor property.)

7.2 The first example

Worked Example 7.1 – Sum of the first n natural numbers

Prove that for every $n \in \mathbb{N}$, $\sum_{j=1}^n j = \frac{n(n+1)}{2}$.

[L1, slides 33–34; L2, slides 5–6]

Solution. For $n = 1, 2, 3, \dots$ let P_n be the proposition

$$P_n : \quad \sum_{j=1}^n j = \frac{n(n+1)}{2}.$$

Put differently, P_n states that the sum of all natural numbers from 1 to n is equal to $\frac{n(n+1)}{2}$.

Induction base. The first proposition, P_1 , is $\sum_{j=1}^1 j = \frac{1 \cdot (1+1)}{2}$. The left-hand side is 1 and the right-hand side is 1, so P_1 is true.

Induction step. We must show that for any natural n , $P_n \Rightarrow P_{n+1}$. Here P_{n+1} is the proposition

$$\sum_{j=1}^{n+1} j = \frac{(n+1)((n+1)+1)}{2}.$$

Assume P_n (the induction hypothesis). Then

$$\begin{aligned}\sum_{j=1}^{n+1} j &= \sum_{j=1}^n j + (n+1) = \frac{n(n+1)}{2} + (n+1) \\ &= \left(\frac{n}{2} + 1\right)(n+1) = \frac{(n+2)(n+1)}{2} = \frac{(n+1)((n+1)+1)}{2},\end{aligned}$$

which is P_{n+1} . Note that we used P_n , the induction hypothesis, in the second equality.

Conclusion. By the principle of induction, P_n is true for every $n \in \mathbb{N}$.

The first equality in the step, $\sum_{j=1}^{n+1} j = \sum_{j=1}^n j + (n+1)$, is property P3 with $m = n+1$: we *peel off the last term*. This is the standard first move in any induction proof about a sum, because it exposes the expression $\sum_{j=1}^n j$ about which the hypothesis tells us something.

Key Formula 7.1 – Sum of the first n natural numbers

$$\sum_{j=1}^n j = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

Symbols

$n \in \mathbb{N}$ is the number of terms; j is the summation index.

What it means

The total of the first n natural numbers. Equivalently, their average is $\frac{n+1}{2}$, the average of the first and last terms.

Where it comes from

Proved by induction in Worked Example 7.1; recalled quickly by pairing the first and last terms (Section 6.4).

When to use it

To evaluate sums of linear expressions $\sum (aj + b)$ together with P1, P2 and the constant-sum rule; in economics, for totals of quantities growing by a fixed amount each period.

Step-by-step example

$\sum_{j=1}^{100} j = \frac{100 \cdot 101}{2} = 5050$. And $\sum_{j=1}^{12} (1000 + 50j) = 12000 + 50 \cdot 78 = 15900$.

Common mistakes

Applying the formula when the sum does not start at 1; use P3: $\sum_{j=k}^n j = \frac{n(n+1)}{2} - \frac{(k-1)k}{2}$.

Writing $\frac{n(n-1)}{2}$, which is $\sum_{j=1}^{n-1} j$.

Rearrangements

Doubling: $2 \sum_{j=1}^n j = n^2 + n$. Shifting: $\sum_{j=0}^{n-1} j = \frac{(n-1)n}{2}$.

Effect of changing the inputs

Adding one more term adds $n+1$; the sum grows roughly like $\frac{n^2}{2}$, i.e. quadratically in n .

How it appears in questions

As the standard first example of induction, and as a given fact (“you may take as given that...”) in computations with $\sum$.

7.3 The structure of an induction proof

CORE CONCEPT | Template for a proof by induction

1. **State** P_n . Write down precisely the proposition P_n to be proved for each n .
2. **Base**. Verify P_1 , evaluating both sides separately.
3. **Hypothesis**. Fix an arbitrary $n \in \mathbb{N}$ and assume P_n .
4. **Target**. Write out P_{n+1} , so you know what you are aiming for.
5. **Step**. Starting from one side of P_{n+1} , use the hypothesis (and say where) to reach the other side.
6. **Conclusion**. "By the principle of induction, P_n holds for every $n \in \mathbb{N}$."

The hypothesis must be used

If your argument for the step never uses P_n , then either the statement could have been proved directly (so induction was unnecessary), or, far more often, there is an error. The hypothesis is the only extra information the step is entitled to use.

The base is not a formality

The step alone proves nothing. Consider the false statement P_n : " $n = n + 1$ ". Its step is perfectly valid: if $n = n + 1$, then adding 1 gives $n + 1 = n + 2$, which is P_{n+1} . But P_1 (" $1 = 2$ ") is false, so no domino ever falls.

7.4 An equivalent form of the step

The slides note that the step can equally be written with the indices shifted down by one.

Worked Example 7.2 – The step written as $P_{n-1} \Rightarrow P_n$

Equivalently, the induction step is the proposition that for any natural $n \geq 2$, $P_{n-1} \Rightarrow P_n$. In this formulation the induction hypothesis is P_{n-1} . Carry out the step for Worked Example 7.1 in this form.

[L1, slide 35; L2, slide 7]

Solution. The hypothesis P_{n-1} is $\sum_{j=1}^{n-1} j = \frac{(n-1)((n-1)+1)}{2} = \frac{(n-1)n}{2}$. Then

$$\sum_{j=1}^n j = \sum_{j=1}^{n-1} j + n = \frac{(n-1)n}{2} + n = \left(\frac{n-1}{2} + 1\right)n = \frac{(n+1)n}{2},$$

so P_n holds. Since the argument works for all $n \geq 2$, $P_{n-1} \Rightarrow P_n$ for all $n \geq 2$, proving the step.

Both forms say the same thing: every proposition in the chain implies the next one. Use whichever makes the algebra tidier.

7.5 More examples

Worked Example 7.3 – Sum of the first n odd numbers

Prove that $\sum_{j=1}^n (2j-1) = n^2$ for every $n \in \mathbb{N}$.

Solution. Let P_n be $\sum_{j=1}^n (2j-1) = n^2$. *Base:* for $n = 1$ the left side is $2 \cdot 1 - 1 = 1$ and the right side is $1^2 = 1$. *Step:* assume P_n . Then

$$\sum_{j=1}^{n+1} (2j-1) = \sum_{j=1}^n (2j-1) + (2(n+1)-1) = n^2 + 2n + 1 = (n+1)^2,$$

using P_n in the second equality. This is P_{n+1} , so by induction P_n holds for all $n \in \mathbb{N}$.

Worked Example 7.4 – Sum of squares

Prove that $\sum_{j=1}^n j^2 = \frac{n(n+1)(2n+1)}{6}$ for every $n \in \mathbb{N}$.

[SG, mock Q5(a)]

Solution. Let P_n be the displayed identity. *Base:* $n = 1$ gives 1 on the left and $\frac{1 \cdot 2 \cdot 3}{6} = 1$ on the right. *Hypothesis:* assume P_n for some $n \in \mathbb{N}$. *Target:* $\sum_{j=1}^{n+1} j^2 = \frac{(n+1)(n+2)(2n+3)}{6}$. *Step:*

$$\begin{aligned} \sum_{j=1}^{n+1} j^2 &= \sum_{j=1}^n j^2 + (n+1)^2 = \frac{n(n+1)(2n+1)}{6} + (n+1)^2 \quad (\text{hypothesis used here}) \\ &= \frac{(n+1)[n(2n+1) + 6(n+1)]}{6} = \frac{(n+1)(2n^2 + 7n + 6)}{6} \\ &= \frac{(n+1)(n+2)(2n+3)}{6}, \end{aligned}$$

since $2n^2 + 7n + 6 = (n+2)(2n+3)$. This is P_{n+1} , because $(n+1)+1 = n+2$ and $2(n+1)+1 = 2n+3$. By induction the identity holds for all $n \in \mathbb{N}$.

Worked Example 7.5 – A geometric sum

Let $r \neq 1$ be a real number. Prove that for every $n \in \mathbb{N}$,

$$\sum_{t=0}^{n-1} r^t = 1 + r + r^2 + \cdots + r^{n-1} = \frac{1-r^n}{1-r}.$$

Solution. *Base:* for $n = 1$ the left side is $r^0 = 1$ and the right side is $\frac{1-r}{1-r} = 1$. *Step:* assume the formula for n . Then

$$\sum_{t=0}^n r^t = \frac{1-r^n}{1-r} + r^n = \frac{1-r^n + r^n - r^{n+1}}{1-r} = \frac{1-r^{n+1}}{1-r},$$

which is the formula for $n+1$. By induction it holds for every $n \in \mathbb{N}$.

Why the geometric sum matters in economics

Present values of constant payment streams, annuity formulas and the Keynesian multiplier are all geometric sums. For example, an asset paying £1 at the end of each of n years, at interest rate $i > 0$, has present value $\sum_{t=1}^n (1+i)^{-t} = \frac{1-(1+i)^{-n}}{i}$, obtained from Worked Example 7.5 with $r = \frac{1}{1+i}$.

7.5.1 Inequalities by induction, and other starting points

Induction proves inequalities as well as identities, and the chain of dominoes need not start at 1: to prove P_n for all $n \geq n_0$, prove P_{n_0} and $P_n \Rightarrow P_{n+1}$ for all $n \geq n_0$.

Worked Example 7.6 – Bernoulli's inequality

Let $x \geq -1$. Prove that $(1+x)^n \geq 1+nx$ for every $n \in \mathbb{N}$. Deduce that $2^n \geq n+1$.

Solution. *Base:* $(1+x)^1 = 1+x$, so equality holds. *Step:* assume $(1+x)^n \geq 1+nx$. Since $x \geq -1$, the number $1+x$ is non-negative, so multiplying the hypothesis by it keeps the direction (M6, or trivially if $1+x=0$):

$$(1+x)^{n+1} = (1+x)^n(1+x) \geq (1+nx)(1+x) = 1 + (n+1)x + nx^2 \geq 1 + (n+1)x,$$

since $nx^2 \geq 0$. By induction the inequality holds for all $n \in \mathbb{N}$. With $x=1$: $2^n \geq n+1$.

Worked Example 7.7 – Starting from a later value

Prove that $2^n > n^2$ for every integer $n \geq 5$.

Solution. *Base* ($n=5$): $32 > 25$. *Step:* assume $2^n > n^2$ for some $n \geq 5$. Then $2^{n+1} = 2 \cdot 2^n > 2n^2$. It remains to show $2n^2 \geq (n+1)^2$, i.e. $n^2 - 2n - 1 \geq 0$, i.e. $(n-1)^2 \geq 2$, which holds for $n \geq 3$. So $2^{n+1} > (n+1)^2$, and by induction the inequality holds for all $n \geq 5$. (It fails for $n=2, 3, 4$: $4=4, 8<9, 16=16$.)

7.6 The alternative (strong) formulation

Principle 7.2 (Induction, alternative formulation)

To prove the propositions $P_1, P_2, \dots, P_n, \dots$:

- the induction base is the first proposition P_1 ;
- the induction step is the proposition that for any index n , $(P_1 \text{ and } \dots \text{ and } P_n) \Rightarrow P_{n+1}$.

When proving this step, the statement that the propositions $P_1, \dots, P_n$ are true is the induction hypothesis. The two versions of the principle are, in fact, equivalent. [L1, slide 36; L2,

slide 8]

The hypothesis is now stronger — all the earlier propositions, not only the last one — so the step is easier to prove. This is useful when P_{n+1} depends on a case other than P_n , for example on

two earlier cases, or on a case with roughly half the size. The two formulations are equivalent: any proof using only P_n is also a proof using all of $P_1, \dots, P_n$; and conversely, applying ordinary induction to the statement Q_n : “ P_1 and $\dots$ and P_n ” recovers the strong form. The proof of the factorisation theorem for polynomials (Theorem 9.1) uses this formulation.

Worked Example 7.8 – A recursively defined sequence

A sequence is defined by $x_1 = 1$, $x_2 = 3$ and $x_{n+1} = x_n + 2x_{n-1}$ for $n \geq 2$. Prove that $x_n = \frac{2^{n+1} + (-1)^n}{3}$ for every $n \in \mathbb{N}$.

Solution. Let P_n be the formula for x_n . Since x_{n+1} depends on *two* earlier terms, we use the strong form and need two base cases. *Base:* $n = 1$: $\frac{4-1}{3} = 1 = x_1$. $n = 2$: $\frac{8+1}{3} = 3 = x_2$. (The case $n = 2$ must be checked separately because the recursion only starts at $n \geq 2$ and would require x_0 , which is not defined.) *Step:* let $n \geq 2$ and assume $P_1, \dots, P_n$; in particular P_n and P_{n-1} . Then

$$\begin{aligned} x_{n+1} = x_n + 2x_{n-1} &= \frac{2^{n+1} + (-1)^n}{3} + \frac{2(2^n + (-1)^{n-1})}{3} \\ &= \frac{2 \cdot 2^{n+1} + (-1)^n - 2(-1)^n}{3} = \frac{2^{n+2} + (-1)^{n+1}}{3}, \end{aligned}$$

using $(-1)^{n-1} = -(-1)^n$. This is P_{n+1} . By (strong) induction, P_n holds for all $n \in \mathbb{N}$.

7.7 What induction is not

Checking cases is not a proof

A student verifies $\sum_{j=1}^n j^2 = \frac{n(n+1)(2n+1)}{6}$ for $n = 1, \dots, 100$ and concludes that it holds for all n . This establishes one hundred propositions and says nothing about P_{101} ; by the successor property there is always a further case. Only the general step, proved for an arbitrary n , covers them all.

A famous faulty induction

“All horses have the same colour.” Let P_n be “in any group of n horses, all have the same colour”. P_1 is true. For the step, take $n + 1$ horses; remove one, and by the hypothesis the remaining n have the same colour; remove a different one, and again the remaining n share a colour; the two groups overlap, so all $n + 1$ share a colour. The flaw is in the step from $n = 1$ to $n = 2$: the two groups of size 1 do not overlap. The step must work for *every* n , including the smallest.

7.8 Chapter review

Summary

Induction proves an infinite sequence of propositions $P_1, P_2, \dots$ by proving the base P_1 and the step $P_n \Rightarrow P_{n+1}$ for arbitrary n ; in the step, P_n is the induction hypothesis and must be used. For sums, the step starts by peeling off the last term (P3). The step may equally be written $P_{n-1} \Rightarrow P_n$ for $n \geq 2$; the chain may start at any n_0 ; and in the equivalent strong formulation the hypothesis is P_1 and $\dots$ and P_n , with as many base cases as the argument needs. Checking finitely many cases is never a proof.

Key definitions

- **Induction base:** the proposition P_1 .
- **Induction step:** for every n , $P_n \Rightarrow P_{n+1}$ (equivalently, $P_{n-1} \Rightarrow P_n$ for $n \geq 2$).
- **Induction hypothesis:** the assumption that P_n (or $P_1, \dots, P_n$) is true, made while proving the step.

Key equations

$$\sum_{j=1}^n j = \frac{n(n+1)}{2}, \quad \sum_{j=1}^n (2j-1) = n^2, \quad \sum_{j=1}^n j^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{t=0}^{n-1} r^t = \frac{1-r^n}{1-r} \quad (r \neq 1).$$

Connections

Induction rests on the successor property of $\mathbb{N}$ (Section 3.1) and proves $\forall n \in \mathbb{N}$ statements (Section 1.6). It is used to prove the binomial formula (Chapter 8) and the factorisation theorem for polynomials (Chapter 9), and Bernoulli's inequality gives $2^n \geq n+1$, which is useful for limits (Chapter 10).

Common mistakes

- Omitting or not actually checking the base.
- Not using the hypothesis, or using P_{n+1} (what is to be proved) as if it were known.
- Not stating P_n clearly, so that the target P_{n+1} is unclear.
- Using too few base cases in the strong form.
- Treating verification of many cases as a proof.

Exam checklist

- ☐ State the principle of induction, naming the base, the step and the hypothesis.
- ☐ Prove summation identities by induction, marking where the hypothesis is used.
- ☐ Write the step in the form $P_{n-1} \Rightarrow P_n$.
- ☐ State the alternative formulation and say when it is useful.
- ☐ Explain why checking finitely many cases is not a proof.

7.9 Practice questions

Question 7.1 [basic]

For the claim $\sum_{j=1}^n a_j = b_n$ for all $n \in \mathbb{N}$, write down (a) the base, (b) the induction hypothesis, (c) the statement to be proved in the step.

Question 7.2 [mathematical]

Prove by induction that $\sum_{j=1}^n j^3 = \left(\frac{n(n+1)}{2}\right)^2$ for every $n \in \mathbb{N}$.

Question 7.3 [mathematical]

Prove by induction that $\sum_{j=1}^n \frac{1}{j(j+1)} = \frac{n}{n+1}$ for every $n \in \mathbb{N}$.

Question 7.4 [mathematical]

Prove by induction that $n^3 - n$ is divisible by 3 for every $n \in \mathbb{N}$.

Question 7.5 [conceptual]

A student proves the step $P_n \Rightarrow P_{n+1}$ for the claim $\sum_{j=1}^n (2j-1) = n^2 + 1$ and concludes that the claim is true. Explain the error.

Question 7.6 [application]

A loan of L pounds accrues interest at rate i per year, and nothing is repaid. Let D_n be the debt after n years, so $D_1 = L(1+i)$ and $D_{n+1} = (1+i)D_n$. Prove by induction that $D_n = L(1+i)^n$.

Question 7.7 [multi-step]

A sequence is defined by $a_1 = 5$, $a_2 = 13$ and $a_{n+1} = 5a_n - 6a_{n-1}$ for $n \geq 2$. (a) Explain why ordinary induction with a single base case is not sufficient, and how many base cases are needed. (b) Prove that $a_n = 2^n + 3^n$ for every $n \in \mathbb{N}$.

Question 7.8 [exam-style]

(a) Prove by mathematical induction that $\sum_{j=1}^n j(j+1) = \frac{n(n+1)(n+2)}{3}$ for every $n \in \mathbb{N}$. Identify the base, the hypothesis and the step, and state where the hypothesis is used. (b) Using (a) and Key Formula 7.1, deduce the formula for $\sum_{j=1}^n j^2$.

7.10 Solutions to practice questions

Solution to Question 7.1

(a) $a_1 = b_1$. (b) For some $n \in \mathbb{N}$, $\sum_{j=1}^n a_j = b_n$. (c) $\sum_{j=1}^{n+1} a_j = b_{n+1}$, i.e. (after peeling off the last term) $b_n + a_{n+1} = b_{n+1}$.

Solution to Question 7.2

Base: $1 = \left(\frac{1 \cdot 2}{2}\right)^2 = 1$. Step: assume the formula for n . Then $\sum_{j=1}^{n+1} j^3 = \frac{n^2(n+1)^2}{4} + (n+1)^3 = \frac{(n+1)^2(n^2+4n+4)}{4} = \frac{(n+1)^2(n+2)^2}{4} = \left(\frac{(n+1)(n+2)}{2}\right)^2$, which is the formula for $n+1$.

Solution to Question 7.3

Base: $\frac{1}{1 \cdot 2} = \frac{1}{2} = \frac{1}{1+1}$. Step: assume the formula for n . Then $\sum_{j=1}^{n+1} \frac{1}{j(j+1)} = \frac{n}{n+1} + \frac{1}{(n+1)(n+2)} = \frac{n(n+2)+1}{(n+1)(n+2)} = \frac{(n+1)^2}{(n+1)(n+2)} = \frac{n+1}{n+2}$.

Solution to Question 7.4

Base: $1 - 1 = 0 = 3 \cdot 0$. Step: assume $n^3 - n = 3k$ for an integer k . Then $(n + 1)^3 - (n + 1) = n^3 + 3n^2 + 3n + 1 - n - 1 = (n^3 - n) + 3(n^2 + n) = 3(k + n^2 + n)$, a multiple of 3.

Solution to Question 7.5

The base fails: for $n = 1$ the left side is 1 and the right side is 2. The step is valid ($n^2 + 1 + 2n + 1 = (n + 1)^2 + 1$), but without a true base it establishes nothing — the correct identity is $\sum(2j - 1) = n^2$. Both the base and the step are needed.

Solution to Question 7.6

Base: $D_1 = L(1 + i) = L(1 + i)^1$. Step: if $D_n = L(1 + i)^n$ then $D_{n+1} = (1 + i)D_n = L(1 + i)^{n+1}$. By induction $D_n = L(1 + i)^n$ for all n .

Solution to Question 7.7

(a) a_{n+1} is determined by a_n and a_{n-1} , so the hypothesis P_n alone does not give enough information; the strong form supplies both P_n and P_{n-1} . Two base cases ($n = 1$ and $n = 2$) are needed, because the recursion expresses a_{n+1} only for $n \geq 2$; a_2 cannot be obtained from it (that would need a_0).
(b) Base: $a_1 = 5 = 2 + 3$, $a_2 = 13 = 4 + 9$. Step: for $n \geq 2$, assume the formula for all indices up to n . Then $a_{n+1} = 5(2^n + 3^n) - 6(2^{n-1} + 3^{n-1}) = 2^{n-1}(10 - 6) + 3^{n-1}(15 - 6) = 4 \cdot 2^{n-1} + 9 \cdot 3^{n-1} = 2^{n+1} + 3^{n+1}$.

Solution to Question 7.8

(a) Let P_n be the identity. Base: $1 \cdot 2 = 2 = \frac{1 \cdot 2 \cdot 3}{3}$. Hypothesis: assume P_n . Step: $\sum_{j=1}^{n+1} j(j+1) = \frac{n(n+1)(n+2)}{3} + (n+1)(n+2)$ (hypothesis used here) $= \frac{(n+1)(n+2)(n+3)}{3}$, which is P_{n+1} . (b) $j^2 = j(j+1) - j$, so by P1, P2: $\sum j^2 = \frac{n(n+1)(n+2)}{3} - \frac{n(n+1)}{2} = \frac{n(n+1)(2(n+2)-3)}{6} = \frac{n(n+1)(2n+1)}{6}$.

The Binomial Formula

Expanding a power of a sum, such as $(1 + r)^n$, is one of the most common calculations in economics and finance: it appears whenever interest compounds or a quantity grows at a constant rate. The binomial formula gives the expansion of $(x + y)^n$ for every n in one line. This chapter introduces factorials and binomial coefficients, states the formula, proves it by induction, and uses it to estimate powers without a calculator, as in a question set on the January 2026 paper.

Before you start: prerequisites

Summation notation (Chapter 6); proof by induction (Chapter 7); the distributive law M5 (Chapter 3).

8.1 Powers of a sum

Consider $(x + y)^n$, where $x, y \in \mathbb{R}$ and $n \in \{0\} \cup \mathbb{N}$. Multiplying out repeatedly gives

$$(x + y)^0 = 1$$

$$(x + y)^1 = x + y$$

$$(x + y)^2 = x^2 + 2xy + y^2$$

$$(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$

$$(x + y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$$

$$(x + y)^5 = x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5$$

[L2, slide 13]

Three patterns are visible. In $(x + y)^n$ every term has total degree n : the power of x falls from n to 0 while the power of y rises from 0 to n . There are $n + 1$ terms. And the coefficients are symmetric and follow a pattern in which each is the sum of the two above it (Figure 8.1).

Proposition 8.1 (Properties of binomial coefficients)

For $n \in \{0\} \cup \mathbb{N}$ and $0 \leq j \leq n$:

- (i) $C_n^0 = C_n^n = 1$ and, for $n \geq 1$, $C_n^1 = C_n^{n-1} = n$;
- (ii) symmetry: $C_n^j = C_n^{n-j}$;
- (iii) Pascal's rule: for $0 \leq j \leq n-1$, $C_n^j + C_n^{j+1} = C_{n+1}^{j+1}$.

Proof. (i) $C_n^0 = \frac{n!}{n!0!} = 1$ and $C_n^1 = \frac{n!}{(n-1)!1!} = n$; the others follow from (ii). (ii) Replacing j by $n-j$ swaps the two factorials in the denominator: $C_n^{n-j} = \frac{n!}{j!(n-j)!} = C_n^j$. (iii) Bring the two fractions to the common denominator $(j+1)!(n-j)!$:

$$\begin{aligned} \frac{n!}{(n-j)!j!} + \frac{n!}{(n-j-1)!(j+1)!} &= \frac{n!(j+1) + n!(n-j)}{(n-j)!(j+1)!} = \frac{n!(n+1)}{(n-j)!(j+1)!} \\ &= \frac{(n+1)!}{((n+1)-(j+1))!(j+1)!} = C_{n+1}^{j+1}. \quad \square \end{aligned}$$

What binomial coefficients count

C_n^j is the number of ways of choosing j objects from n distinct objects when order does not matter. For example there are $C_5^2 = 10$ ways to choose 2 of 5 candidates for a committee. This is why the coefficients appear in $(x+y)^n$: expanding the product of n brackets $(x+y)(x+y)\cdots(x+y)$, the term $x^{n-j}y^j$ arises once for each way of choosing the j brackets that contribute a y .

8.3 The binomial formula

Key Formula 8.1 – The binomial formula

$$(x+y)^n = \sum_{j=0}^n C_n^j x^{n-j} y^j, \quad C_n^j = \frac{n!}{(n-j)!j!}$$

Symbols

$x, y \in \mathbb{R}$; $n \in \{0\} \cup \mathbb{N}$ is the power; j is the summation index, the power of y in each term; C_n^j are the binomial coefficients. [L2, slide 13]

What it means

$(x+y)^n$ is a sum of $n+1$ terms, one for each $j = 0, 1, \dots, n$. The j -th term contains y to the power j , x to the power $n-j$, and the coefficient C_n^j .

Where it comes from

From repeated use of the distributive law; formally by induction on n using Pascal's rule (Theorem 8.2).

When to use it

To expand powers of sums exactly, to compute particular coefficients, and to bound or approximate powers such as $(1+r)^n$ by their first few terms.

Step-by-step example

$(2a - 1)^3$: with $x = 2a$, $y = -1$, $n = 3$: $C_3^0(2a)^3 + C_3^1(2a)^2(-1) + C_3^2(2a)(-1)^2 + C_3^3(-1)^3 = 8a^3 - 12a^2 + 6a - 1$.

Common mistakes

Forgetting to raise the coefficient inside x or y to the power (writing $2a^3$ for $(2a)^3$); losing the signs when y is negative ($(-1)^j$ alternates); counting n terms instead of $n + 1$.

Rearrangements

Exchanging the roles of x and y : $(x + y)^n = \sum_{j=0}^n C_n^j x^j y^{n-j}$ (by symmetry of C_n^j). Special cases: $(1 + y)^n = \sum_{j=0}^n C_n^j y^j$; $x = y = 1$ gives $\sum_{j=0}^n C_n^j = 2^n$; $x = 1$, $y = -1$ gives $\sum_{j=0}^n (-1)^j C_n^j = 0$ for $n \geq 1$.

Effect of changing the inputs

Increasing n by one adds a term and updates the coefficients by Pascal's rule. If y is small relative to x , the terms shrink rapidly as j grows (each contains a further factor y/x), which is why the first few terms give a good approximation.

How it appears in questions

"Write the binomial formula for $(x + 1)^n$ and the first four terms explicitly", followed by an estimate such as "show that $1.02^{102} > 6$ " (January 2026, Question 1).

Theorem 8.2 (Binomial formula)

For all $x, y \in \mathbb{R}$ and $n \in \{0\} \cup \mathbb{N}$, $(x + y)^n = \sum_{j=0}^n C_n^j x^{n-j} y^j$.

Proof

The slides state the general formula without proof. The proof by induction below is supplied for completeness.

Proof. For $n = 0$ both sides equal 1 ($C_0^0 = 1$, and $x^0 y^0 = 1$). We prove the formula for $n \in \mathbb{N}$ by induction. *Base* ($n = 1$): $C_1^0 x + C_1^1 y = x + y$. *Step*: assume the formula for n . Then, by M5,

$$(x + y)^{n+1} = (x + y) \sum_{j=0}^n C_n^j x^{n-j} y^j = \sum_{j=0}^n C_n^j x^{n+1-j} y^j + \sum_{j=0}^n C_n^j x^{n-j} y^{j+1}.$$

In the second sum, shift the index ($i = j + 1$) so that the power of y is again the index: $\sum_{i=1}^{n+1} C_n^{i-1} x^{n+1-i} y^i$. Now collect the terms with the same power y^j . The term $j = 0$ appears only in the first sum, with coefficient $C_n^0 = 1 = C_{n+1}^0$; the term $j = n + 1$ appears only in the second, with coefficient $C_n^n = 1 = C_{n+1}^{n+1}$; and for $1 \leq j \leq n$ the coefficient is $C_n^j + C_n^{j-1} = C_{n+1}^j$ by Pascal's rule. Hence $(x + y)^{n+1} = \sum_{j=0}^{n+1} C_{n+1}^j x^{n+1-j} y^j$, which is the formula for $n + 1$. $\square$

8.4 Using the binomial formula

Worked Example 8.2 – The first four terms of $(x + 1)^n$

Write the binomial formula for $(x + 1)^n$ and the first four terms explicitly, computing $C_n^0, C_n^1, C_n^2, C_n^3$. [SG: Jan 2026 Q1(a)]

Solution. With $y = 1$, every power of y equals 1:

$$(x + 1)^n = \sum_{j=0}^n C_n^j x^{n-j}.$$

The first four coefficients are

$$C_n^0 = 1, \quad C_n^1 = n, \quad C_n^2 = \frac{n(n-1)}{2}, \quad C_n^3 = \frac{n(n-1)(n-2)}{6},$$

so (for $n \geq 3$)

$$(x + 1)^n = x^n + n x^{n-1} + \frac{n(n-1)}{2} x^{n-2} + \frac{n(n-1)(n-2)}{6} x^{n-3} + \dots$$

Worked Example 8.3 – Estimating a power without a calculator

Show that $1.02^{102} > 6$. Hint: $1.02 = \frac{2}{100} + 1$; four terms of the expansion suffice. [SG: Jan 2026 Q1(b)]

Solution. Use the binomial formula in the form $(1 + y)^n = \sum_{j=0}^n C_n^j y^j$ with $y = \frac{2}{100} = 0.02$ and $n = 102$. Every term is positive (since $y > 0$), so the sum is greater than the sum of its first four terms:

$$1.02^{102} > C_{102}^0 + C_{102}^1(0.02) + C_{102}^2(0.02)^2 + C_{102}^3(0.02)^3.$$

Compute each term:

$$\begin{aligned} C_{102}^0 &= 1, \\ C_{102}^1(0.02) &= 102 \times 0.02 = 2.04, \\ C_{102}^2(0.02)^2 &= \frac{102 \cdot 101}{2} \times 0.0004 = 5151 \times 0.0004 = 2.0604, \\ C_{102}^3(0.02)^3 &= \frac{102 \cdot 101 \cdot 100}{6} \times 0.000008 = 171700 \times 0.000008 = 1.3736. \end{aligned}$$

The total is $1 + 2.04 + 2.0604 + 1.3736 = 6.474 > 6$. Hence $1.02^{102} > 6.474 > 6$.

Recognising this in an exam

The key observation is that all omitted terms are positive, so dropping them gives a lower bound. State this explicitly: it is the logical step that turns a partial sum into a proof. Keep the arithmetic exact (calculators were not permitted on any of the past papers).

Worked Example 8.4 – A specific coefficient

Find the coefficient of x^3 in $(2x - 3)^5$.

Solution. The general term is $C_5^j (2x)^{5-j} (-3)^j$. The power of x is $5 - j = 3$ when $j = 2$. The coefficient is $C_5^2 \cdot 2^3 \cdot (-3)^2 = 10 \cdot 8 \cdot 9 = 720$.

8.5 The binomial formula in economics and finance

Worked Example 8.5 – Compound interest

A deposit of £1000 earns interest at 3% per year, compounded annually. Use the binomial formula to approximate its value after 10 years, and compare with simple interest.

Solution. The value is $1000(1 + 0.03)^{10}$. By the binomial formula,

$$(1.03)^{10} = 1 + 10(0.03) + 45(0.03)^2 + 120(0.03)^3 + \cdots = 1 + 0.3 + 0.0405 + 0.00324 + \cdots \approx 1.3437,$$

so the deposit is worth about £1343.7 (the exact value is £1343.92). Simple interest would give only $1000(1 + 10 \times 0.03) = £1300$: that is the first two terms of the expansion. The remaining terms are the “interest on interest”.

Bernoulli's inequality again

For $y \geq 0$, dropping all terms of $(1 + y)^n$ after the second gives $(1 + y)^n \geq 1 + ny$, a special case of Bernoulli's inequality (Worked Example 7.6). It says that compound growth is at least as large as simple growth.

8.6 Chapter review

Summary

$k!$ is defined by $0! = 1$ and $k! = (k - 1)! \cdot k$. The binomial coefficient $C_n^j = \frac{n!}{(n-j)!j!}$ (also written $\binom{n}{j}$) is defined for $0 \leq j \leq n$; it is symmetric, equals 1 at the ends, and satisfies Pascal's rule $C_n^j + C_n^{j+1} = C_{n+1}^{j+1}$. The binomial formula $(x + y)^n = \sum_{j=0}^n C_n^j x^{n-j} y^j$ expands any power of a sum into $n + 1$ terms; it is proved by induction using Pascal's rule. When all terms are positive, keeping only the first few gives a lower bound, which is how $1.02^{102} > 6$ is shown.

Key definitions

- $0! = 1, k! = (k - 1)! \cdot k$.
- $C_n^j = \binom{n}{j} = \frac{n!}{(n-j)!j!}$ for $j, n \in \{0\} \cup \mathbb{N}, n \geq j$.

Key equations

$$(x + y)^n = \sum_{j=0}^n C_n^j x^{n-j} y^j, \quad C_n^j = C_n^{n-j}, \quad C_n^j + C_n^{j+1} = C_{n+1}^{j+1}, \quad \sum_{j=0}^n C_n^j = 2^n.$$

Connections

The formula is written with $\sum$ (Chapter 6) and proved by induction (Chapter 7). Expanding $(x - \bar{x})^n$ and similar expressions is needed when working with polynomials (Chapter 9), and binomial expansions are used to estimate terms of sequences such as $(1 + \frac{1}{n})^n$ in Topic 3.

Common mistakes

- $(x + y)^n = x^n + y^n$ (true only for $n = 1$).
- Forgetting the power on a coefficient, e.g. writing $2x^3$ for $(2x)^3$.
- Losing signs in $(x - y)^n$.
- Arithmetic slips in C_n^j : cancel $(n - j)!$ first.
- Dropping terms to get a lower bound without noting that they are positive.

Exam checklist

- ☐ Compute factorials and binomial coefficients exactly.
- ☐ State the binomial formula with the definition of C_n^j .
- ☐ Write the first few terms of $(x + 1)^n$ for general n .
- ☐ Use a partial sum of a binomial expansion to prove a lower bound.
- ☐ Find a specified coefficient in an expansion.

8.7 Practice questions

Question 8.1 [basic]

Compute $6!$, $\frac{10!}{8!}$, C_6^2 , C_7^3 and C_{20}^{18} .

Question 8.2 [basic]

Expand $(1 + 2t)^4$ and $(a - b)^5$.

Question 8.3 [conceptual]

Explain why $(x + y)^n$ has $n + 1$ terms, and why the sum of the exponents of x and y is n in each term.

Question 8.4 [mathematical]

Use the binomial formula to show that $\sum_{j=0}^n C_n^j = 2^n$ and that $\sum_{j=0}^n (-1)^j C_n^j = 0$ for $n \geq 1$.

Question 8.5 [mathematical]

Find the term independent of x in $(x + \frac{2}{x})^6$.

Question 8.6 [application]

An economy's real output grows by 2% per year. Use the first three terms of a binomial expansion to estimate the percentage growth over 5 years, and state whether your estimate is an under- or over-estimate.

Question 8.7 [multi-step]

Show that $1.01^{100} > 2.7$ using the binomial formula. How many terms do you need?

Question 8.8 [exam-style]

(a) Write the binomial formula for $(x + 1)^n$, $n \in \mathbb{N}$, and write the first four terms explicitly, computing $C_n^0, C_n^1, C_n^2, C_n^3$. (b) Use your answer to show that $1.03^{50} > 4$.

8.8 Solutions to practice questions

Solution to Question 8.1

$$6! = 720; \frac{10!}{8!} = 10 \cdot 9 = 90; C_6^2 = \frac{6 \cdot 5}{2} = 15; C_7^3 = \frac{7 \cdot 6 \cdot 5}{6} = 35; C_{20}^{18} = C_{20}^2 = \frac{20 \cdot 19}{2} = 190.$$

Solution to Question 8.2

$$(1 + 2t)^4 = 1 + 4(2t) + 6(2t)^2 + 4(2t)^3 + (2t)^4 = 1 + 8t + 24t^2 + 32t^3 + 16t^4. (a - b)^5 = a^5 - 5a^4b + 10a^3b^2 - 10a^2b^3 + 5ab^4 - b^5.$$

Solution to Question 8.3

The index j (the power of y) runs from 0 to n , giving $n + 1$ values. Each term comes from choosing either x or y from each of the n brackets, so the powers of x and y add to n ; the term with y^j therefore has x^{n-j} .

Solution to Question 8.4

Put $x = y = 1$: $2^n = (1 + 1)^n = \sum_j C_n^j$. Put $x = 1, y = -1$: $0 = 0^n = (1 - 1)^n = \sum_j C_n^j (-1)^j$ for $n \geq 1$.

Solution to Question 8.5

The general term is $C_6^j x^{6-j} (2/x)^j = C_6^j 2^j x^{6-2j}$. It is independent of x when $j = 3$: $C_6^3 \cdot 2^3 = 20 \cdot 8 = 160$.

Solution to Question 8.6

$(1.02)^5 \approx 1 + 5(0.02) + 10(0.02)^2 = 1 + 0.1 + 0.004 = 1.104$, i.e. about 10.4%. The omitted terms are positive, so this is an underestimate (the exact value is 1.10408..., about 10.41%).

Solution to Question 8.7

$(1 + 0.01)^{100} > 1 + 100(0.01) + \frac{100 \cdot 99}{2}(0.0001) + \frac{100 \cdot 99 \cdot 98}{6}(0.000001) = 1 + 1 + 0.495 + 0.1617 = 2.6567$, which is not yet enough. Adding the fifth term $C_{100}^4 (0.01)^4 = 3921225 \times 10^{-8} = 0.0392$ gives 2.6959, still short. The sixth term $C_{100}^5 (0.01)^5 = 75287520 \times 10^{-10} = 0.0075$ gives 2.7034 > 2.7 . So six terms suffice, and since all omitted terms are positive, $1.01^{100} > 2.7$.

Solution to Question 8.8

(a) $(x + 1)^n = \sum_{j=0}^n C_n^j x^{n-j} = x^n + nx^{n-1} + \frac{n(n-1)}{2}x^{n-2} + \frac{n(n-1)(n-2)}{6}x^{n-3} + \dots$ (b) $1.03^{50} = (1 + 0.03)^{50} > 1 + 50(0.03) + \frac{50 \cdot 49}{2}(0.0009) + \frac{50 \cdot 49 \cdot 48}{6}(0.000027) = 1 + 1.5 + 1.1025 + 0.5292 = 4.1317 > 4$, since all omitted terms are positive.

Functions and Polynomials

Economic models are built from relationships: quantity demanded depends on price, cost depends on output, utility depends on consumption. The mathematical object that captures a relationship of this kind is a function. This chapter gives the informal and formal definitions of a function, then studies the simplest important family of numerical functions, the polynomials, and proves that a polynomial can be factorised through each of its roots.

Before you start: prerequisites

Sets and set-builder notation (Chapter 2); summation notation (Chapter 6); the strong form of induction (Section 7.6); factorising quadratics through their roots (Section 4.4).

9.1 The concept of a function

Definition 9.1 (Function (informal))

A **function** $f: D \rightarrow R$ is a rule that associates each element of a set D (the **domain**) with an element of a set R (the **codomain**).

[L2, slide 16]

Notation

In the slides the codomain is called R , an ordinary capital letter for an arbitrary set. Do not confuse it with $\mathbb{R}$, the set of real numbers. When the codomain is the real numbers, we write $f: D \rightarrow \mathbb{R}$.

Two features of the definition matter. *Every* element of the domain is associated with something: a function is defined everywhere on its domain. And each element is associated with exactly *one* element of the codomain: a function never gives two outputs for one input. Several inputs may share an output, and some elements of the codomain may not be outputs at all.

Definition 9.2 (Function (formal))

Given sets D and R , a **function** f is a set of ordered pairs (x, y) , with $x \in D$ and $y \in R$, such that for every $x \in D$ there is a unique $y \in R$ with $(x, y) \in f$. In this case we write $y = f(x)$.

[L2, slide 16]

The formal definition identifies a function with its *graph*: the set of all input–output pairs. It shows that a function is a set (Chapter 2) with a special property, and it removes the vague word “rule”. “For every $x \in D$ there is a unique y ” combines a $\forall$, an $\exists$ and a uniqueness condition (Section 1.6).

Worked Example 9.1 – Two functions

- (a) Let D be a set of physical objects, R a set of masses, and $g(x)$ the mass of object x . For instance, $g(\text{Jupiter}) \approx 1.898 \times 10^{27}$ kg.
 (b) Let $D = R = \mathbb{R}$ and $h(x) = x^2$. For instance $h(10) = 10^2 = 100$ and $h(-1) = (-1)^2 = 1$.

[L2, slide 16] In (b), the inputs 1 and -1 share the output 1, which is allowed; and negative numbers in the codomain are never outputs, which is also allowed.

Definition 9.3 (Equal functions)

Two functions f and g are **equal** if they have the same domain D and the same codomain R , and for any $x \in D$, $f(x) = g(x)$.

[L2, slide 16]

Worked Example 9.2 – Same formula, different functions

- (a) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, and $g: [0, \infty) \rightarrow \mathbb{R}$, $g(x) = x^2$, are *not* equal: their domains differ.
 (b) $p: \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}$, $p(x) = \frac{x^2-1}{x-1}$, and $q: \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}$, $q(x) = x + 1$, are equal: on their common domain $\frac{x^2-1}{x-1} = \frac{(x-1)(x+1)}{x-1} = x + 1$. But p is not equal to $r: \mathbb{R} \rightarrow \mathbb{R}$, $r(x) = x + 1$, because p is not defined at $x = 1$.

Definition 9.4 (Numerical function)

When $D, R \subseteq \mathbb{R}$, we call $f: D \rightarrow R$ a **numerical function**.

[L2, slide 16]

Almost every function in ECN115 is numerical, or (in Topic 6) a function of several real variables.

Functions in economics

- A *demand function* $Q = D(p)$ gives the quantity demanded at each price $p \geq 0$, for instance $D(p) = 100 - 2p$ on the domain $[0, 50]$.
- A *cost function* $C(q)$ gives the total cost of producing $q \geq 0$ units.
- A *production function* $Y = F(K, L)$ gives output from capital K and labour L ; it is a function of two variables.

The domain is part of the model: a negative price or quantity usually has no meaning, so the domain is restricted to non-negative values.

9.2 Polynomials

Definition 9.5 (Polynomial)

Given $n \in \{0\} \cup \mathbb{N}$ and numbers $a_0, a_1, \dots, a_n \in \mathbb{R}$, define a function $f: \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n = \sum_{k=0}^n a_kx^k.$$

- The function f is called a **polynomial**.
- The numbers $a_0, a_1, \dots, a_n$ are called the **coefficients** of f .
- If at least one coefficient is not zero, the largest number k such that $a_k \neq 0$ is called the **degree** of f .
- The functions $a_0, a_1x, a_2x^2, \dots, a_nx^n$ are called the **monomials** (terms) of f .
- If $f(\bar{x}) = 0$, we call $\bar{x}$ a **root** of the polynomial f .

[L2, slide 17]

Worked Example 9.3 – Degrees and roots

- $f(x) = 3x^2 - 5x + 2$ has degree 2 and roots $\frac{2}{3}$ and 1 (Worked Example 4.9).
- $f(x) = 7$ has degree 0 (the only coefficient is $a_0 = 7 \neq 0$) and no roots.
- $f(x) = 0$ (all coefficients zero) has no degree under the definition, and every real number is a root.
- $f(x) = 0x^3 + 2x + 1$ has degree 1, not 3: the degree is the largest power with a *non-zero* coefficient.

Polynomials of degree 1, 2 and 3 are called *linear*, *quadratic* and *cubic*. A non-zero constant polynomial has degree 0 and no root, which matters in the proof below.

9.3 Factorising a polynomial through a root

For a quadratic with roots x_1, x_2 we found $ax^2 + bx + c = a(x - x_1)(x - x_2)$. The next theorem shows that every root of every polynomial produces a linear factor in the same way.

Theorem 9.1 (Polynomial factorisation via roots (Theorem L2.1))

Suppose a polynomial p of degree $n \geq 1$, $p(x) = a_nx^n + \dots + a_1x + a_0$, has a root $\bar{x} \in \mathbb{R}$; that is, $p(\bar{x}) = 0$. Then there is a unique polynomial q of degree $n - 1$ such that

$$p(x) = (x - \bar{x}) \cdot q(x) \quad \text{for all } x \in \mathbb{R}.$$

[L2, slide 18]

CORE CONCEPT | Root $\Leftrightarrow$ linear factor

$\bar{x}$ is a root of p exactly when $x - \bar{x}$ is a factor of p . (If $p(x) = (x - \bar{x})q(x)$ then clearly $p(\bar{x}) = 0$; the theorem gives the converse.) Finding roots and finding linear factors are the same problem.

Proof (by induction on the degree n).

[L2, slides 18–19]

Induction base, $n = 1$. Then $p(x) = a_1x + a_0$ with $a_1 \neq 0$. The polynomial p has the unique root $\bar{x} = -a_0/a_1$, and

$$p(x) = a_1 \left(x + \frac{a_0}{a_1} \right) = (x - \bar{x}) \cdot a_1 = (x - \bar{x}) \cdot q(x),$$

where $q(x) = a_1$ is a polynomial of degree $1 - 1 = 0$. This q is unique, since $(x - \bar{x}) \cdot q(x) = a_1x + a_0$ implies $q(x) = a_1$ (compare the coefficients of x).

Induction step. Consider an arbitrary $n > 1$ and assume that the statement of the theorem is true for all degrees k with $1 \leq k < n$ (this is the alternative formulation of induction, Section 7.6). Let p have degree n and root $\bar{x}$.

Case 1: $p(x) = (x - \bar{x}) \cdot a_n x^{n-1}$. Then $q(x) = a_n x^{n-1}$, a polynomial of degree $n - 1$, does the job.

Case 2: $p(x) \neq a_n x^{n-1}(x - \bar{x})$. Consider the difference

$$s(x) = p(x) - a_n x^{n-1}(x - \bar{x}) = (a_{n-1} + a_n \bar{x})x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_0.$$

The x^n terms cancel, so s is a polynomial whose degree is at most $n - 1$, and s is not the zero polynomial (this is what Case 2 means). Moreover

$$s(\bar{x}) = p(\bar{x}) - a_n \bar{x}^{n-1}(\bar{x} - \bar{x}) = 0 - 0 = 0,$$

so $\bar{x}$ is a root of s . A non-zero polynomial of degree 0 is a non-zero constant and has no root, so the degree k of s satisfies $1 \leq k \leq n - 1$. By the induction hypothesis, $s(x) = (x - \bar{x}) \cdot r(x)$ for a (unique) polynomial r of degree $k - 1 \leq n - 2$. In this case

$$p(x) = s(x) + a_n x^{n-1}(x - \bar{x}) = (x - \bar{x}) \cdot (a_n x^{n-1} + r(x)).$$

The polynomial $q(x) = a_n x^{n-1} + r(x)$ has degree exactly $n - 1$, because $a_n \neq 0$ and r has degree at most $n - 2$. This completes the existence part of the step. $\square$

! Uniqueness in the induction step

The slides label the final q “a unique $q(x)$ of degree $n - 1$ ” but do not show why no other polynomial works. The uniqueness of r from the induction hypothesis does not by itself rule out a different factorisation of p . The missing argument is supplied below; it is short and uses only the existence part.

Proof of uniqueness. Suppose $p(x) = (x - \bar{x})q_1(x) = (x - \bar{x})q_2(x)$ for all x , where q_1, q_2 are polynomials. Let $d = q_1 - q_2$, a polynomial with $(x - \bar{x})d(x) = 0$ for every x . For every $x \neq \bar{x}$ we may divide by $x - \bar{x} \neq 0$, so $d(x) = 0$: every real number other than $\bar{x}$ is a root of d . If d were a non-zero polynomial, of degree m say, then applying the existence part repeatedly would factor out one linear factor for each of $m + 1$ distinct roots, leaving a non-zero polynomial of negative degree, which is impossible; equivalently, a non-zero polynomial of degree m has at most m roots (Proposition 9.2). So d is the zero polynomial and $q_1 = q_2$. $\square$

Proposition 9.2 (At most n roots)

A polynomial of degree $n \geq 0$ has at most n distinct roots.

Proof. By induction on n . A polynomial of degree 0 is a non-zero constant and has no roots. Suppose the claim holds for degree $n - 1$, and let p have degree $n \geq 1$. If p has no root we are done. Otherwise, let $\bar{x}$ be a root; by Theorem 9.1 (existence part), $p(x) = (x - \bar{x})q(x)$ with q of degree $n - 1$. If $y \neq \bar{x}$ is another root, then $0 = p(y) = (y - \bar{x})q(y)$ with $y - \bar{x} \neq 0$, so $q(y) = 0$ (Proposition 3.3). Hence every root of p other than $\bar{x}$ is a root of q , of which there are at most $n - 1$; so p has at most n roots. $\square$

9.3.1 Finding the factor in practice

Worked Example 9.4 – Factorising a cubic

Factorise $p(x) = x^3 - 6x^2 + 11x - 6$ completely.

Solution. Try small integers: $p(1) = 1 - 6 + 11 - 6 = 0$, so 1 is a root and, by Theorem 9.1, $p(x) = (x - 1)q(x)$ with q quadratic. Write $q(x) = ax^2 + bx + c$ and compare coefficients in $(x - 1)(ax^2 + bx + c) = ax^3 + (b - a)x^2 + (c - b)x - c$:

$$a = 1, \quad b - a = -6, \quad c - b = 11, \quad -c = -6,$$

giving $a = 1$, $b = -5$, $c = 6$ (and the third equation, $6 - (-5) = 11$, checks). So $p(x) = (x - 1)(x^2 - 5x + 6) = (x - 1)(x - 2)(x - 3)$.

Where to look for integer roots

If a polynomial has integer coefficients and leading coefficient 1, any integer root must divide the constant term. In Worked Example 9.4 the candidates were $\pm 1, \pm 2, \pm 3, \pm 6$.

Worked Example 9.5 – Using the factorisation to solve an inequality

Solve $x^3 - 6x^2 + 11x - 6 > 0$.

Solution. By Worked Example 9.4 the expression is $(x - 1)(x - 2)(x - 3)$. Critical points 1, 2, 3. For $x > 3$ all three factors are positive; crossing each critical point changes the sign of exactly one factor. So the product is positive on $(3, \infty)$, negative on $(2, 3)$, positive on $(1, 2)$ and negative on $(-\infty, 1)$. Solution: $x \in (1, 2) \cup (3, \infty)$.

Worked Example 9.6 – Break-even points of a cubic profit function

A firm's profit is $\pi(q) = -q^3 + 9q^2 - 23q + 15$ (in thousands of pounds, with q in thousands of units). Find all outputs at which the firm breaks even ($\pi = 0$), and where profit is positive.

Solution. $\pi(1) = -1 + 9 - 23 + 15 = 0$, so $q = 1$ is a root. Dividing as in Worked Example 9.4: $\pi(q) = -(q - 1)(q^2 - 8q + 15) = -(q - 1)(q - 3)(q - 5)$. The firm breaks even at $q = 1, 3, 5$. The product $(q - 1)(q - 3)(q - 5)$ is negative on $(3, 5)$ and on $(-\infty, 1)$, so $\pi(q) > 0$ for $q \in [0, 1) \cup (3, 5)$.

(restricting to $q \geq 0$).

9.4 Chapter review

Summary

A function $f: D \rightarrow R$ associates each element of its domain D with exactly one element of its codomain R ; formally it is a set of ordered pairs (x, y) with a unique y for each x . Two functions are equal only if their domains, codomains and values all agree. Numerical functions have $D, R \subseteq \mathbb{R}$. A polynomial is $f(x) = \sum_{k=0}^n a_k x^k$; its degree is the largest k with $a_k \neq 0$, and a root is a value where it vanishes. Theorem L2.1 says that a root $\bar{x}$ of a polynomial p of degree $n \geq 1$ yields a unique factorisation $p(x) = (x - \bar{x})q(x)$ with q of degree $n - 1$; consequently a polynomial of degree n has at most n roots.

Key definitions

- **Function:** a rule (formally, a set of ordered pairs) giving each $x \in D$ a unique $f(x) \in R$.
- **Domain, codomain:** the sets D and R in $f: D \rightarrow R$.
- **Equal functions:** same domain, same codomain, same value at every point.
- **Numerical function:** $D, R \subseteq \mathbb{R}$.
- **Polynomial, coefficients, degree, monomials, root:** as in Definition 9.5.

Key results

$$p(\bar{x}) = 0 \iff p(x) = (x - \bar{x})q(x) \text{ with } \deg q = \deg p - 1;$$

a polynomial of degree n has at most n roots.

Connections

Functions are sets of ordered pairs (Chapter 2), and polynomials are written with $\sum$ (Chapter 6). The proof of Theorem L2.1 uses the strong form of induction (Section 7.6), and the theorem generalises the factorisation of quadratics used to solve inequalities (Chapter 4). Sequences, studied next, are functions with domain $\mathbb{N}$ (Chapter 10). Topic 2 continues the study of functions.

Common mistakes

- Regarding two formulas as the same function when their domains differ.
- Reading the degree from the largest power written rather than the largest power with a non-zero coefficient.
- Confusing the codomain R in $f: D \rightarrow R$ with the real numbers $\mathbb{R}$.
- Forgetting that the factor theorem requires $\bar{x}$ to be a root.

Exam checklist

- ☐ Give the informal and formal definitions of a function, and the definition of equal functions.
- ☐ Define polynomial, coefficient, degree, monomial and root.
- ☐ State Theorem L2.1 and outline its proof by induction.
- ☐ Factorise a polynomial given (or after finding) a root, by comparing coefficients.

9.5 Practice questions

Question 9.1 [basic]

Which of the following define a function from $\mathbb{R}$ to $\mathbb{R}$? (a) $f(x) = x^3$; (b) $f(x) = \sqrt{x}$; (c) $f(x) = \frac{1}{x^2+1}$; (d) $f(x)$ is “a number whose square is x^2 ”.

Question 9.2 [basic]

State the degree and the roots of: (a) $2x - 8$; (b) $x^2 + 1$; (c) $(x - 1)^2(x + 4)$; (d) $0 \cdot x^2 + 5$.

Question 9.3 [conceptual]

Are the functions $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x|$, and $g: \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = \sqrt{x^2}$, equal? Are $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x$, and $h: \mathbb{R} \rightarrow [0, \infty)$, $h(x) = |x|$, equal? Justify.

Question 9.4 [mathematical]

Show that 2 is a root of $p(x) = x^3 - 3x^2 - 4x + 12$, and factorise p completely.

Question 9.5 [mathematical]

Use Proposition 9.2 to show that if two polynomials of degree at most n agree at $n + 1$ distinct points, they are equal.

Question 9.6 [application]

A demand function is $D(p) = 120 - 3p$. State a sensible domain, find the price at which demand is zero, and explain why D cannot be defined with domain $\mathbb{R}$ in an economic model.

Question 9.7 [multi-step]

Find all real x satisfying $x^3 - 2x^2 - 5x + 6 \leq 0$.

Question 9.8 [exam-style]

(a) State Theorem L2.1 on the factorisation of a polynomial via its roots. (b) Explain where the proof uses the induction hypothesis and why the alternative (strong) formulation of induction is convenient. (c) Use the theorem to factorise $p(x) = 2x^3 - 3x^2 - 11x + 6$, given that $p(3) = 0$.

9.6 Solutions to practice questions

Solution to Question 9.1

(a) Yes. (b) No: $\sqrt{x}$ is not defined for $x < 0$, so not every element of the domain $\mathbb{R}$ has an output. (c) Yes: the denominator is never 0. (d) No: for $x \neq 0$ there are two such numbers (x and $-x$), so the output is not unique.

Solution to Question 9.2

(a) Degree 1, root 4. (b) Degree 2, no real roots. (c) Degree 3, roots 1 and -4 . (d) Degree 0 (the polynomial is the constant 5), no roots.

Solution to Question 9.3

f and g are equal: same domain and codomain, and $\sqrt{x^2} = |x|$ for every real x . f and h are not equal: their codomains differ, and in any case $f(-1) = -1 \neq 1 = h(-1)$.

Solution to Question 9.4

$p(2) = 8 - 12 - 8 + 12 = 0$. Then $p(x) = (x - 2)(x^2 + bx + c)$; comparing, $b - 2 = -3$, so $b = -1$, and $-2c = 12$, so $c = -6$. Thus $p(x) = (x - 2)(x^2 - x - 6) = (x - 2)(x - 3)(x + 2)$.

Solution to Question 9.5

If f and g have degree at most n and agree at $n + 1$ points, then $d = f - g$ is a polynomial of degree at most n (or the zero polynomial) with at least $n + 1$ roots. By Proposition 9.2 a non-zero polynomial of degree at most n has at most n roots, so d is the zero polynomial and $f = g$.

Solution to Question 9.6

Quantities and prices cannot be negative: $p \geq 0$ and $D(p) \geq 0 \Leftrightarrow p \leq 40$. A sensible domain is $[0, 40]$, with demand zero at $p = 40$. With domain $\mathbb{R}$, the formula would assign negative quantities to prices above 40 and to negative prices, which have no economic meaning.

Solution to Question 9.7

$p(1) = 1 - 2 - 5 + 6 = 0$, so $p(x) = (x - 1)(x^2 - x - 6) = (x - 1)(x - 3)(x + 2)$. Critical points $-2, 1, 3$. For $x > 3$ the product is positive; it is negative on $(1, 3)$, positive on $(-2, 1)$, negative on $(-\infty, -2)$. Including the roots: $x \in (-\infty, -2] \cup [1, 3]$.

Solution to Question 9.8

(a) If a polynomial p of degree $n \geq 1$ has a root $\bar{x}$, there is a unique polynomial q of degree $n - 1$ with $p(x) = (x - \bar{x})q(x)$. (b) In Case 2 the auxiliary polynomial $s(x) = p(x) - a_n x^{n-1}(x - \bar{x})$ has the root $\bar{x}$ and degree k with $1 \leq k \leq n - 1$, but k need not equal $n - 1$; the hypothesis is applied to s , of degree k . The strong form, which assumes the theorem for *all* degrees below n , covers every possible k ; the ordinary form would cover only $k = n - 1$. (c) $p(x) = (x - 3)(2x^2 + bx + c)$; comparing, $b - 6 = -3$ so $b = 3$, and $-3c = 6$ so $c = -2$. Hence $p(x) = (x - 3)(2x^2 + 3x - 2)$, and $2x^2 + 3x - 2 = (2x - 1)(x + 2)$, so $p(x) = (x - 3)(2x - 1)(x + 2)$.

Sequences and Limits

Economics is full of processes that unfold over time: output in successive quarters, the balance of a savings account year after year, the price in successive rounds of an auction. A sequence records such a process, and its limit describes where the process is heading in the long run. This chapter defines sequences, gives the precise definition of the limit of a sequence — one of the most important definitions in the whole module — and shows how to prove that a sequence converges to a given number, or that it has no limit at all.

Before you start: prerequisites

Quantifiers and their order (Section 1.6); proof by contradiction (Section 1.7); functions (Section 9.1); absolute value as distance and the triangle inequality (Chapter 5); solving inequalities (Chapter 4).

10.1 Sequences

Definition 10.1 (Sequence)

A **sequence** is an infinite ordered list $x = (x_1, x_2, x_3, \dots)$, whose elements x_n are indexed by $n \in \mathbb{N}$. More formally, a sequence x is a function whose domain is the set of natural numbers; we write $x_n = x(n)$.

[L2, slide 22]

Worked Example 10.1 – Sequences in applications

- (a) Writing H for heads and T for tails, a sequence of coin-flip results might appear as $(T, T, T, T, H, T, T, T, H, H, \dots)$.
- (b) A macroeconomic series might record four numbers in each period,

$$x_t = (\text{output}(t), \text{employment}(t), \text{debt}(t), \text{inflation}(t)),$$

where $t = 1, 2, 3, \dots$ denotes the time period.

[L2, slide 22]

⚠ Codomain of a sequence

The slide's formal definition reads "a function $x: \mathbb{N} \rightarrow \mathbb{R}$ ", but its own examples have values that are not real numbers (the letters H and T; four-number vectors). The general definition allows any codomain; sequences whose values are real numbers are singled out next as *numerical* sequences.

Definition 10.2 (Numerical sequence)

A sequence $x = (x_1, x_2, x_3, \dots)$ with each $x_k \in \mathbb{R}$ is called a **numerical sequence**. [L2, slide 23]

Worked Example 10.2 – Numerical sequences

- (a) The sequence $a = (7, 7, 7, 7, \dots)$ is *constant*.
- (b) The n -th element of $b = (3, 3.1, 3.14, 3.141, \dots)$ gives the first n digits of the decimal expansion of π .
- (c) The sequence $c = (-0.5, 0.11, 0.3596, 0.365881, \dots)$ is given by $c_n = f(b_n)$, where $f(x) = x^2 - 9.5$. For example $c_2 = 3.1^2 - 9.5 = 9.61 - 9.5 = 0.11$.
- (d) The n -th element of $d = (0, 0.69, 1.09, 1.38, 1.60, 1.79, 1.94, \dots)$ gives $\ln(n)$ truncated to two decimal places.

[L2, slide 23]

Notation. All of the following may denote a sequence: x , $x_1, x_2, x_3, \dots$, $(x_1, x_2, x_3, \dots)$, (x_n) , $(x_n)_{n \in \mathbb{N}}$, $(x_n)_{n=1}^\infty$. The index need not be n ; letters such as i, j, k, l, m are also used. [L2, slide 23]

10.2 Behaviour at infinity

What happens to a_n when n grows arbitrarily large? Consider $a_n = 2^{-n}$:

$$(a_n) = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{64}, \frac{1}{128}, \frac{1}{256}, \frac{1}{512}, \frac{1}{1024}, \frac{1}{2048}, \dots \right).$$

As n grows, a_n stays positive but becomes smaller and smaller, and closer and closer to zero. So 0 is a good description of what happens "at infinity". If instead the sequence were

$$(b_n) = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{64}, \frac{1}{128}, \frac{1}{256}, \frac{1}{256}, \frac{1}{256}, \frac{1}{256}, \dots \right),$$

then the number $\frac{1}{256}$ would better describe its behaviour at infinity.

[L2, slide 24]

CORE CONCEPT | Why a precise definition is needed

The behaviour "at infinity" cannot be determined from any finite number of initial elements: (a_n) and (b_n) agree in their first eight terms but head to different places. A definition is needed that speaks about *all sufficiently late* terms at once.

10.3 The definition of a limit

Definition 10.3 (Limit of a sequence)

A number $x \in \mathbb{R}$ is a **limit** of a sequence (a_n) if for any real number $\varepsilon > 0$ there exists a natural number N such that for all $n > N$, $|a_n - x| < \varepsilon$. In this case we say that the sequence (a_n) has a **finite limit**, or that it **converges**. In symbols:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} : \forall n > N, |a_n - x| < \varepsilon.$$

[L2, slide 25]

⚠ A missing subscript

On the slide, the symbolic version of the definition is printed with “ $|a - x| < \varepsilon$ ”; the subscript n is missing. The correct condition is $|a_n - x| < \varepsilon$, as in the verbal definition above it.

10.3.1 Reading the definition

CORE CONCEPT | The limit as a game

Read the definition from left to right, as a challenge and a response.

1. $\forall \varepsilon > 0$: a challenger names a tolerance ε , as small as they like.
2. $\exists N \in \mathbb{N}$: you must reply with a point N in the sequence. Your N may depend on ε (it comes after ε ; Worked Example 1.5).
3. $\forall n > N$: every term after position N must then pass the test:
4. $|a_n - x| < \varepsilon$: a_n is within distance ε of x (Section 5.3), i.e. $x - \varepsilon < a_n < x + \varepsilon$.

x is a limit if you can win this game whatever ε the challenger chooses. Equivalently: however narrow a band is drawn around x , the sequence eventually enters the band and never leaves it.

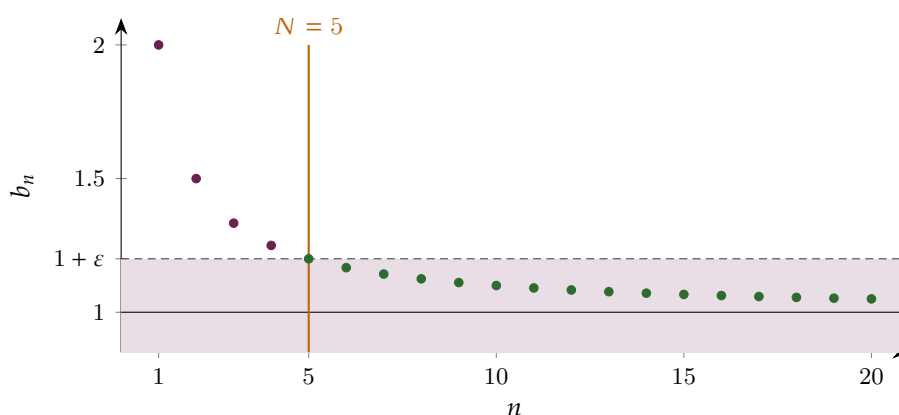


Figure 10.1: The sequence $b_n = 1 + \frac{1}{n}$ with the band $|b_n - 1| < \varepsilon$ for $\varepsilon = 0.2$ (shaded). The terms $b_1, \dots, b_4$ lie outside the band; $b_5 = 1.2$ lies on its edge (since $|b_5 - 1| = 0.2 \not< 0.2$); every term after $N = 5$ lies strictly inside. A smaller ε requires a larger N , but for every $\varepsilon > 0$ some N works.

10.4 Proving convergence

Worked Example 10.3 – Finding N for particular ε

Let $a_n = 2^{-n}$ and $x = 0$, so $|a_n - x| = |2^{-n} - 0| = 2^{-n}$. For each ε below, find an N such that $|a_n - 0| < \varepsilon$ for all $n > N$. [L2, slide 26]

Solution.

ε	first term inside the band	a valid N
1	all terms ($a_1 = \frac{1}{2} < 1$)	$N = 1$
0.1	$a_4 = \frac{1}{16} = 0.0625$ (since $a_3 = 0.125$)	$N = 3$
0.01	$a_7 = \frac{1}{128} \approx 0.0078$ (since $a_6 \approx 0.0156$)	$N = 6$
0.003	$a_9 = \frac{1}{512} \approx 0.00195$ (since $a_8 \approx 0.0039$)	$N = 8$

Because 2^{-n} decreases as n increases, once one term is inside the band, all later terms are too.

Any working N will do

$N = 100$ works for $\varepsilon = 1, 0.1, 0.01, 0.003$ as well. It does not matter which particular N we choose for which ε : the important thing is that for *any* $\varepsilon > 0$ there is *some* N such that $|a_n - x| < \varepsilon$ whenever $n > N$. In words, people sometimes say that $|a_n - x| < \varepsilon$ “for large enough n ”, meaning the same statement. [L2, slide 27]

A table, however long, does not prove convergence: it deals with finitely many values of ε . A proof must handle an *arbitrary* $\varepsilon > 0$.

Worked Example 10.4 – 2^{-n} converges to 0

Prove that $x = 0$ is a limit of $a_n = 2^{-n}$. [L2, slide 28]

Solution. Consider an arbitrary $\varepsilon > 0$. We want to show that for large enough n , $|2^{-n} - 0| < \varepsilon$, that is, $2^{-n} < \varepsilon$. Since $\log_2$ is increasing,

$$2^{-n} < \varepsilon \iff \log_2(2^{-n}) < \log_2 \varepsilon \iff -n < \log_2 \varepsilon \iff n > -\log_2 \varepsilon.$$

Hence we can take any natural number $N > -\log_2 \varepsilon$; then for all $n > N$,

$$n > N > -\log_2 \varepsilon \implies |2^{-n} - 0| < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, 0 is a limit of (2^{-n}) . (There are many other proofs.) $\square$

A proof without logarithms

By Bernoulli's inequality (Worked Example 7.6), $2^n \geq n + 1 > n$, so $0 < 2^{-n} < \frac{1}{n}$. Given $\varepsilon > 0$, choose a natural number $N \geq \frac{1}{\varepsilon}$. For $n > N$: $2^{-n} < \frac{1}{n} < \frac{1}{N} \leq \varepsilon$. This uses only the properties of the real numbers and induction, and it shows a useful technique: bound the distance $|a_n - x|$ by something simpler (here $\frac{1}{n}$) whose behaviour is easy to control.

Worked Example 10.5 – $1 + \frac{1}{n}$ converges to 1

Consider $b_n = 1 + \frac{1}{n}$, i.e. $(b_n) = (2, \frac{3}{2}, \frac{4}{3}, \frac{5}{4}, \frac{6}{5}, \frac{7}{6}, \dots)$. Show that $x = 1$ is a limit of (b_n) . [L2, slide 29]

Solution.

- Consider an arbitrary $\varepsilon > 0$. (We need to prove that the statement holds for *any* $\varepsilon > 0$.)
- Let $N = \lceil 1/\varepsilon \rceil$ be the smallest natural number not less than $1/\varepsilon$. (It is enough to find *one* N that works for the given ε .)
- Consider an arbitrary $n > N$. (We again need to show that the inequality holds for *any* $n > N$, given the N we have picked.) Then

$$|b_n - 1| = \left| \frac{1}{n} \right| = \frac{1}{n} < \frac{1}{N} \leq \varepsilon.$$

- Therefore $|b_n - 1| < \varepsilon$ for all $n > N$, which we wanted to prove. $\square$

Each line of the proof corresponds to one quantifier of the definition: an arbitrary ε ($\forall$), a chosen N ($\exists$), an arbitrary n ($\forall$), and finally the inequality.

CORE CONCEPT | Template for proving $a_n \rightarrow x$ from the definition

1. *Scratch work:* simplify $|a_n - x|$ and find a condition on n that makes it less than ε (solve the inequality, or bound $|a_n - x|$ by something simpler first).
2. *Proof:* “Let $\varepsilon > 0$ be arbitrary. Choose $N = \dots$ (a natural number depending on ε). Let $n > N$. Then $|a_n - x| = \dots < \varepsilon$. Hence x is a limit of (a_n) .”

Worked Example 10.6 – A rational sequence

Prove from the definition that $a_n = \frac{n^2 - 1}{n^2 + 1}$ converges to 1.

Solution. *Scratch work.* $|a_n - 1| = \left| \frac{n^2 - 1 - (n^2 + 1)}{n^2 + 1} \right| = \frac{2}{n^2 + 1} < \frac{2}{n^2} \leq \frac{2}{n}$ for $n \geq 1$. So it is enough to make $\frac{2}{n} < \varepsilon$, i.e. $n > \frac{2}{\varepsilon}$.

Proof. Let $\varepsilon > 0$ be arbitrary and choose a natural number $N \geq \frac{2}{\varepsilon}$. For $n > N$,

$$|a_n - 1| = \frac{2}{n^2 + 1} < \frac{2}{n} < \frac{2}{N} \leq \varepsilon.$$

Hence $a_n \rightarrow 1$. $\square$

10.5 Showing that a sequence has no limit

To show that a sequence does not converge, we must show that *no* real number x is a limit. Negating the definition (Section 1.5): x is *not* a limit of (a_n) when

$$\exists \varepsilon > 0 : \forall N \in \mathbb{N}, \exists n > N : |a_n - x| \geq \varepsilon.$$

Some tolerance ε defeats every N : however late we start, some later term is at least ε away from x .

Worked Example 10.7 – An oscillating sequence has no limit

Consider the sequence $1, 0, 1, 0, 1, 0, \dots$, that is, $c_n = 1$ if n is odd and $c_n = 0$ if n is even. Show that (c_n) does not have a (finite) limit. [L2, slide 30]

Solution.

- Towards a contradiction, assume that x is a limit of (c_n) . (We should consider an *arbitrary* x .)
- Consider $\varepsilon = \frac{1}{4}$. (It is enough to find *some* $\varepsilon > 0$ that leads to a contradiction.)
- Then there should exist $N \in \mathbb{N}$ such that for all $n > N$, $|c_n - x| < \varepsilon$. (We should consider an arbitrary such N .)
- One of the numbers c_{N+1}, c_{N+2} is zero and the other is one; since $N+1, N+2 > N$, it should be that $|1 - x| < \frac{1}{4}$ and $|0 - x| < \frac{1}{4}$. Then, by the triangle inequality (Theorem 5.2),

$$1 = |1 - 0| \leq |1 - x| + |x - 0| < \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

- We get $1 < \frac{1}{2}$, a contradiction. $\square$

The choice $\varepsilon = \frac{1}{4}$ is not special: any $\varepsilon \leq \frac{1}{2}$ works. The idea is that the terms keep returning to both 0 and 1, which are 1 apart, so no single number can be within less than $\frac{1}{2}$ of both; the triangle inequality turns this picture into a proof.

10.6 Further properties

A sequence has at most one limit

The slides say “ x is a limit”. In fact a sequence cannot have two different limits, so one may speak of *the* limit and write $\lim_{n \rightarrow \infty} a_n = x$ or $a_n \rightarrow x$. *Proof.* Suppose $x \neq y$ are both limits and take $\varepsilon = \frac{|x-y|}{2} > 0$. There are N_1 and N_2 with $|a_n - x| < \varepsilon$ for $n > N_1$ and $|a_n - y| < \varepsilon$ for $n > N_2$. For $n > \max\{N_1, N_2\}$ both hold, and by the triangle inequality $|x - y| \leq |x - a_n| + |a_n - y| < 2\varepsilon = |x - y|$, a contradiction.

The following results are in the style of the first question of the January 2025 paper (as summarised in the study guide). They show how the definition is used to prove general facts.

Worked Example 10.8 – Limits preserve non-negativity

Let $a_n \rightarrow a$ and suppose $a_n \geq 0$ for every n . Prove that $a \geq 0$. Hint: assume, towards a contradiction, that $a < 0$. [SG: Jan 2025 Q1(b)]

Solution. Suppose, for a contradiction, that $a < 0$. Take $\varepsilon = -a > 0$ (the distance from a to 0). By the definition of the limit there is N with $|a_n - a| < -a$ for all $n > N$. Take $n = N + 1$. Then $a_n - a \leq |a_n - a| < -a$, so $a_n < 0$. This contradicts $a_n \geq 0$. Hence $a \geq 0$. (The negation of $a \geq 0$ is $a < 0$ by trichotomy, Section 1.5.)

Worked Example 10.9 – Comparing limits

Let $b_n \rightarrow b$ and $c_n \rightarrow c$ with $c_n \geq b_n$ for every n . Prove that $c \geq b$.

[SG: Jan 2025 Q1(c)]

Solution. Let $a_n = c_n - b_n \geq 0$. First we show $a_n \rightarrow c - b$. Given $\varepsilon > 0$, choose N_1 with $|b_n - b| < \varepsilon/2$ for $n > N_1$, and N_2 with $|c_n - c| < \varepsilon/2$ for $n > N_2$. For $n > N = \max\{N_1, N_2\}$, by the triangle inequality,

$$|a_n - (c - b)| = |(c_n - c) - (b_n - b)| \leq |c_n - c| + |b_n - b| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

So $a_n \rightarrow c - b$. Since $a_n \geq 0$ for all n , Worked Example 10.8 gives $c - b \geq 0$, i.e. $c \geq b$.

Strict inequalities need not survive the limit

If $a_n > 0$ for every n and $a_n \rightarrow a$, it does *not* follow that $a > 0$: the sequence $a_n = \frac{1}{n}$ is always positive but its limit is 0. Only the non-strict inequality $a \geq 0$ is guaranteed.

10.7 Limits in economics

Worked Example 10.10 – Convergence to a steady state

A simple model of capital accumulation sets $K_{t+1} = 0.8K_t + 20$, with $K_1 = 50$. Show that $K_t = 100 - 50 \cdot (0.8)^{t-1}$ and that $K_t \rightarrow 100$.

Solution. The formula holds for $t = 1$ ($100 - 50 = 50$). If it holds for t , then $K_{t+1} = 0.8(100 - 50(0.8)^{t-1}) + 20 = 100 - 50(0.8)^t$, which is the formula for $t + 1$; so by induction it holds for all t . Then $|K_t - 100| = 50(0.8)^{t-1}$. Given $\varepsilon > 0$, this is less than ε once $(0.8)^{t-1} < \frac{\varepsilon}{50}$, i.e. $t - 1 > \frac{\ln(\varepsilon/50)}{\ln 0.8}$ (dividing by $\ln 0.8 < 0$ reverses the inequality). Choosing N larger than $1 + \frac{\ln(\varepsilon/50)}{\ln 0.8}$ completes the proof. The capital stock approaches the *steady state* $K^* = 100$, the value at which $K^* = 0.8K^* + 20$.

Infinite sums are limits

The present value of a perpetuity paying £1 per year forever at interest rate $i > 0$ is defined as the limit of the present values of finite streams, $\sum_{t=1}^n (1+i)^{-t} = \frac{1-(1+i)^{-n}}{i}$ (Worked Example 7.5). Since $(1+i)^{-n} \rightarrow 0$, the limit is $\frac{1}{i}$. Infinite sums of this kind are the subject of Lecture 3.

10.8 Chapter review

Summary

A sequence is an infinite ordered list indexed by $\mathbb{N}$, formally a function with domain $\mathbb{N}$; a numerical sequence has real values. Its behaviour at infinity cannot be read from finitely many terms, which is why limits are defined with quantifiers: x is a limit of (a_n) if for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that $|a_n - x| < \varepsilon$ for all $n > N$. To prove convergence, take an arbitrary ε , produce an

N (which may depend on ε), take an arbitrary $n > N$ and verify the inequality. To prove that there is no limit, assume a limit x exists and use one well-chosen ε and the triangle inequality to reach a contradiction. Limits are unique, preserve non-strict inequalities, and describe long-run behaviour such as convergence to a steady state.

Key definitions

- **Sequence:** $(x_1, x_2, \dots)$, formally a function on $\mathbb{N}$ with $x_n = x(n)$; **numerical** if $x_n \in \mathbb{R}$.
- **Limit:** x is a limit of (a_n) if $\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n > N : |a_n - x| < \varepsilon$; the sequence then **converges** (has a finite limit).

Key statements

$$\begin{aligned} x \text{ is a limit: } & \forall \varepsilon > 0, \exists N \in \mathbb{N} : \forall n > N, |a_n - x| < \varepsilon; \\ x \text{ is not a limit: } & \exists \varepsilon > 0 : \forall N \in \mathbb{N}, \exists n > N : |a_n - x| \geq \varepsilon; \\ 2^{-n} \rightarrow 0, & \quad 1 + \frac{1}{n} \rightarrow 1, \quad (1, 0, 1, 0, \dots) \text{ has no limit.} \end{aligned}$$

Connections

The definition combines nearly everything in the book: quantifiers and their order (Chapter 1), sets and functions (Chapters 2 and 9), the properties of $\mathbb{R}$ (Chapter 3), solving inequalities for n (Chapter 4), distance and the triangle inequality (Chapter 5) and induction (Chapter 7). It is the foundation of Topic 3: properties and calculation of limits, infinite sums, continuity and derivatives are all defined through limits.

Common mistakes

- Choosing N before ε , or a single N meant to work for every ε .
- “Proving” convergence with a table of values for a few ε .
- Fixing a particular n instead of an arbitrary $n > N$.
- Writing $|a_n - x| \leq \varepsilon$ in place of $< \varepsilon$ without care (harmless in practice, but match the definition).
- In a non-existence proof, choosing a particular x instead of an arbitrary one.
- Concluding $a > 0$ from $a_n > 0$.

Exam checklist

- ☐ Define a sequence and a numerical sequence; give examples.
- ☐ State the definition of a limit exactly, with quantifiers in the right order.
- ☐ For a given sequence and ε , find a valid N .
- ☐ Prove from the definition that a given sequence converges to a given number.
- ☐ Prove that a given sequence has no limit.
- ☐ Prove that limits preserve $\geq$, using contradiction.

10.9 Practice questions

Question 10.1 [basic]

Write down the first five terms of $a_n = \frac{(-1)^n}{n}$, $b_n = \frac{n}{n+1}$ and $c_n = n^2 - 3n$.

Question 10.2 [basic]

For $a_n = \frac{1}{n}$ and $x = 0$, find an N that works for $\varepsilon = 0.1$, for $\varepsilon = 0.01$ and for $\varepsilon = 0.003$.

Question 10.3 [conceptual]

Explain why the following is *not* a valid definition of $a_n \rightarrow x$: “there is $N \in \mathbb{N}$ such that for every $\varepsilon > 0$ and every $n > N$, $|a_n - x| < \varepsilon$.” Which sequences satisfy it?

Question 10.4 [mathematical]

Prove from the definition that $\frac{3n+1}{n} \rightarrow 3$.

Question 10.5 [mathematical]

Prove from the definition that $\frac{(-1)^n}{n} \rightarrow 0$.

Question 10.6 [mathematical]

Prove that the sequence $a_n = (-1)^n$ has no limit.

Question 10.7 [application]

A savings balance evolves as $B_{t+1} = 0.5B_t + 10$ with $B_1 = 30$. Show by induction that $B_t = 20 + 10 \cdot (0.5)^{t-1}$, and prove from the definition that $B_t \rightarrow 20$.

Question 10.8 [multi-step]

Let $a_n \rightarrow a$. Prove that $3a_n \rightarrow 3a$, directly from the definition.

Question 10.9 [exam-style]

(a) Define what it means that a sequence (a_n) has a finite limit a . (b) Let $a_n \rightarrow a$, and suppose $a_n \leq 5$ for all n . Prove that $a \leq 5$. Hint: assume, towards a contradiction, that $a > 5$. (c) Give an example of a sequence with $a_n < 5$ for all n whose limit is not less than 5.

10.10 Solutions to practice questions

Solution to Question 10.1

a : $-1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, -\frac{1}{5}$. b : $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}$. c : $-2, -2, 0, 4, 10$.

Solution to Question 10.2

We need $\frac{1}{n} < \varepsilon$ for $n > N$, i.e. $n > \frac{1}{\varepsilon}$. So $N = 10$ works for $\varepsilon = 0.1$ (for $n \geq 11$, $\frac{1}{n} < 0.1$); $N = 100$ for $\varepsilon = 0.01$; and $N = 333$ for $\varepsilon = 0.003$ (since $\frac{1}{334} < 0.003 < \frac{1}{333}$). Any larger N also works.

Solution to Question 10.3

Here N is chosen *before* ε and must work for every ε simultaneously. If $|a_n - x| < \varepsilon$ for every $\varepsilon > 0$, then $|a_n - x|$ is a non-negative number smaller than every positive number, so it is 0. The condition therefore says that $a_n = x$ for all $n > N$: only sequences that are eventually constant equal to x satisfy it. The correct definition lets N depend on ε (Worked Example 1.5).

Solution to Question 10.4

$\left| \frac{3n+1}{n} - 3 \right| = \frac{1}{n}$. Given $\varepsilon > 0$, choose a natural number $N \geq \frac{1}{\varepsilon}$. For $n > N$: $\frac{1}{n} < \frac{1}{N} \leq \varepsilon$. Hence the limit is 3.

Solution to Question 10.5

$\left| \frac{(-1)^n}{n} - 0 \right| = \frac{1}{n}$. Given $\varepsilon > 0$, choose $N \geq \frac{1}{\varepsilon}$; for $n > N$, $\frac{1}{n} < \varepsilon$.

Solution to Question 10.6

Suppose, for a contradiction, that x is a limit, and take $\varepsilon = \frac{1}{2}$. There is N with $|a_n - x| < \frac{1}{2}$ for $n > N$. Among $N+1$ and $N+2$ one is even and one is odd, so $|1 - x| < \frac{1}{2}$ and $|-1 - x| < \frac{1}{2}$. By the triangle inequality, $2 = |1 - (-1)| \leq |1 - x| + |x - (-1)| < \frac{1}{2} + \frac{1}{2} = 1$, a contradiction. So (a_n) has no limit.

Solution to Question 10.7

Base: $20 + 10 = 30 = B_1$. Step: if $B_t = 20 + 10(0.5)^{t-1}$ then $B_{t+1} = 0.5(20 + 10(0.5)^{t-1}) + 10 = 20 + 10(0.5)^t$. Limit: $|B_t - 20| = 10(0.5)^{t-1} = 20 \cdot 2^{-t}$. By Bernoulli, $2^t > t$, so $|B_t - 20| < \frac{20}{t}$. Given $\varepsilon > 0$, choose $N \geq \frac{20}{\varepsilon}$; for $t > N$, $|B_t - 20| < \frac{20}{t} < \frac{20}{N} \leq \varepsilon$.

Solution to Question 10.8

Let $\varepsilon > 0$. Since $a_n \rightarrow a$, there is N with $|a_n - a| < \frac{\varepsilon}{3}$ for all $n > N$. For such n , $|3a_n - 3a| = 3|a_n - a| < 3 \cdot \frac{\varepsilon}{3} = \varepsilon$. Hence $3a_n \rightarrow 3a$. (The trick is to apply the definition for a_n with the smaller tolerance $\varepsilon/3$.)

Solution to Question 10.9

(a) (a_n) has finite limit $a \in \mathbb{R}$ if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|a_n - a| < \varepsilon$ for all $n > N$. (b) Suppose $a > 5$ and let $\varepsilon = a - 5 > 0$. There is N with $|a_n - a| < a - 5$ for all $n > N$. For $n = N+1$: $a - a_n \leq |a_n - a| < a - 5$, so $a_n > 5$, contradicting $a_n \leq 5$. Hence $a \leq 5$. (c) $a_n = 5 - \frac{1}{n}$: every term is less than 5, but $a_n \rightarrow 5$ (since $|a_n - 5| = \frac{1}{n}$), and 5 is not less than 5.

Final Master Review

This final part draws the whole book together. It is designed for revision: read the module summary and the concept map to see the structure, use the formula sheet and the glossary as references, work through the checklist honestly, and then attempt the final practice paper under timed conditions (about 1.2 minutes per mark, without a calculator) before reading the solutions.

Module summary

The language (Chapters 1 and 2). Mathematics is written in propositions — statements that are true or false — linked by implication, equivalence and negation, and quantified with “for all” ($\forall$) and “there exists” ($\exists$). Axioms are accepted without proof; theorems are proved from them. Sets collect objects; they are compared by $\subseteq$ and $=$ (two sets are equal exactly when each is a subset of the other) and combined by $\cap$, $\cup$ and $\setminus$. Set-builder notation $\{x \in A \mid P\}$ describes a set by a property, and solving an inequality amounts to describing such a set simply.

The numbers (Chapter 3). $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$. The real numbers satisfy thirteen properties: trichotomy and transitivity of the order (O1–O2), five properties of addition (A1–A5) and six of multiplication (M1–M6). The rationals satisfy them too; the real numbers are distinguished by the (non-examinable) Least Upper Bound property, which says the number line has no gaps. Intervals name the sets of numbers between two endpoints.

Calculation (Chapters 4 to 6). Inequalities are solved by chains of equivalences: adding is always safe, multiplying by a positive number keeps the direction, by a negative number reverses it (M6). Products are handled by sign analysis; quadratics are factorised through their roots using the discriminant; fractions are handled by bringing everything to one side. The absolute value measures size and $|x - y|$ measures distance; the triangle inequality says a detour is never shorter than the direct route. The summation operator $\sum$ writes long sums compactly and obeys P1–P3, with the conventions that a one-term sum is that term and an empty sum is 0.

Proof and structure (Chapters 7 to 9). Induction proves a proposition for every natural number by proving a base and a step. It establishes summation formulas, the binomial formula $(x + y)^n = \sum_j C_n^j x^{n-j} y^j$, and — in its strong form — the factorisation $p(x) = (x - \bar{x})q(x)$ of a polynomial through any root. Functions associate each element of a domain with a unique element of a codomain; two functions are equal only if domains, codomains and values agree.

Limits (Chapter 10). A sequence is a function on $\mathbb{N}$. A number x is its limit if for every $\varepsilon > 0$ there is N such that $|a_n - x| < \varepsilon$ for all $n > N$. Convergence is proved by producing N for an arbitrary ε ;

non-convergence by contradiction, using the triangle inequality. Limits are unique and preserve non-strict inequalities. This definition is the gateway to the rest of the module.

How the ideas connect

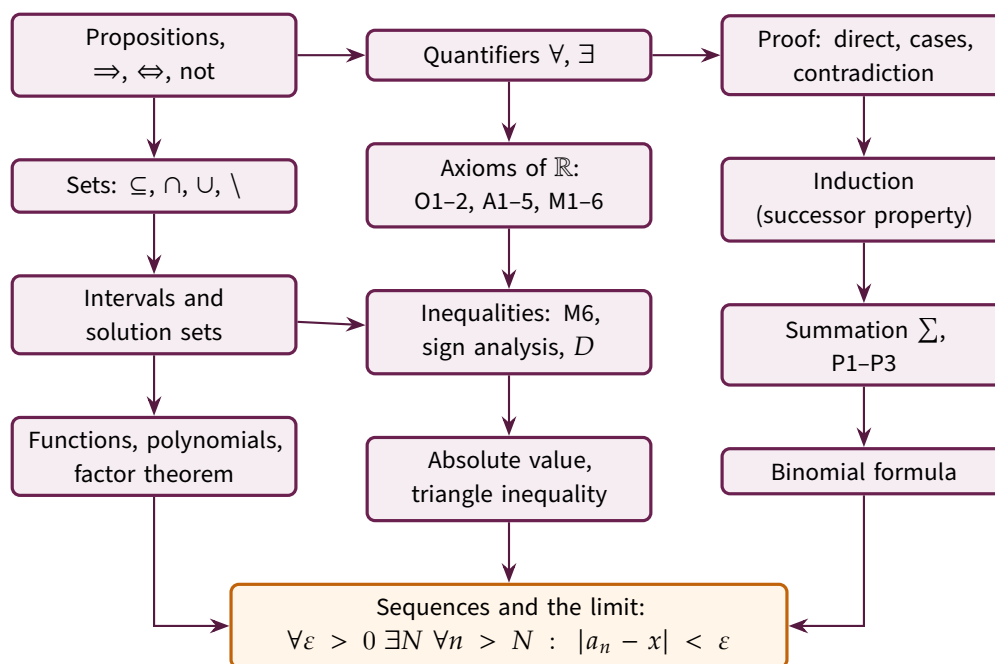


Figure 10.2: Concept map of the book. Arrows point from ideas to the ideas that use them. Every route ends at the definition of the limit, which needs quantifiers, inequalities, distance and induction together. (Induction is also used to prove the factor theorem for polynomials, Theorem 9.1.)

Five threads that run through everything.

1. *Equivalence preserves solution sets.* Whether solving an inequality (Chapter 4) or proving set equality (Chapter 2), the goal is a two-way statement.
2. *The sign of a multiplier.* M6 drives linear inequalities, sign analysis, parameter cases, quadratic and fractional inequalities, and every manipulation of inequalities inside limit proofs.
3. *Arbitrary versus particular.* $\forall$ statements are proved for an arbitrary object, $\exists$ statements by an example; this structures subset proofs, induction steps and limit proofs alike.
4. *Distance.* $|x - y|$ and the triangle inequality turn “close to” into algebra.
5. *Building on the previous case.* Induction, the summation property P3 (peeling off the last term) and Pascal’s rule all express a structure in terms of a smaller one.

Master formula sheet

Formula	Meaning and variables	Conditions / when to use
Logic and sets		
$P \Leftrightarrow Q$ iff $(P \Rightarrow Q)$ and $(Q \Rightarrow P)$	Equivalence is implication both ways	Proving “iff”; solving
$\text{not}(\forall x \in A, P) \Leftrightarrow \exists x \in A : \text{not } P$	Negating a universal claim	Counterexamples
$\text{not}(\exists x \in A : P) \Leftrightarrow \forall x \in A, \text{not } P$	Negating an existential claim	Contradiction proofs
$A = B \Leftrightarrow (A \subseteq B \text{ and } B \subseteq A)$	Set equality via two inclusions	Proving sets equal
$A \setminus B = \{x \in A \mid x \notin B\}$	Set difference	Definition questions
$A \subseteq B \Rightarrow A \setminus B = \emptyset$	Removing a superset leaves nothing	(Converse also true)
Real numbers and inequalities		
$a > b, c > 0 \Rightarrow ac > bc; a > b, c < 0 \Rightarrow ac < bc$	M6: sign of multiplier decides direction	Every multiplication/division
$a > b \Leftrightarrow a + c > b + c$	Adding preserves order (A5 twice)	Moving terms across
$ax + b > 0: x > -\frac{b}{a} (a > 0), x < -\frac{b}{a} (a < 0)$	Linear inequality	Case $a = 0$ separately
$uv > 0 \Leftrightarrow$ same signs; $uv < 0 \Leftrightarrow$ opposite	Sign of a product	Sign analysis
$D = b^2 - 4ac, x_{1,2} = \frac{-b \pm \sqrt{D}}{2a}$	Discriminant and roots of $ax^2 + bx + c$	$a \neq 0$; real roots iff $D \geq 0$
$ax^2 + bx + c = a(x - x_1)(x - x_2)$	Factorisation through roots	$D \geq 0$
$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 - \frac{D}{4a}$	Completing the square	$D \leq 0$; minimum value
$a > 0$: quadratic > 0 outside roots, < 0 between	Sign of a quadratic	Quadratic inequalities
Absolute value		
$ x = x (x \geq 0), -x (x < 0); x = \max\{x, -x\}$	Size of x	Definition
$- a \leq a \leq a , -a = a , ab = a b $	Basic properties	Simplifying
$ x - z \leq x - y + y - z ; a + b \leq a + b $	Triangle inequality	Bounding distances; limits
$ x - a < r \Leftrightarrow a - r < x < a + r$	Within distance r of a	$r > 0$
Sums and induction		
$\sum_{j=k}^n a_j = a_k + \dots + a_n$	$n - k + 1$ terms	Integers $k \leq n$
$\sum_{j=k}^{k-1} a_j = 0, \sum_{j=k}^k a_j = a_k$	Empty and one-term sums	Conventions
$\sum_{j=k}^n a = (n - k + 1)a$	Constant summand	Count terms carefully
P1, P2, P3	Split sums; constants out; join ranges	Manipulating sums
$\sum_{j=1}^n j = \frac{n(n+1)}{2}; \sum_{j=1}^n j^2 = \frac{n(n+1)(2n+1)}{6}$	Standard sums	Closed forms
$\sum_{t=0}^{n-1} r^t = \frac{1-r^n}{1-r}$	Geometric sum	$r \neq 1$
Base P_1 + step $P_n \Rightarrow P_{n+1}$	Induction	Statements for all $n \in \mathbb{N}$
Binomial formula		

Formula	Meaning and variables	Conditions / when to use
$0! = 1, k! = (k-1)!k; C_n^j = \frac{n!}{(n-j)!j!}$	Factorial; binomial coefficient	$0 \leq j \leq n$
$(x+y)^n = \sum_{j=0}^n C_n^j x^{n-j} y^j$	Expansion of a power of a sum	$n \in \{0\} \cup \mathbb{N}$
$C_n^j = C_n^{n-j}, C_n^j + C_n^{j+1} = C_{n+1}^{j+1}$	Symmetry; Pascal's rule	Computing coefficients
Functions, polynomials, limits		
$p(\bar{x}) = 0 \Leftrightarrow p(x) = (x - \bar{x})q(x)$	Factor theorem (L2.1), $\deg q = n - 1$	$\deg p = n \geq 1$
$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n > N : a_n - x < \varepsilon$	x is a limit of (a_n)	Proving convergence
$a_n \geq 0, a_n \rightarrow a \Rightarrow a \geq 0$	Limits preserve $\geq$	Not $>$

Glossary

Absolute value

Definition. $|x| = x$ if $x \geq 0$ and $|x| = -x$ if $x < 0$; equivalently $|x| = \max\{x, -x\}$.

In plain terms. How big a number is, ignoring its sign; $|x - y|$ is how far apart x and y are.

Axiom

Definition. A proposition accepted without proof.

In plain terms. A starting rule of the game; in ECN115, the properties O1–M6 of the real numbers.

Binomial coefficient

Definition. $C_n^j = \binom{n}{j} = \frac{n!}{(n-j)!j!}$ for integers $0 \leq j \leq n$.

In plain terms. The number in front of $x^{n-j}y^j$ when $(x+y)^n$ is multiplied out.

Binomial formula

Definition. $(x+y)^n = \sum_{j=0}^n C_n^j x^{n-j} y^j$.

In plain terms. A one-line recipe for multiplying out any power of a sum.

Codomain

Definition. The set R in $f: D \rightarrow R$, in which the values of f lie.

In plain terms. Where the outputs are allowed to land.

Coefficient

Definition. In $f(x) = \sum_{k=0}^n a_k x^k$, the numbers $a_0, \dots, a_n$.

In plain terms. The fixed numbers multiplying the powers of x .

Completing the square

Definition. Writing $ax^2 + bx + c = a(x + \frac{b}{2a})^2 - \frac{D}{4a}$.

In plain terms. Rewriting a quadratic as a square plus a constant, so its sign and minimum are visible.

Contrapositive

Definition. Of $P \Rightarrow Q$: the implication $(\text{not } Q) \Rightarrow (\text{not } P)$.

In plain terms. The same promise read backwards; always equivalent to the original.

Converge

Definition. A sequence converges if it has a finite limit.

In plain terms. It settles down towards a single number.

Converse

Definition. Of $P \Rightarrow Q$: the implication $Q \Rightarrow P$.

In plain terms. The reversed claim; it may be false even when the original is true.

Counterexample

Definition. An element for which a claimed $\forall$ statement fails.

In plain terms. One case that breaks a general claim.

Critical point (of a sign analysis)

Definition. A value at which a factor of the numerator or denominator is zero.

In plain terms. A place where an expression can change sign.

Degree

Definition. For a polynomial with at least one non-zero coefficient, the largest k with $a_k \neq 0$.

In plain terms. The highest power of x that actually appears.

Discriminant

Definition. $D = b^2 - 4ac$ for $ax^2 + bx + c$.

In plain terms. Tells you whether a quadratic has two, one or no real roots.

Domain

Definition. The set D in $f: D \rightarrow R$ of inputs.

In plain terms. The values the function accepts.

Element

Definition. An object in a set; written $x \in A$.

In plain terms. One of the things in the bag.

Empty set

Definition. The set with no elements, $\emptyset$ (or $\{\}$).

In plain terms. An empty bag.

Empty sum

Definition. $\sum_{j=k}^{k-1} a_j$, defined to be 0.

In plain terms. Adding up nothing gives nothing.

Equal functions

Definition. Functions with the same domain, the same codomain and the same value at every point.

In plain terms. Not just the same formula: the same inputs, outputs and allowed outputs.

Equal sets

Definition. Sets with exactly the same elements; equivalently, each is a subset of the other.

In plain terms. Same contents, however described.

Equivalence

Definition. $P \Leftrightarrow Q$: $P \Rightarrow Q$ and $Q \Rightarrow P$.

In plain terms. Two statements that are true in exactly the same situations.

Existential quantifier

Definition. $\exists x \in A : P$ — there is at least one $x \in A$ with property P .

In plain terms. “Some” or “there is”; proved by giving an example.

Factorial

Definition. $0! = 1$ and $k! = (k - 1)! \cdot k$.

In plain terms. $1 \times 2 \times \cdots \times k$.

Function

Definition. A rule (formally, a set of ordered pairs) assigning to each element of a domain a unique element of a codomain.

In plain terms. A machine that takes each allowed input to exactly one output.

Implication

Definition. $P \Rightarrow Q$: if P , then Q .

In plain terms. A one-way promise; broken only if P happens and Q does not.

Induction (mathematical)

Definition. Proving P_n for all $n \in \mathbb{N}$ by proving P_1 (base) and $P_n \Rightarrow P_{n+1}$ for all n (step).

In plain terms. Knock over the first domino and make sure each knocks over the next.

Induction hypothesis

Definition. The assumption that P_n (or $P_1, \dots, P_n$) holds, made while proving the step.

In plain terms. What you are allowed to use about the previous case.

Integers

Definition. $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$.

In plain terms. Whole numbers, positive, negative and zero.

Intersection

Definition. $A \cap B$: the objects in both A and B .

In plain terms. What two sets have in common.

Interval

Definition. A set of real numbers between two endpoints, e.g. $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$.

In plain terms. A stretch of the number line; square brackets include the endpoint, round ones exclude it.

Least upper bound

Definition. An upper bound z of A such that every $w < z$ is exceeded by some element of A .

In plain terms. The smallest number that is at least every element of the set (non-examinable).

Limit (of a sequence)

Definition. x such that $\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n > N : |a_n - x| < \varepsilon$.

In plain terms. The number the terms eventually stay as close to as you like.

Monomial

Definition. A single term $a_k x^k$ of a polynomial.

In plain terms. One piece of a polynomial.

Natural numbers

Definition. $\mathbb{N} = \{1, 2, 3, \dots\}$ (not including 0 in this module).

In plain terms. The counting numbers.

Negation

Definition. “not P ”: true exactly when P is false.

In plain terms. The opposite claim.

Numerical function

Definition. A function $f : D \rightarrow R$ with $D, R \subseteq \mathbb{R}$.

In plain terms. A function from numbers to numbers.

Numerical sequence

Definition. A sequence whose terms are real numbers.

In plain terms. An endless list of numbers.

Parameter

Definition. A letter standing for a fixed but unspecified number.

In plain terms. A number in the problem that you are not told, so the answer depends on it.

Polynomial

Definition. A function $f(x) = \sum_{k=0}^n a_k x^k$.

In plain terms. A sum of multiples of powers of x .

Proof

Definition. A demonstration that one proposition follows from one or more others.

In plain terms. A step-by-step argument that leaves no room for doubt.

Proposition

Definition. A statement that may be either true or false.

In plain terms. A claim.

Rational numbers

Definition. $\mathbb{Q}$: numbers p/q with $p \in \mathbb{Z}, q \in \mathbb{N}$.

In plain terms. Fractions.

Real numbers

Definition. $\mathbb{R}$: informally, the points of the number line.

In plain terms. All the numbers on the line, including $\sqrt{2}$ and π .

Root

Definition. A value $\bar{x}$ with $f(\bar{x}) = 0$.

In plain terms. Where the graph meets the horizontal axis.

Sequence

Definition. An infinite ordered list $(x_1, x_2, \dots)$; formally a function with domain $\mathbb{N}$.

In plain terms. A list of values, one for each period $1, 2, 3, \dots$

Set

Definition. A collection of objects (taken as a basic notion).

In plain terms. A bag of things; only membership matters.

Set-builder notation

Definition. $\{x \in A \mid x \text{ has property } P\}$.

In plain terms. "The members of A that satisfy P ."

Set difference

Definition. $A \setminus B = \{x \in A \mid x \notin B\}$.

In plain terms. A with everything in B thrown out.

Solution set

Definition. The set of all values for which an equation or inequality holds.

In plain terms. Every answer, collected together.

Subset

Definition. $A \subseteq B$: every element of A is an element of B .

In plain terms. A fits inside B .

Successor property

Definition. For any $n \in \mathbb{N}, n + 1 \in \mathbb{N}$.

In plain terms. There is always a next natural number.

Summation operator

Definition. $\sum_{j=k}^n a_j = a_k + a_{k+1} + \cdots + a_n$.

In plain terms. Shorthand for adding a list of terms.

Superset

Definition. B is a superset of A if $A \subseteq B$.

In plain terms. The bigger set that contains A .

Theorem

Definition. A proposition proved on the basis of axioms and previously established theorems.

In plain terms. A claim that has been shown to follow from the rules.

Triangle inequality

Definition. $|x - z| \leq |x - y| + |y - z|$; equivalently $|a + b| \leq |a| + |b|$.

In plain terms. A detour is never shorter than the direct route.

Trichotomy

Definition. Property O1: for real a, b exactly one of $a > b$, $a = b$, $a < b$ holds.

In plain terms. Any two numbers can be compared, in exactly one way.

Union

Definition. $A \cup B$: the objects in A , in B , or in both.

In plain terms. Everything in either set.

Universal quantifier

Definition. $\forall x \in A, P$ — every $x \in A$ has property P .

In plain terms. “All”; proved for an arbitrary element.

Upper bound

Definition. y with $x \leq y$ for every $x \in A$.

In plain terms. A ceiling for the set.

What do I actually need to know?

Tick each item only when you can do it without notes.

Chapter 1: The language of mathematics

- ☐ Define proposition, proof, axiom and theorem.
- ☐ Explain $P \Rightarrow Q$, $P \Leftrightarrow Q$ and “not P ”; give the converse and contrapositive of an implication.
- ☐ Negate inequalities, “and”/“or” statements and quantified statements.
- ☐ Prove $\exists$ statements by example and disprove $\forall$ statements by counterexample.
- ☐ Explain why $\forall n \exists m$ and $\exists m \forall n$ differ.

Chapter 2: Sets

- Define subset, set equality, intersection, union, set difference and set-builder notation.
- Calculate $A \cap B$, $A \cup B$, $A \setminus B$ for given sets.
- Prove a subset relation for arbitrary sets.
- Answer “find all sets D such that. . .” and “are there sets such that. . .” questions.

Chapter 3: Number systems

- State $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, their inclusions and the successor property.
- State O1–O2, A1–A5, M1–M6, especially both clauses of M6.
- Explain why $\mathbb{R}$ is needed beyond $\mathbb{Q}$.
- Write intervals in set-builder notation and back.

Chapter 4: Inequalities

- Solve linear inequalities, reversing the direction when dividing by a negative number.
- Solve parameter inequalities by cases $a > 0$, $a = 0$, $a < 0$.
- Solve product, quadratic and fractional inequalities by sign analysis.
- Use the discriminant and completing the square.

Chapter 5: Absolute value

- Define $|x|$ both ways and prove its basic properties.
- State and apply the triangle inequality; derive $|a + b| \leq |a| + |b|$ and $|a| \leq |a - b| + |b|$.
- Solve $|x - a| < r$ and $|x - a| > r$.

Chapter 6: Summation

- Evaluate sums with any integer limits; count $n - k + 1$ terms.
- State the one-term and empty-sum conventions and P1–P3.
- Find closed forms using P1, P2 and the constant-sum rule.

Chapter 7: Induction

- State the principle of induction with base, step and hypothesis.
- Prove summation formulas and simple inequalities by induction.
- State the alternative (strong) formulation and use it with several base cases.

Chapter 8: Binomial formula

- Compute $k!$ and C_n^j ; state the binomial formula.
- Expand powers and find specific coefficients.

- Prove lower bounds such as $1.02^{102} > 6$ using partial sums.

Chapter 9: Functions and polynomials

- Give the informal and formal definitions of a function and of equal functions.
- Define polynomial, degree, monomial and root.
- State Theorem L2.1 and use it to factorise polynomials.

Chapter 10: Sequences and limits

- Define sequence and numerical sequence; state the definition of a limit exactly.
- Find N for a given ε ; prove convergence from the definition.
- Prove that a sequence such as $1, 0, 1, 0, \dots$ has no limit.
- Prove that limits preserve non-strict inequalities.

Final practice paper

This paper is written in the style of the past ECN115 papers (definitions, “if yes give an example, if no argue why”, parameter questions and “explain why the argument is wrong”), but uses only the material in this book. Total: 100 marks.

Question F.1 [18 marks]

Let A, B be sets. (a) Define the set difference $A \setminus B$ and state what it means for A and B to be equal in terms of $\subseteq$. [5] (b) Are there non-empty sets of natural numbers with $A \cup B = A$ and $A \neq B$? If yes, give an example; if no, argue why. [4] (c) Find all sets $D \subseteq \mathbb{N}$ such that $\{1, 2\} \cup D = \{1, 2, 3, 4\}$ and $1 \notin D$. [5] (d) Prove that $A \setminus B \subseteq A$ for all sets A, B . [4]

Question F.2 [22 marks]

(a) Solve $4 - \frac{3}{2}x \geq x + 9$, naming the property that decides the direction at the step where you divide. [5] (b) Solve $(2 - x)(3x + 1) < 0$. [5] (c) Let $a \in \mathbb{R}$ be a parameter. Find all x with $(a - 2)x \leq a - 2$. Your answer should depend on a . [6] (d) Solve $\frac{x + 2}{x - 1} \leq 2$. [6]

Question F.3 [16 marks]

(a) State the triangle inequality for real numbers x, y, z . [2] (b) Prove that $|a| - |b| \leq |a - b|$ for all real a, b . [4] (c) Solve $|3x - 2| \leq 4$ and $|x + 1| \geq |x - 3|$. [6] (d) Show that if $|x - 2| < 1$ then $|x^2 - 4| < 5$. [4]

Question F.4 [16 marks]

(a) Evaluate $\sum_{j=-2}^3 (2j + 1)$ and $\sum_{j=4}^3 j^5$. [3] (b) Prove by induction that $\sum_{j=1}^n (3j - 1) = \frac{n(3n+1)}{2}$ for all $n \in \mathbb{N}$, identifying the base, the hypothesis and where it is used. [8] (c) A student checks the identity in (b) for $n = 1, \dots, 20$ and says this proves it. Explain why this is not a proof. [2] (d) Find $\sum_{j=1}^n (3j - 1)$ using P1, P2 and $\sum_{j=1}^n j = \frac{n(n+1)}{2}$ instead. [3]

Question F.5 [14 marks]

(a) Write the binomial formula for $(1 + x)^n$ and the first four terms explicitly. [5] (b) Show that $1.05^{40} > 6$. [6] (c) Find the coefficient of x^4 in $(1 - 2x)^7$. [3]

Question F.6 [14 marks]

(a) Define what it means that a number x is a limit of a sequence (a_n) . [4] (b) Prove from the definition that $\frac{2n+3}{n+1} \rightarrow 2$. [5] (c) A student argues: "The sequence $a_n = (-1)^n \frac{n}{n+1}$ has terms that get closer and closer to 1 and to -1 , so it converges to both 1 and -1 ." Explain why this argument is wrong and prove that (a_n) has no limit. [5]

Solutions to the final practice paper**Solution to Question F.1**

(a) $A \setminus B = \{x \in A \mid x \notin B\}$, the elements of A that are not elements of B . $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$. (b) Yes. $A \cup B = A$ holds exactly when $B \subseteq A$. Take $A = \{1, 2\}$, $B = \{1\}$: both non-empty, $A \cup B = \{1, 2\} = A$ and $A \neq B$. (c) $3, 4 \in D$ (they are in the union but not in $\{1, 2\}$); $D \subseteq \{1, 2, 3, 4\}$; $1 \notin D$; 2 is free. So $D \in \{\{3, 4\}, \{2, 3, 4\}\}$. (d) Let $x \in A \setminus B$. By definition $x \in A$ and $x \notin B$; in particular $x \in A$. As x was arbitrary, $A \setminus B \subseteq A$.

Solution to Question F.2

(a) $4 - \frac{3}{2}x \geq x + 9 \Leftrightarrow -\frac{5}{2}x \geq 5$. Multiplying by $-\frac{2}{5} < 0$ reverses the inequality (M6, second clause): $x \leq -2$. Solution $(-\infty, -2]$. (b) Critical points $x = 2$ and $x = -\frac{1}{3}$. The product is negative when the signs differ: $2 - x > 0$ and $3x + 1 < 0$ gives $x < -\frac{1}{3}$; $2 - x < 0$ and $3x + 1 > 0$ gives $x > 2$. Solution $(-\infty, -\frac{1}{3}) \cup (2, \infty)$. (c) If $a > 2$: divide by $a - 2 > 0$: $x \leq 1$, i.e. $(-\infty, 1]$. If $a < 2$: divide by $a - 2 < 0$, reversing: $x \geq 1$, i.e. $[1, \infty)$. If $a = 2$: $0 \leq 0$, true for all x : $\mathbb{R}$. (d) $\frac{x+2}{x-1} - 2 \leq 0 \Leftrightarrow \frac{x+2-2x+2}{x-1} \leq 0 \Leftrightarrow \frac{4-x}{x-1} \leq 0$. Critical points 1 (excluded) and 4 (included). For $x < 1$: numerator +, denominator -: quotient -, included. For $1 < x < 4$: both +: excluded. For $x > 4$: numerator -, denominator +: quotient -, included. Solution $(-\infty, 1) \cup [4, \infty)$.

Solution to Question F.3

(a) $|x - z| \leq |x - y| + |y - z|$. (b) With $x = a$, $y = b$, $z = 0$: $|a| \leq |a - b| + |b|$, so $|a| - |b| \leq |a - b|$. (c) $|3x - 2| \leq 4 \Leftrightarrow -4 \leq 3x - 2 \leq 4 \Leftrightarrow -\frac{2}{3} \leq x \leq 2$: $[-\frac{2}{3}, 2]$. For the second, square (both sides ≥ 0): $(x + 1)^2 \geq (x - 3)^2 \Leftrightarrow 2x + 1 \geq -6x + 9 \Leftrightarrow 8x \geq 8 \Leftrightarrow x \geq 1$: $[1, \infty)$ (points at least as far from -1 as from 3). (d) If $|x - 2| < 1$ then $1 < x < 3$, so $3 < x + 2 < 5$ and $|x + 2| < 5$; hence $|x^2 - 4| = |x - 2| |x + 2| < 1 \cdot 5 = 5$.

Solution to Question F.4

(a) $-3 - 1 + 1 + 3 + 5 + 7 = 12$; the second is an empty sum, 0. (b) Let P_n be the identity. Base: $3 - 1 = 2 = \frac{1 \cdot 4}{2}$. Hypothesis: assume P_n . Step: $\sum_{j=1}^{n+1} (3j - 1) = \frac{n(3n+1)}{2} + (3n + 2)$ (hypothesis used) $= \frac{3n^2 + 7n + 4}{2} = \frac{(n+1)(3n+4)}{2} = \frac{(n+1)(3(n+1)+1)}{2}$, which is P_{n+1} . By induction P_n holds for all $n \in \mathbb{N}$. (c) It verifies twenty propositions $P_1, \dots, P_{20}$; the claim concerns infinitely many, and nothing is said about P_{21} onwards. A proof needs the general step. (d) $3 \sum j - \sum 1 = \frac{3n(n+1)}{2} - n = \frac{3n^2 + n}{2} = \frac{n(3n+1)}{2}$.

Solution to Question F.5

(a) $(1 + x)^n = \sum_{j=0}^n C_n^j x^j = 1 + nx + \frac{n(n-1)}{2} x^2 + \frac{n(n-1)(n-2)}{6} x^3 + \dots$ (b) With $x = 0.05$, $n = 40$, all terms positive: $1.05^{40} > 1 + 40(0.05) + 780(0.0025) + 9880(0.000125) = 1 + 2 + 1.95 + 1.235 = 6.185 > 6$ (using $C_{40}^2 = 780$, $C_{40}^3 = 9880$). (c) General term $C_7^j (-2x)^j$; for $j = 4$: $C_7^4 \cdot 16 = 35 \cdot 16 = 560$.

Solution to Question F.6

- (a) x is a limit of (a_n) if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|a_n - x| < \varepsilon$ for all $n > N$.
- (b) $\left| \frac{2n+3}{n+1} - 2 \right| = \frac{1}{n+1} < \frac{1}{n}$. Given $\varepsilon > 0$, choose $N \geq \frac{1}{\varepsilon}$. For $n > N$: $\left| \frac{2n+3}{n+1} - 2 \right| < \frac{1}{n} < \frac{1}{N} \leq \varepsilon$.
- (c) A sequence has at most one limit, and being close to two values infinitely often prevents convergence rather than giving two limits. Proof: suppose x is a limit, and take $\varepsilon = \frac{1}{2}$. There is N with $|a_n - x| < \frac{1}{2}$ for $n > N$. Choose an even $m > N$ and an odd $k > N$ with $m, k \geq 3$; then $a_m = \frac{m}{m+1} \geq \frac{3}{4}$ and $a_k = -\frac{k}{k+1} \leq -\frac{3}{4}$. By the triangle inequality, $\frac{3}{2} \leq |a_m - a_k| \leq |a_m - x| + |x - a_k| < \frac{1}{2} + \frac{1}{2} = 1$, a contradiction. So (a_n) has no limit.